5-1-1

1. If F and f are differentiable functions such that $F(x) = \int_0^x f(t)dt$, and if $F(a) = -2$ and $F(b) = -2$ where $a<b$, which of the following must be true?

(A) $f(x)=0$ for some x such that $a<x<b$. ✓

(B) $f(x)>0$ for all x such that $a<x<b$.

(C) $f(x)<0$ for all x such that $a<x<b$.

(D) $F(x) \leq 0$ for all x such that $a<x<b$.

5-1-1

2. Let f be a twice-differentiable function such that $f(2) = 5$ and $f(5) = 2$. Let g be the function given by $g(x) = f(f(x))$.

(a) Explain why there must be a value c for $2 < c < 5$ such that $f'(c) = -1$.

(b) Show that $g'(2) = g'(5)$. Use this result to explain why there must be a value k for $2 < k < 5$ such that $g''(k) = 0$.

(c) Show that if $f''(x) = 0$ for all x, then the graph of g does not have a point of inflection.

(d) Let $h(x) = f(x) - x$. Explain why there must be a value r for $2 < r < 5$ such that $h(r) = 0$.

Part A

1 point is earned for $\dfrac{f(5) - f(2)}{5 - 2}$

1 point is earned for conclusion, using MVT

The Mean Value Theorem guarantees that there is a value c, with $2 < c < 5$, so that

$$f'(c) = \frac{f(5) - f(2)}{5 - 2} = \frac{2 - 5}{5 - 2} = -1.$$

0	1	2

The student response earns all of the following points:

1 point is earned for $\dfrac{f(5) - f(2)}{5 - 2}$

1 point is earned for conclusion, using MVT

The Mean Value Theorem guarantees that there is a value c, with $2 < c < 5$, so that

$$f'(c) = \frac{f(5) - f(2)}{5 - 2} = \frac{2 - 5}{5 - 2} = -1.$$

Part B

1 point is earned for $g'(x)$

1 point is earned for $g'(2) = f'(5) \cdot f'(2) = g'(5)$

1 point is earned if uses MVT with g'

5-1-1

$g'(x) = f'(f(x)) \cdot f'(x)$
$g'(2) = f'(f(2)) \cdot f'(2) = f'(5) \cdot f'(2)$
$g'(5) = f'(f(5)) \cdot f'(5) = f'(2) \cdot f'(5)$
Thus, $g'(2) = g'(5)$.

Since f is twice-differentiable, g' is differentiable everywhere, so the Mean Value Theorem applied to g' on [2, 5] guarantees there is a value k, with $2 < k < 5$, such that $g''(k) = \dfrac{g'(5) - g'(2)}{5 - 2} = 0$.

0	1	2	3 ✓

The student response earns all of the following points:

1 point is earned for $g'(x)$

1 point is earned for $g'(2) = f'(5) \cdot f'(2) = g'(5)$

1 point is earned if uses MVT with g'

$g'(x) = f'(f(x)) \cdot f'(x)$
$g'(2) = f'(f(2)) \cdot f'(2) = f'(5) \cdot f'(2)$
$g'(5) = f'(f(5)) \cdot f'(5) = f'(2) \cdot f'(5)$
Thus, $g'(2) = g'(5)$.

Since f is twice-differentiable, g' is differentiable everywhere, so the Mean Value Theorem applied to g' on [2, 5] guarantees there is a value k, with $2 < k < 5$, such that $g''(k) = \dfrac{g'(5) - g'(2)}{5 - 2} = 0$.

Part C

1 point is earned if considers g''

1 point is earned for $g''(x) = 0$ for all x

OR

1 point is earned for f is linear

1 point is earned for g is linear

$g''(x) = f''(f(x)) \cdot f'(x) \cdot f'(x) + f'(f(x)) \cdot f''(x)$
If $f''(x) = 0$ for all x, then
$g''(x) = 0 \cdot f'(x) \cdot f'(x) + f'(f(x)) \cdot 0 = 0$ for all x.

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Thus, there is no x-value at which $g''(x)$ changes sign, so the graph of g has no inflection points. OR

If $f''(x) = 0$ for all x, then f is linear, so $g = f \circ f$ is linear and the graph of g has no inflection points.

0	1	2

The student response earns all of the following points:

1 point is earned if considers g''

1 point is earned for $g''(x) = 0$ for all x

OR

1 point is earned for f is linear

1 point is earned for g is linear

$g''(x) = f''(f(x)) \cdot f'(x) \cdot f'(x) + f'(f(x)) \cdot f''(x)$
If $f''(x) = 0$ for all x, then
$g''(x) = 0 \cdot f'(x) \cdot f'(x) + f'(f(x)) \cdot 0 = 0$ for all x.

Thus, there is no x-value at which $g''(x)$ changes sign, so the graph of g has no inflection points. OR

If $f''(x) = 0$ for all x, then f is linear, so $g = f \circ f$ is linear and the graph of g has no inflection points.

Part D

1 point is earned for $h(2)$ and $h(5)$

1 point is earned for conclusion, using IVT

Let $h(x) = f(x) - x$.

$h(2) = f(2) - 2 = 5 - 2 = 3$

$h(5) = f(5) - 5 = 2 - 5 = -3$

Since $h(2) > 0 > h(5)$, the Intermediate Value Theorem guarantees that there is a value r, with $2 < r < 5$, such that $h(r) = 0$.

0	1	2

The student response earns all of the following points:

1 point is earned for $h(2)$ and $h(5)$

1 point is earned for conclusion, using IVT

Let $h(x) = f(x) - x$.

$h(2) = f(2) - 2 = 5 - 2 = 3$

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$h(5) = f(5) - 5 = 2 - 5 = -3$

Since $h(2) > 0 > h(5)$, the Intermediate Value Theorem guarantees that there is a value r, with $2 < r < 5$, such that $h(r) = 0$.

Let f be a twice-differentiable function such that $f(2)=5$ and $f(5)=2$. Let g be the function given by $g(x)=f(f(x))$.

3. Explain why there must be a value c for $2 < c < 5$ such that $f'(c)=-1$.

Part A

1 point is earned for $\dfrac{f(5)-f(2)}{5-2}$

1 point is earned for conclusion, using MVT

The Mean Value Theorem guarantees that there is a value c, with $2<c<5$, so that

$$f'(c)=\frac{f(5)-f(2)}{5-2}=\frac{2-5}{5-2}=-1.$$

0	1	2

The student response earns all of the following points:

1 point is earned for $\dfrac{f(5)-f(2)}{5-2}$

1 point is earned for conclusion, using MVT

The Mean Value Theorem guarantees that there is a value c, with $2<c<5$, so that

$$f'(c)=\frac{f(5)-f(2)}{5-2}=\frac{2-5}{5-2}=-1.$$

5-1-1

4. Show that $g'(2)=g'(5)$. Use this result to explain why there must be a value k for $2 < k < 5$ such that $g''(k)=0$.

Part B

1 point is earned for $g'(x)$

1 point is earned for $g'(2) = f'(5)*f'(2)=g'(5)$

1 point is earned if uses MVT with g'

$g'(x)=f'(f(x))*f'(x)$

$g'(2)=f'(f(2))*f'(2)=f'(5)*f'(2)$

$g'(5)=f'(f(5))*f'(5)=f'(2)*f'(5)$

Thus, $g'(2)=g'(5)$. Since f is twice-differentiable, g' is differentiable everywhere, so the Mean Value Theorem applied to g' on [2, 5] guarantees there is a value k, with $2<k<5$, such that $g''(k) = \dfrac{g'(5) - g'(2)}{5 - 2} = 0$.

0	1	2	3

The student response earns all of the following points:

1 point is earned for $g'(x)$

1 point is earned for $g'(2) = f'(5)*f'(2)=g'(5)$

1 point is earned if uses MVT with g'

$g'(x)=f'(f(x))*f'(x)$

$g'(2)=f'(f(2))*f'(2)=f'(5)*f'(2)$

$g'(5)=f'(f(5))*f'(5)=f'(2)*f'(5)$

Thus, $g'(2)=g'(5)$.

Since f is twice-differentiable, g' is differentiable everywhere, so the Mean Value Theorem applied to g' on [2, 5] guarantees there is a value k, with $2<k<5$, such that $g''(k) = \dfrac{g'(5) - g'(2)}{5 - 2} = 0$.

5-1-1

5.

x	-1.5	-1.0	-0.5	0	0.5	1.0	1.5
$f(x)$	-1	-4	-6	-7	-6	-4	-1
$f'(x)$	-7	-5	-3	0	3	5	7

Let f be a function that is differentiable for all real numbers. The table above gives the values of f and its derivative f' for selected points x in the closed interval $-1.5 \le x \le 1.5$. The second derivative of f has the property that $f''(x) > 0$ for $-1.5 \le x \le 1.5$.

(a) Evaluate $\int_0^{1.5} \left(3f'(x) + 4\right) dx$. Show the work that leads to your answer.

(b) Write an equation of the line tangent to the graph of f at the point where $x = 1$. Use this line to approximate the value of $f(1.2)$. Is this approximation greater than or less than the actual value of $f(1.2)$? Give a reason for your answer.

(c) Find a positive real number r having the property that there must exist a value c with $0 < c < 0.5$ and $f''(c) = r$. Give a reason for your answer.

(d) Let g be the function given by $g(x) = \begin{cases} 2x^2 - x - 7 & \text{for } x < 0 \\ 2x^2 + x - 7 & \text{for } x \ge 0. \end{cases}$

The graph of g passes through each of the points $(x, f(x))$ given in the table above. Is it possible that f and g are the same function? Give a reason for your answer.

Part A

1 point is earned for the antiderivative

1 point is earned for the answer

$$\int_0^{1.5} \left(3f'(x) + 4\right) dx = 3\int_0^{1.5} f'(x)dx + \int_0^{1.5} 4dx$$
$$= 3f(x) + 4x\Big|_0^{1.5} = 3(-1 - (-7)) + 4(1.5) = 24$$

0	1	2

The student response earns all of the following points:

1 point is earned for the antiderivative

1 point is earned for the answer

$$\int_0^{1.5} \left(3f'(x) + 4\right) dx = 3\int_0^{1.5} f'(x)dx + \int_0^{1.5} 4dx$$
$$= 3f(x) + 4x\Big|_0^{1.5} = 3(-1 - (-7)) + 4(1.5) = 24$$

Part B

1 point is earned for the tangent line

5-1-1

1 point is earned for computing y on tangent line at $x = 1.2$

1 point is earned for the answer with reason

$y = 5(x-1) - 4$
$f(1.2) \approx 5(0.2) - 4 = -3$

The approximation is less than $f(1.2)$ because the graph of f is concave up on the interval

$1 < x < 1.2$.

0	1	2	3 ✓

The student response earns all of the following points:

1 point is earned for the tangent line

1 point is earned for computing y on tangent line at $x = 1.2$

1 point is earned for the answer with reason

$y = 5(x-1) - 4$
$f(1.2) \approx 5(0.2) - 4 = -3$

The approximation is less than $f(1.2)$ because the graph of f is concave up on the interval

$1 < x < 1.2$.

Part C

1 point is earned for the reference to MVT for f' (or differentiability of f')

1 point is earned for the value of r for interval $0 \leq x \leq 0.5$

By the Mean Value Theorem there is a c with $0 < c < 0.5$ such that

$$f''(c) = \frac{f'(0.5) - f'(0)}{0.5 - 0} = \frac{3 - 0}{0.5} = 6 = r$$

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for the reference to MVT for f' (or differentiability of f')

5-1-1

1 point is earned for the value of r for interval $0 \leq x \leq 0.5$

By the Mean Value Theorem there is a c with $0 < c < 0.5$ such that

$$f''(c) = \frac{f'(0.5) - f'(0)}{0.5 - 0} = \frac{3 - 0}{0.5} = 6 = r$$

Part D

1 point is earned for the answers "no" with reference to g' or g''

1 point is earned for the correct reason

$$\lim_{x \to 0^-} g'(x) = \lim_{x \to 0^-} (4x - 1) = -1$$
$$\lim_{x \to 0^+} g'(x) = \lim_{x \to 0^+} (4x + 1) = +1$$

Thus g' is not continuous at $x = 0$, but f' is continuous at $x = 0$, so $f \neq g$.

OR

$g''(x) = 4$ for all $x \neq 0$, but it was shown in part (c) that $f''(c) = 6$ for some $c \neq 0$, so $f \neq g$.

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for the answers "no" with reference to g' or g''

1 point is earned for the correct reason

$$\lim_{x \to 0^-} g'(x) = \lim_{x \to 0^-} (4x - 1) = -1$$
$$\lim_{x \to 0^+} g'(x) = \lim_{x \to 0^+} (4x + 1) = +1$$

Thus g' is not continuous at $x = 0$, but f' is continuous at $x = 0$, so $f \neq g$.

OR

$g''(x) = 4$ for all $x \neq 0$, but it was shown in part (c) that $f''(c) = 6$ for some $c \neq 0$, so $f \neq g$.

5-1-1

6. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Let f be a twice-differentiable function such that $f'(1) = 0$. The second derivative of f is given by $f''(x) = x^2 \cos\left(x^2 + \pi\right)$ for $-1 \le x \le 3$.

(a) On what open intervals contained in $-1 < x < 3$ is the graph of f concave up? Give a reason for your answer.

(b) Does f have a relative minimum, a relative maximum, or neither at $x = 1$? Justify your answer.

(c) Use the Mean Value Theorem on the closed interval $[-1, 1]$ to show that $f'(-1)$ cannot equal 2.5.

(d) Does the graph of f have a point of inflection at $x = 0$? Give a reason for your answer.

Part A

A maximum of 1 out of 2 points is earned for one correct interval with reason and no incorrect intervals.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

- ☐ intervals
- ☐ reason

Solution:

The graph of f is concave up on $(1.253,\ 2.170\ (\text{or } 2.171))$ and $(2.802,\ 3)$ because $f''(x)$ is positive on those intervals.

Part B

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

5-1-1

0	1	2

The student response accurately includes both of the criteria below.

- ☐ answer
- ☐ justification

Solution:

Because $f'(1) = 0$ and $f''(1) = -0.540302 < 0$, by the Second Derivative Test f has a relative maximum at $x = 1$.

Part C

The first point requires use of average rate of change of f' on $[-1,\ 1]$.

One point is earned for using MVT to conclude that there would be a number c in the interval $(-1,\ 1)$ such that $f''(c) = -1.25$. Two points are earned for showing that this is not possible.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- ☐ considers average rate of change
- ☐ conclusion using Mean Value Theorem

Solution:

If $f'(-1) = 2.5$, then the average rate of change of f' on the closed interval $[-1,\ 1]$ would be $\frac{f'(1)-f'(-1)}{1-(-1)} = \frac{0-2.5}{2} = -1.25$.

Since f' is continuous on $[-1,\ 1]$ and differentiable on $(-1,\ 1)$, by the Mean Value Theorem there would be a number c in the interval $(-1,\ 1)$ such that $f''(c) = -1.25$.

However, since $f''(x) = x^2 \cos\left(x^2 + \pi\right)$, it follows that $-1 \le f''(x) \le 1$ on the interval $(-1,\ 1)$. Therefore, there is no number c in the interval such that $f''(c) = -1.25$.

Thus, it is not possible that $f'(-1) = 2.5$.

Part D

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

5-1-1

0	1	2

The student response accurately includes both of the criteria below.

- [ ] answer
- [ ] reason

Solution:

Because $f''(x)$ does not change sign at $x = 0$, the graph of f does not have a point of inflection there.

5-1-1

7. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Let f be a twice-differentiable function such that $f'(2) = 0$. The second derivative of f is given by $f''(x) = x^2 e^{2-x} - 1$ for $0 \le x \le 6$.

(a) On what open intervals contained in $0 < x < 6$ is the graph of f concave down? Give a reason for your answer.

(b) Does f have a relative minimum, a relative maximum, or neither at $x = 2$? Justify your answer.

(c) Use the Mean Value Theorem on the closed interval $[2, 4]$ to show that $f'(4)$ cannot equal 8.5.

(d) Does the graph of f have a point of inflection at $x = 5$? Give a reason for your answer.

Part A

A maximum of 1 out of 2 points is earned for one correct interval with reason and no incorrect intervals.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

- [ ] intervals
- [ ] reason

Solution:

The graph of f is concave down on $(0,\ 0.464)$ (or $(0,\ 0.463)$) and $(5.357,\ 6)$ (or $(5.356,\ 6)$) because $f''(x)$ is negative on those intervals.

Part B

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

5-1-1

0	1	2

The student response accurately includes both of the criteria below.

- ☐ answer
- ☐ justification

Solution:

Because $f'(2) = 0$ and $f''(2) = 3 > 0$, by the Second Derivative Test f has a relative minimum at $x = 2$.

Part C

The first point requires use of average rate of change of f' on $[2, 4]$.

The second point is earned for using MVT to conclude that there would be a number c in the interval $(2, 4)$ such that $f''(c) = 4.25$. The third point is earned for showing that this is not possible.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- ☐ considers average rate of change
- ☐ uses Mean Value Theorem to determine $f''(c) = 4.25$
- ☐ conclusion of "not possible" using Mean Value Theorem

Solution:

If $f'(4) = 8.5$, then the average rate of change of f' on the closed interval $[2, 4]$ would be $\frac{f'(4)-f'(2)}{4-2} = \frac{8.5-0}{2} = 4.25$.

Since f' is continuous on $[2, 4]$ and differentiable on $(2, 4)$, by the Mean Value Theorem there would be a number c in the interval $(2, 4)$ such that $f''(c) = 4.25$.

However, since $f''(x) = x^2e^{2-x} - 1$, it follows that $f''(x) = 4.25$ has no solution on the interval $[2, 4]$. Therefore, there is no number c in the interval $(2, 4)$ such that $f''(c) = 4.25$.

Thus, it is not possible that $f'(4) = 8.5$.

Part D

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

5-1-1

0	1	2

The student response accurately includes both of the criteria below.

- [ ] answer
- [ ] reason

Solution:

Because $f''(x)$ does not change sign at $x = 5$, the graph of f does not have a point of inflection there. $f''(x) > 0$ on $(0.464,\ 5.357)$.

5-1-1

8. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Let f be a twice-differentiable function with $f(0) = 4$. The derivative of f is given by $f'(x) = \sin\left(x^2 - 2x + 1\right)$ for $-2 \leq x \leq 2$.

(a) Find all values of x in the interval $-2 < x < 2$ at which f has a critical point. Classify each as the location of a relative minimum, a relative maximum, or neither. Justify your answers.

(b) Use the line tangent to the graph of f at $x = 0$ to approximate $f(0.25)$.

(c) On the interval $0 \leq x \leq 0.25$, $f'(x) > 0$ and $f''(x) < 0$. Is the approximation found in part (b) an overestimate or an underestimate for $f(0.25)$? Give a reason for your answer.

(d) Using the Mean Value Theorem, explain why the average rate of change of f over the interval $-2 \leq x \leq 2$ cannot equal 1.25.

Part A

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

A maximum of 1 out of 3 points is earned for reference $f'(x) = 0$ to and the three critical points without identification and/or justification, and no incorrect critical points are included.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- ☐ relative maximum at $x = -1.507$ with justification
- ☐ relative minimum at $x = -0.772$ with justification
- ☐ neither at $x = 1$ with justification

Solution:

$f'(x) = 0 \Rightarrow x = -1.507$ (or -1.506), $x = -0.772$, $x = 1$

5-1-1

f has a relative maximum at $x = -1.507$ (or -1.506) because f' changes from positive to negative there.

f has a relative minimum at $x = -0.772$ because f' changes from negative to positive there.

f has neither a relative minimum nor a relative maximum at $x = 1$ because f' does not change sign there.

Part B

A response of $4 + (0.841471)(0.25)$ or $4 + (\sin(1))(0.25)$ earns both points.

The second point requires substitution of values without a simplified answer. Trigonometric function values do not need to be evaluated.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

✓

0	1	2

The student response accurately includes both of the criteria below.

- ☐ form of tangent line approximation
- ☐ answer

Solution:

$f'(0) = \sin(1) = 0.841471$

$f(0.25) \approx f(0) + f'(0)(0.25) = 4.210$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

✓

0	1	2

The student response accurately includes both of the criteria below.

- ☐ answer
- ☐ reason

Solution:

Because $f''(x) < 0$ for $0 \leq x \leq 0.25$, the graph of f is concave down for $0 \leq x \leq 0.25$.

Therefore, the tangent line approximation from part (b) is an overestimate for $f(0.25)$.

Part D

One point is earned for using MVT to conclude that there should be a number c for $-2 < c < 2$ such that $f'(c)$ equals the average rate of change of f over the interval $-2 \leq x \leq 2$. Two points are earned for showing that this is not possible.

5-1-1

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes a correct explanation.

Solution:

f' is differentiable for $-2 \le x \le 2$.

$\Rightarrow f$ is continuous for $-2 \le x \le 2$.

Because the Mean Value Theorem can be applied to f on the interval $-2 \le x \le 2$, there should be a number c for $-2 < c < 2$ such that $f'(c)$ equals the average rate of change of f over the interval $-2 \le x \le 2$.

However, since $-1 \le f'(x) \le 1$, there can be no number c such that $f'(c) = 1.25$. Therefore, the average rate of change cannot equal 1.25.

5-1-1

9. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Let f be a twice-differentiable function with $f(1.5) = 3$. The derivative of f is given by $f'(x) = (7x - 19)\sin\left(x^2 - 4x + 4\right)$ for $1 \leq x \leq 4$.

(a) Find all values of x in the interval $1 < x < 4$ at which f has a critical point. Classify each as the location of a relative minimum, a relative maximum, or neither. Justify your answers.

(b) Use the line tangent to the graph of f at $x = 1.5$ to approximate $f(1.8)$.

On the interval $1.5 \leq x \leq 1.8$, $f'(x) < 0$ and $f''(x) > 0$. Is the approximation found in part (b) an overestimate or an underestimate for $f(1.8)$? Give a reason for your answer.

Using the Mean Value Theorem, explain why the average rate of change of f over the interval $1 \leq x \leq 4$ cannot equal 6.5.

Part A

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

A maximum of 1 out of 3 points is earned for reference to $f'(x) = 0$ and the three critical points without identification and/or justification, and no incorrect critical points are included.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- ☐ neither at $x = 2$ with justification
- ☐ relative minimum at $x = 2.714$ with justification
- ☐ relative maximum at $x = 3.772$ with justification

Solution:

$f'(x) = 0 \;\Rightarrow\; x = 2,\ x = 2.714,\ x = 3.772$

5-1-1

f has neither a relative minimum nor a relative maximum at $x = 2$ because f'does not change sign there.

f has a relative minimum at $x = 2.714$ because f' changes from negative to positive there.

f has a relative maximum at $x = 3.772$ because f' changes from positive to negative there.

Part B

A response of $3 + (-2.102934)(0.3)$ or $3 + (-8.5\sin(0.25))(0.3)$ earns both points.

The second point requires substitution of values without a simplified answer. Trigonometric function values do not need to be evaluated.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] form of tangent line approximation
- [ ] answer

Solution:

$f'(1.5) = -8.5\sin(0.25) = -2.102934$

$f(1.8) \approx f(1.5) + f'(1.5)(0.3) = 2.369$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] answer
- [ ] reason

Solution:

Because $f''(x) > 0$ for $1.5 \le x \le 1.8$, the graph of f is concave up for $1.5 \le x \le 1.8$.

Therefore, the tangent line approximation from part (b) is an underestimate for $f(1.8)$.

Part D

The first point is earned for using MVT to conclude that there should be a number c for $1 < c < 4$ such that $f'(c)$ equals the average rate of change of f over the interval $1 \le x \le 4$. The second point is earned for showing that this is not possible.

5-1-1

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] uses Mean Value Theorem to determine there should be a c such that $f'(c)$ equals the average rate of change of f
- [ ] explanation using Mean Value Theorem

Solution:

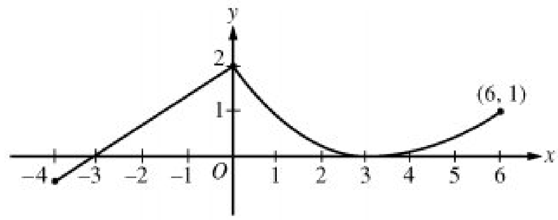
f' is differentiable for $1 \le x \le 4$. $\Rightarrow f$ is continuous for $1 \le x \le 4$.

Because the Mean Value Theorem can be applied to f on the interval $1 \le x \le 4$, there should be a number c for $1 < c < 4$ such that $f'(c)$ equals the average rate of change of f over the interval $1 \le x \le 4$.

However, because the equation $f'(x) = 6.5$ has no solution on the interval $1 \le x \le 4$, there can be no number c such that $f'(c) = 6.5$. Therefore, the average rate of change cannot equal 6.5.

5-1-1

10.



Graph of f

A continuous function f is defined on the closed interval $-4 \leq x \leq 6$. The graph of f consists of a line segment and a curve that is tangent to the x -axis at $x = 3$, as shown in the figure above. On the interval $0 < x < 6$, the function f is twice differentiable, with $f''(x) > 0$.

(a) Is f differentiable at $x = 0$? Use the definition of the derivative with one-sided limits to justify your answer.

(b) For how many values of a , $-4 \leq a < 6$, is the average rate of change of f on the interval [a , 6] equal to 0? Give a reason for your answer.

(c) Is there a value of a, $-4 \leq a < 6$, for which the Mean Value Theorem, applied to the interval [a, 6], guarantees a value c, $a < c < 6$, at which $f'(c) = \frac{1}{3}$? Justify your answer.

(d) The function g is defined by $g(x) = \int_0^x f(t)dt$ for $-4 \leq x \leq 6$. On what intervals contained in [-4, 6] is the graph of g concave up? Explain your reasoning.

Part A

1 point is earned if sets up difference quotient at $x = 0$

1 point is earned for the answer with justification

$$\lim_{h \to 0^-} \frac{f(h) - f(0)}{h} = \frac{2}{3}$$
$$\lim_{h \to 0^+} \frac{f(h) - f(0)}{h} < 0$$

Since the one-sided limits do not agree, f is not differentiable at $x = 0$.

0	1	2

The student response earns all of the following points:

1 point is earned if sets up difference quotient at $x = 0$

1 point is earned for the answer with justification

$$\lim_{h \to 0^-} \frac{f(h) - f(0)}{h} = \frac{2}{3}$$
$$\lim_{h \to 0^+} \frac{f(h) - f(0)}{h} < 0$$

Since the one-sided limits do not agree, f is not differentiable at $x = 0$.

5-1-1

Part B

1 point is earned for expression of average rate of change

1 point is earned for answer with reason

$\frac{f(6)-f(a)}{6-a} = 0$ when $f(a) = f(6)$. There are two values of a for which this is true.

		✓
0	1	2

The student response earns all of the following points:

1 point is earned for expression of average rate of change

1 point is earned for answer with reason

$\frac{f(6)-f(a)}{6-a} = 0$ when $f(a) = f(6)$. There are two values of a for which this is true.

Part C

1 point is earned for answering "yes" and identifying $a = 3$

1 point is earned for the justification

Yes, $a = 3$. The function f is differentiable on the interval $3 < x < 6$ and continuous on $3 \le x \le 6$.

Also, $\frac{f(6)-f(3)}{6-3} = \frac{1-0}{6-3} = \frac{1}{3}$.

By the Mean Value Theorem, there is a value c, $3 < c < 6$, such that $f'(c) = \frac{1}{3}$.

		✓
0	1	2

The student response earns all of the following points:

1 point is earned for answering "yes" and identifying $a = 3$

1 point is earned for the justification

Yes, $a = 3$. The function f is differentiable on the interval $3 < x < 6$ and continuous on $3 \le x \le 6$.

Also, $\frac{f(6)-f(3)}{6-3} = \frac{1-0}{6-3} = \frac{1}{3}$.

By the Mean Value Theorem, there is a value c, $3 < c < 6$, such that $f'(c) = \frac{1}{3}$.

Part D

1 point is earned for $g'(x) = f(x)$

5-1-1

1 point is earned for considering $g''(x) > 0$

1 point is earned for the answer

$g'(x) = f(x)$, $g''(x) = f'(x)$

$g''(x) > 0$ when $f'(x) > 0$

This is true for $-4 < x < 0$ and $3 < x < 6$.

0	1	2	3

The student response earns all of the following points:

1 point is earned for $g'(x) = f(x)$

1 point is earned for considering $g''(x) > 0$

1 point is earned for the answer

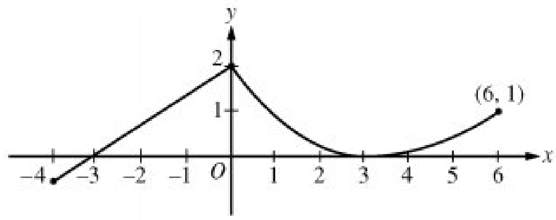
$g'(x) = f(x)$, $g''(x) = f'(x)$

$g''(x) > 0$ when $f'(x) > 0$

This is true for $-4 < x < 0$ and $3 < x < 6$.

5-1-1

11.



Graph of f

A continuous function f is defined on the closed interval $-4 \leq x \leq 6$. The graph of f consists of a line segment and a curve that is tangent to the x -axis at x =3, as shown in the figure above. On the interval 0< x <6, the function f is twice differentiable, with $f''(x)>0$.

Is there a value of a,$-4 \leq a < 6$, for which the Mean Value Theorem, applied to the interval [a, 6], guarantees a value c, $a<c<6$, at which $f'(c) = \frac{1}{3}$? Justify your answer.

Part C

1 point is earned for the answers "yes" and identifies a=3

1 point is earned for the justification

Yes, a=3. The function f is differential on the

interval $3<x<6$ and continuous on $3 \leq x \leq 6$.

Also, $\frac{f(6)-f(3)}{6-3} = \frac{1-0}{6-3} = \frac{1}{3}$.

By the Mean Value Theorem, there is a value c,

$3<c<6$, such that $f'(c) = \frac{1}{3}$.

0	1	2

The student response earns two of the following points:

1 point is earned for the answers "yes" and identifies a=3

1 point is earned for the justification

Yes, a=3. The function f is differentiable on the

interval $3<x<6$ and continuous on $3 \leq x \leq 6$.

Also, $\frac{f(6)-f(3)}{6-3} = \frac{1-0}{6-3} = \frac{1}{3}$.

By the Mean Value Theorem, there is a value c,

5-1-1

$3<c<6$, such that $f'(c) = \frac{1}{3}$.

12. The function h is differentiable, and for all values of x, $h(x) = h(2-x)$. Which of the following statements must be true?

I. $\int_0^2 h(x)dx > 0$

II. $h'(1) = 0$

III. $h'(0) = h'(2) = 1$

(A) I only

(B) II only ✓

(C) III only

(D) II and III only

(E) I, II, and III

5-1-1

13.

t (seconds)	0	10	40	60
$B(t)$ (meters)	100	136	9	49
$v(t)$ (meters per second)	2.0	2.3	2.5	4.6

Ben rides a unicycle back and forth along a straight east-west track. The twice-differentiable function B models Ben's position on the track, measured in meters from the western end of the track, at time t, measured in seconds from the start of the ride. The table above gives values for $B(t)$ and Ben's velocity, $v(t)$, measured in meters per second, at selected times t.

(a) Use the data in the table to approximate Ben's acceleration at time $t = 5$ seconds. Indicate units of measure.

(b) Using correct units, interpret the meaning of $\int_0^{60} |v(t)|dt$ in the context of this problem. Approximate $\int_0^{60} |v(t)|dt$ using a left Riemann sum with the subintervals indicated by the data in the table.

(c) For $40 \le t \le 60$, must there be a time t when Ben's velocity is 2 meters per second? Justify your answer.

(d) A light is directly above the western end of the track. Ben rides so that at time t, the distance $L(t)$ between Ben and the light satisfies $(L(t))^2 = 12^2 + (B(t))^2$. At what rate is the distance between Ben and the light changing at time $t = 40$?

Part A

1 point is earned for answer

$$a(5) \approx \frac{v(10) - v(0)}{10 - 0} = \frac{0.3}{10} = 0.03 \text{ meters/sec}^2$$

1 point is earned for units in (a) or (b)

0	1	2

The student response earns all of the following points:

1 point is earned for answer

$$a(5) \approx \frac{v(10) - v(0)}{10 - 0} = \frac{0.3}{10} = 0.03 \text{ meters/sec}^2$$

1 point is earned for units in (a) or (b)

Part B

1 point is earned for meaning of the integral

1 point is earned for the approximation

$\int_0^{60} |v(t)|dt$ is the total distance, in meters, that Ben rides over the 60-second interval $t = 0$ to $t = 60$.

5-1-1

$$\int_0^{60} |v(t)|dt \approx 2.0 \cdot 10 + 2.3(40 - 10) + 2.5(60 - 40) = 139 \text{ meters}$$

1 point is earned for units in (a) or (b)

0	1	2	3

The student response earns all of the following points:

1 point is earned for meaning of the integral

1 point is earned for the approximation

$\int_0^{60} |v(t)|dt$ is the total distance, in meters, that Ben rides over the 60-second interval $t = 0$ to $t = 60$.

$$\int_0^{60} |v(t)|dt \approx 2.0 \cdot 10 + 2.3(40 - 10) + 2.5(60 - 40) = 139 \text{ meters}$$

1 point is earned for units in (a) or (b)

Part C

1 point is earned for difference quotient

1 point is earned for conclusion with justification

Because $\dfrac{B(60) - B(40)}{60 - 40} = \dfrac{49 - 9}{20} = 2$, the Mean Value Theorem implies there is a time t, $40 < t < 60$, such that $v(t) = 2$.

0	1	2

The student response earns all of the following points:

1 point is earned for difference quotient

1 point is earned for conclusion with justification

Because $\dfrac{B(60) - B(40)}{60 - 40} = \dfrac{49 - 9}{20} = 2$, the Mean Value Theorem implies there is a time t, $40 < t < 60$, such that $v(t) = 2$.

Part D

5-1-1

1 point is earned for the derivatives

1 point is earned if uses $B'(t) = v(t)$

1 point is earned for the answer

$$2L(t)L'(t) = 2B(t)B'(t)$$
$$L'(40) = \frac{B(40)v(40)}{L(40)} = \frac{9 \cdot 2.5}{\sqrt{144+81}} = \frac{3}{2} \text{ meters/sec}$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for the derivatives

1 point is earned if uses $B'(t) = v(t)$

1 point is earned for the answer

$$2L(t)L'(t) = 2B(t)B'(t)$$
$$L'(40) = \frac{B(40)v(40)}{L(40)} = \frac{9 \cdot 2.5}{\sqrt{144+81}} = \frac{3}{2} \text{ meters/sec}$$

5-1-1

14.

t (seconds)	0	10	40	60
$B(t)$ (meters)	100	136	9	49
$v(t)$ (meters per second)	2.0	2.3	2.5	4.6

Ben rides a unicycle back and forth along a straight east-west track. The twice-differentiable function B models Ben's position on the track, measured in meters from the western end of the track, at time t, measured in seconds from the start of the ride. The table above gives values for $B(t)$ and Ben's velocity, $v(t)$, measured in meters per second, at selected times t.

For $40 \le t \le 60$, must there be a time t when Ben's velocity is 2 meters per second? Justify your answer.

Part C

1 point is earned for difference quotient

1 point is earned for conclusion with justification

Because $\frac{B(60)-B(40)}{60-40} = \frac{49-9}{20} = 2$, the Mean Value Theorem implies there is a time t, $40<t<60$, such that $v(t)=2$.

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for difference quotient

1 point is earned for conclusion with justification

Because $\frac{B(60)-B(40)}{60-40} = \frac{49-9}{20} = 2$, the Mean Value Theorem implies there is a time t, $40<t<60$, such that $v(t)=2$.

5-1-1

15.

t (sec)	0	15	25	30	35	50	60
$v(t)$ (ft/sec)	–20	–30	–20	–14	–10	0	10
$a(t)$ (ft/sec^2)	1	5	2	1	2	4	2

A car travels on a straight track. During the time interval $0 \le t \le 60$ seconds, the car's velocity v, measured in feet per second, and acceleration a, measured in feet per second per second, are continuous functions. The table above shows selected values of these functions.

(a) Using appropriate units, explain the meaning of $\int_{30}^{60} |v(t)|dt$ in terms of the car's motion. Approximate $\int_{30}^{60} |v(t)|dt$ using a trapezoidal approximation with the three subintervals determined by the table.

(b) Using appropriate units, explain the meaning of $\int_{0}^{30} a(t)dt$ in terms of the car's motion. Find the exact value of $\int_{0}^{30} a(t)dt$.

(c) For $0 < t < 60$, must there be a time t when $v(t) = -5$? Justify your answer.

(d) For $0 < t < 60$, must there be a time t when $a(t) = 0$? Justify your answer.

Part A

1 point is earned for the explanation

1 point is earned for the value

$\int_{30}^{60} |v(t)|dt$ is the distance in feet that the car travels from $t = 30$ sec to $t = 60$ sec.

Trapezoidal approximation for $\int_{30}^{60} |v(t)|dt$:

$$A = \frac{1}{2}(14 + 10)5 + \frac{1}{2}(10)(15) + \frac{1}{2}(10)(10) = 185 \text{ ft}$$

1 point is earned for units in both (a) and (b)

Units of ft in (a) and ft/sec in (b)

0	1	2	3

The student response earns all of the following points:

1 point is earned for the explanation

5-1-1

1 point is earned for the value

$\int_{30}^{60} |v(t)| dt$ is the distance in feet that the car travels from $t = 30$ sec to $t = 60$ sec.

Trapezoidal approximation for $\int_{30}^{60} |v(t)| dt$:

$$A = \frac{1}{2}(14 + 10)5 + \frac{1}{2}(10)(15) + \frac{1}{2}(10)(10) = 185 \text{ ft}$$

1 point is earned for units in both (a) and (b)

Units of ft in (a) and ft/sec in (b)

Part B

1 point is earned for the explanation

1 point is earned for the value

$\int_{0}^{30} a(t)\, dt$ is the car's change in velocity in ft/sec from $t = 0$ sec to $t = 30$ sec.

$$\int_{0}^{30} a(t)\, dt = \int_{0}^{30} v'(t)\, dt = v(30) - v(0)$$

$= -14 - (-20) = 6$ ft/sec

1 point is earned for units in both (a) and (b)

Units of ft in (a) and ft/sec in (b)

0	1	2	3

The student response earns all of the following points:

1 point is earned for the explanation

1 point is earned for the value

$\int_{0}^{30} a(t)\, dt$ is the car's change in velocity in ft/sec from $t = 0$ sec to $t = 30$ sec.

5-1-1

$$\int_0^{30} a(t)\,dt = \int_0^{30} v'(t)\,dt = v(30) - v(0)$$

$= -14 - (-20) = 6$ ft/sec

1 point is earned for units in both (a) and (b)

Units of ft in (a) and ft/sec in (b)

Part C

1 point is earned for $v(35) < -5 < v(50)$

1 point is earned for Yes; refers to IVT or hypotheses

Yes. Since $v(35) = -10 < -5 < 0 = v(50)$, the IVT guarantees a t in $(35, 50)$ so that $v(t) = -5$.

0	1	2

The student response earns all of the following points:

1 point is earned for $v(35) < -5 < v(50)$

1 point is earned for Yes; refers to IVT or hypotheses

Yes. Since $v(35) = -10 < -5 < 0 = v(50)$, the IVT guarantees a t in $(35, 50)$ so that $v(t) = -5$.

Part D

1 point is earned for $v(0) = v(25)$

1 point is earned for Yes; refers to MVT or hypotheses

Yes. Since $v(0) = v(25)$, the MVT guarantees a t in $(0,25)$ so that $a(t) = v'(t) = 0$.

0	1	2

The student response earns all of the following points:

1 point is earned for $v(0) = v(25)$

1 point is earned for Yes; refers to MVT or hypotheses

Yes. Since $v(0) = v(25)$, the MVT guarantees a t in $(0,25)$ so that $a(t) = v'(t) = 0$.

5-1-1

16.

t (sec)	0	15	25	30	35	50	60
$v(t)$ (ft/sec)	–20	–30	–20	–14	–10	0	10
$a(t)$ (ft/sec^2)	1	5	2	1	2	4	2

A car travels on a straight track. During the time interval $0 \le t \le 60$ seconds, the car's velocity v, measured in feet per second, and acceleration a, measured in feet per second per second, are continuous functions. The table above shows selected values of these functions.

For $0<t<60$, must there be a time t when $a(t)=0$? Justify your answer.

Part D

1 point is earned for $v(0)=v(25)$

1 point is earned for Yes; refers to MVT or hypotheses

Yes. Since $v(0)=v(25)$, the MVT guarantees a t in $(0,25)$ so that $a(t)=v'(t)=0$.

0	1	2

The student response earns all of the following points:

1 point is earned for $v(0)=v(25)$

1 point is earned for Yes; refers to MVT or hypotheses

Yes. Since $v(0)=v(25)$, the MVT guarantees a t in $(0,25)$ so that $a(t)=v'(t)=0$.

5-1-1

17. The wind chill is the temperature, in degrees Fahrenheit (^{o}F), a human feels based on the air temperature, in degrees Fahrenheit, and the wind velocity v, in miles per hour (mph). If the air temperature is 32 ^{o}F, then the wind chill is given by $W(v)$=55.6-22.1$v^{0.16}$ and is valid for $5 \le v \le 60$.

Find the average rate of change of W over the interval $5 \le v \le 60$. Find the value of v at which the instantaneous rate of change of W is equal to the average rate of change of W over the interval $5 \le v \le 60$ ..

Part A

1 point is earned for the average rate of change.

1 point is earned for $W'(v)$ = average rate of change

1 point is earned for the value of v.

The average rate of change of W over the interval

$5 \le v \le 60$ is $\dfrac{W(60)-W(5)}{60-5}$ =-0.253 or -0.254.

$W'(v)=\dfrac{W(60)-W(5)}{60-5}$ when v=23.011.

0	1	2	3 ✓

The student response earns all of the following points:

1 point is earned for the average rate of change.

1 point is earned for $W'(v)$ = = average rate of change

1 point is earned for the value of v.

The average rate of change of W over the interval

$5 \le v \le 60$ is $\dfrac{W(60)-W(5)}{60-5}$ =-0.253 or -0.254.

$W'(v)=\dfrac{W(60)-W(5)}{60-5}$ when v=23.011.

18. Let f be the function defined by $f(x) = x + \ln x$. What is the value of c for which the instantaneous rate of change of f at $x = c$ is the same as the average rate of change of f over [1,4] ?

(A) 0.456

(B) 1.244

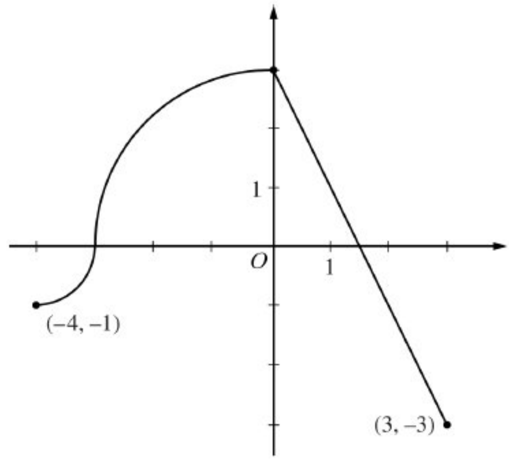
(C) 2.164 ✓

(D) 2.342

(E) 2.452

5-1-1

19.



Graph of f

The continuous function f is defined on the interval $-4 \leq x \leq 3$. The graph of f consists of two quarter circles and one line segment, as shown in the figure above. Let $g(x) = 2x + \int_0^x f(t)dt$.

Find the average rate of change of f on the interval $-4 \leq x \leq 3$. There is no point c, $-4 < c < 3$, for which $f'(c)$ is equal to that average rate of change. Explain why this statement does not contradict the Mean Value Theorem.

Part D

1 point is earned for average rate of change

1 point is earned for explanation

The average rate of change of f on the interval

$$-4 \leq x \leq 3\, is \frac{f(3)-f(-4)}{3-(-4)} = -\frac{2}{7}$$

To apply the Mean Value Theorem, f must be differentiable at each point in the interval $-4<x<3$. However, f is not differentiable at x=-3 and x=0.

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for average rate of change

1 point is earned for explanation

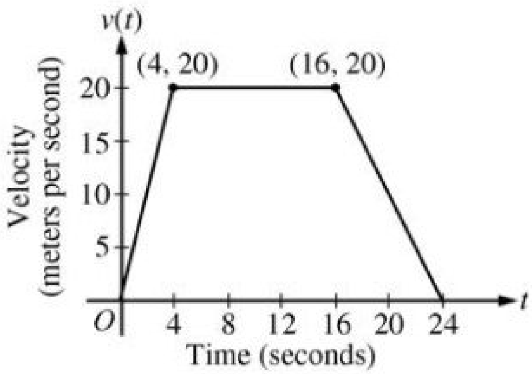
The average rate of change of f on the interval

$$-4 \leq x \leq 3\, is \frac{f(3)-f(-4)}{3-(-4)} = -\frac{2}{7}$$

To apply the Mean Value Theorem, f must be differentiable at each point in the interval $-4<x<3$. However, f is not differentiable at x=-3 and x=0.

5-1-1

20.



A car is traveling on a straight road. For $0 \leq t \leq 24$ seconds, the car's velocity *v(t)*, in meters per second, is modeled by the piecewise-linear function defined by the graph above.

Find the average rate of change of *v* over the interval $8 \leq t \leq 20$. Does the Mean Value Theorem guarantee a value of *c*, for $8 < c < 20$, such that *v′(c)* is equal to this average rate of change? Why or why not?

Part D

1 point is earned for the average rate of change of *v* on [8, 20]

1 point is earned for the answer with explanation

The average rate of change of *v* on [8, 20] is

$$\frac{v(20)-v(8)}{20-8} = -\frac{5}{6} m/\text{sec}^2.$$

No, the Mean Value Theorem does not apply to *v* on [8, 20] because *v* is not differentiable at $t = 16$.

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for the average rate of change of *v* on [8, 20]

1 point is earned for the answer with explanation

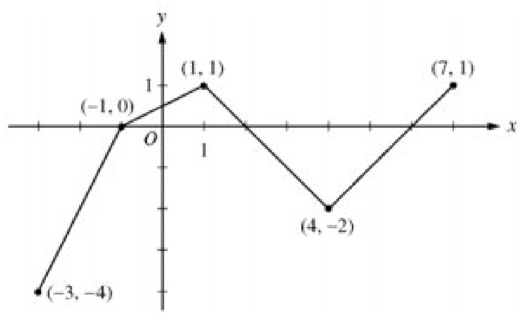
The average rate of change of *v* on [8, 20] is

$$\frac{v(20)-v(8)}{20-8} = -\frac{5}{6} m/\text{sec}^2.$$

No, the Mean Value Theorem does not apply to *v* on [8, 20] because *v* is not differentiable at $t = 16$.

5-1-1

21.



Graph of g'

Let g be a continuous function with $g(2) = 5$. The graph of the piecewise-linear function g', the derivative of g, is shown above for $-3 \le x \le 7$.

Find the average rate of change of $g'(x)$ on the interval $-3 \le x \le 7$. Does the Mean Value Theorem applied on the interval $-3 \le x \le 7$ guarantee a value of c, for $-3 < c < 7$, such that $g''(c)$ is equal to this average rate of change? Why or why not?

Part D

1 point is earned for average value of $g'(x)$

1 point is earned for answer "No" with reason

$$\frac{g\prime(7) - g\prime(-3)}{7 - (-3)} = \frac{1 - (-4)}{10} = \frac{1}{2}$$

No, the MVT does *not* guarantee the existence of a value c with the stated properties because g' is not differentiable for at least one point in $-3<x<7$.

0	1	2

The student response earns all of the following points:

1 point is earned for average value of $g'(x)$

1 point is earned for answer "No" with reason

$$\frac{g\prime(7) - g\prime(-3)}{7 - (-3)} = \frac{1 - (-4)}{10} = \frac{1}{2}$$

No, the MVT does *not* guarantee the existence of a value c with the stated properties because g' is not differentiable for at least one point in $-3<x<7$.

5-1-1

22.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	6	4	2	5
2	9	2	3	1
3	10	−4	4	2
4	−1	3	6	7

The functions f and g are differentiable for all real numbers, and g is strictly increasing. The table above gives values of the functions and their first derivatives at selected values of x. The function h is given by $h(x) = f(g(x)) - 6$.

Explain why there must be a value c for $1 < c < 3$ such that $h'(c) = -5$.

Part B

1 point is earned for: $\frac{h(3)-h(1)}{3-1}$

1 point is earned for the conclusion, using MVT

$$\frac{h(3)-h(1)}{3-1} = \frac{-7-3}{3-1} = -5$$

Since h is continuous and differentiable, bu the Mean Value Theorem, there exists a value c, $1 < c < 3$, such that $h'(c) = -5$.

0	1	2

The student response earns all of the following points:

1 point is earned for: $\frac{h(3)-h(1)}{3-1}$

1 point is earned for the conclusion, using MVT

$$\frac{h(3)-h(1)}{3-1} = \frac{-7-3}{3-1} = -5$$

Since h is continuous and differentiable, bu the Mean Value Theorem, there exists a value c, $1 < c < 3$, such that $h'(c) = -5$.

5-1-1

23.

t (minutes)	0	1	2	3	4	5	6
$C(t)$ (ounces)	0	5.3	8.8	11.2	12.8	13.8	14.5

Hot water is dripping through a coffeemaker, filling a large cup with coffee. The amount of coffee in the cup at time t, $0 \le t \le 6$, is given by a differentiable function C, where t is measured in minutes. Selected values of $C(t)$, measured in ounces, are given in the table above.

(a) Use the data in the table to approximate $C'(3.5)$. Show the computations that lead to your answer, and indicate units of measure.

(b) Is there a time t, $2 \le t \le 4$, at which $C'(t) = 2$? Justify your answer.

(c) Use a midpoint sum with three subintervals of equal length indicated by the data in the table to approximate the value of $\frac{1}{6}\int_0^6 C(t)dt$. Using correct units, explain the meaning of $\frac{1}{6}\int_0^6 C(t)dt$ in the context of the problem.

(d) The amount of coffee in the cup, in ounces, is modeled by $B(t) = 16 - 16e^{-0.4t}$. Using this model, find the rate at which the amount of coffee in the cup is changing when $t = 5$.

Part A

1 point is earned for the approximation

1 point is earned for units

$$C'(3.5) \approx \frac{C(4) - C(3)}{4-3} = \frac{12.8 - 11.2}{1} = 1.6 \text{ ounces/min}$$

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for the approximation

1 point is earned for units

$$C'(3.5) \approx \frac{C(4) - C(3)}{4-3} = \frac{12.8 - 11.2}{1} = 1.6 \text{ ounces/min}$$

Part B

1 point is earned for $\frac{C(4) - C(2)}{4-2}$

1 point is earned for the conclusion, using MVT

C is differentiable $\Rightarrow$ C is continuous (on the closed interval)

5-1-1

$$\frac{C(4)-C(2)}{4-2}=\frac{12.8-8.8}{2}=2$$

Therefore, by the Mean Value Theorem, there is at least one time t, $2<t<4$, for which $C'(t)=2$.

0	1	2

The student response earns all of the following points:

1 point is earned for $\frac{C(4)-C(2)}{4-2}$

1 point is earned for the conclusion, using MVT

C is differentiable ⇒ C is continuous (on the closed interval)

$$\frac{C(4)-C(2)}{4-2}=\frac{12.8-8.8}{2}=2$$

Therefore, by the Mean Value Theorem, there is at least one time t, $2<t<4$, for which $C'(t)=2$.

Part C

1 point is earned for the midpoint sum

1 point is earned for the approximation

1 point is earned for the interpretation

$$\begin{aligned}\tfrac{1}{6}\int_0^6 C(t)dt &\approx \tfrac{1}{6}[2\cdot C(1)+2\cdot C(3)+2\cdot C(5)]\\ &= \tfrac{1}{6}(2\cdot 5.3+2\cdot 11.2+2\cdot 13.8)\\ &= \tfrac{1}{6}(60.6)=10.1 \text{ ounces}\end{aligned}$$

$\frac{1}{6}\int_0^6 C(t)dt$ is the average amount of coffee in the cup, in ounces, over the time interval $0\le t\le 6$ minutes.

0	1	2	3

The student response earns all of the following points:

1 point is earned for the midpoint sum

5-1-1

1 point is earned for the approximation

1 point is earned for the interpretation

$$\frac{1}{6}\int_0^6 C(t)dt \approx \frac{1}{6}[2 \cdot C(1) + 2 \cdot C(3) + 2 \cdot C(5)]$$
$$= \frac{1}{6}(2 \cdot 5.3 + 2 \cdot 11.2 + 2 \cdot 13.8)$$
$$= \frac{1}{6}(60.6) = 10.1 \text{ ounces}$$

$\frac{1}{6}\int_0^6 C(t)dt$ is the average amount of coffee in the cup, in ounces, over the time interval $0 \le t \le 6$ minutes.

Part D

1 point is earned for $B'(t)$

1 point is earned for $B'(5)$

$$B'(t) = -16(-0.4)e^{-0.4t} = 6.4e^{-0.4t}$$
$$B'(5) = 6.4e^{-0.4(5)} = \frac{6.4}{e^2} \text{ ounces/min}$$

		✓
0	1	2

The student response earns all of the following points:

1 point is earned for $B'(t)$

1 point is earned for $B'(5)$

$$B'(t) = -16(-0.4)e^{-0.4t} = 6.4e^{-0.4t}$$
$$B'(5) = 6.4e^{-0.4(5)} = \frac{6.4}{e^2} \text{ ounces/min}$$

5-1-1

24.

t (minutes)	0	1	2	3	4	5	6
$C(t)$ (ounces)	0	5.3	8.8	11.2	12.8	13.8	14.5

Hot water is dripping through a coffeemaker, filling a large cup with coffee. The amount of coffee in the cup at time t, $0 \le t \le 6$, is given by a differentiable function C, where t is measured in minutes. Selected values of $C(t)$, measured in ounces, are given in the table above.

Is there a time $t, 2 \le t \le 4$, at which $C'(t)=2$? Justify your answer.

Part B

1 point is earned for $\dfrac{C(4) - C(2)}{4 - 2}$

1 point is earned for the conclusion, using MVT

C is differentiable ⇒ C is continuous (on the closed interval)

$$\frac{C(4)-C(2)}{4-2} = \frac{12.8-8.8}{2} = 2$$

Therefore, by the Mean Value Theorem, there is at least one time t, $2<t<4$, for which $C'(t)=2$.

0	1	2

The student response earns all of the following points:

1 point is earned for $\dfrac{C(4) - C(2)}{4 - 2}$

1 point is earned for the conclusion, using MVT

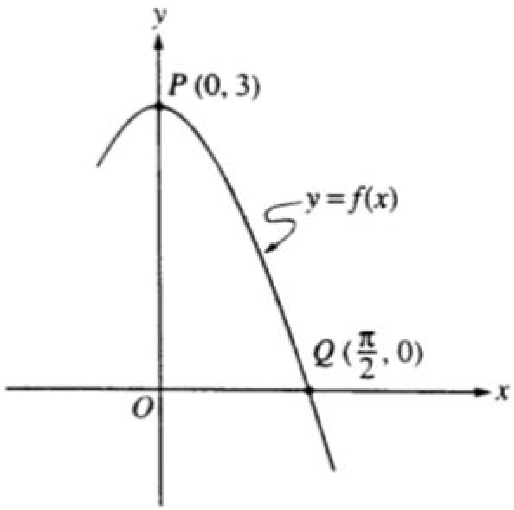
C is differentiable ⇒ C is continuous (on the closed interval)

$$\frac{C(4)-C(2)}{4-2} = \frac{12.8-8.8}{2} = 2$$

Therefore, by the Mean Value Theorem, there is at least one time t, $2<t<4$, for which $C'(t)=2$.

5-1-1

25.



Let f be the function given by $f(x) = 3 \cos x$. As shown above, the graph of f crosses the y-axis at point P and the x-axis at point Q.

(a) Write an equation for the line passing through the points P and Q.

(b) Write an equation for the line tangent to the graph of f at point Q. Show the analysis that leads to your conclusion.

(c) Find the x-coordinate of the point on the graph of f, between points P and Q, at which the line tangent to the graph of f is parallel to line PQ.

(d) Let R be the region in the first quadrant bounded by the graph of f and line segment PQ. Write an integral expression for the volume of the solid generated by revolving the region R about the x-axis. Do not evaluate.

Part A

1 point is earned for the correct slope

$$\text{slope} = \frac{3-0}{0-\pi/2} = -\frac{6}{\pi}$$

1 point is earned for the correct equation

$$y - 3 = -\frac{6}{\pi}(x - 0)$$

0	1	2

The student response earns all of the following points:

1 point is earned for the correct slope

$$\text{slope} = \frac{3-0}{0-\pi/2} = -\frac{6}{\pi}$$

1 point is earned for the correct equation

$$y - 3 = -\frac{6}{\pi}(x - 0)$$

Part B

5-1-1

1 point is earned for the correct statement $f'(\pi/2) = -3$

1 point is earned for an equation using $f'(\pi/2)$ (or using a derivation) and $f(\pi/2)$

1/2 if equations only

$$f'(x) = -3\sin x$$
$$f'(\pi/2) = -3\sin(\pi/2) = -3$$
$$y - 0 = -3(x - \pi/2)$$

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for the correct statement $f'(\pi/2) = -3$

1 point is earned for an equation using $f'(\pi/2)$ (or using a derivation) and $f(\pi/2)$

1/2 if equations only

$$f'(x) = -3\sin x$$
$$f'(\pi/2) = -3\sin(\pi/2) = -3$$
$$y - 0 = -3(x - \pi/2)$$

Part C

1 point is earned for correctly equating the derivative to the slope

$$f'(x) = -3\sin x = -\frac{6}{\pi}$$
$$\sin x = \frac{2}{\pi}$$

1 point is earned for the correct solution in $[0, \frac{\pi}{2}]$

1/2 if solution only

no credit for "no solution exists"

$x = 0.690$

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for correctly equating the derivative to the slope

$$f'(x) = -3\sin x = -\frac{6}{\pi}$$
$$\sin x = \frac{2}{\pi}$$

1 point is earned for the correct solution in $[0, \frac{\pi}{2}]$

5-1-1

1/2 if solution only

no credit for "no solution exists"

$x = 0.690$

Part D

2 points are earned for the correct integrand

0/2 if not difference of 2 squares

1/2 if incorrect but of form $(a\cos x)^2 - (bx+c)^2; a, b \neq 0$

1/2 if reversed

1 point is earned for the correct constant and limits

0/3 for shell around y-axis

$$V = \pi \int_0^{\pi/2} \left[(3\cos x)^2 - \left(-\frac{6}{\pi}x + 3 \right)^2 \right] dx$$

0	1	2	3

The student response earns all of the following points:

2 points are earned for the correct integrand

0/2 if not difference of 2 squares

1/2 if incorrect but of form $(a\cos x)^2 - (bx+c)^2; a, b \neq 0$

1/2 if reversed

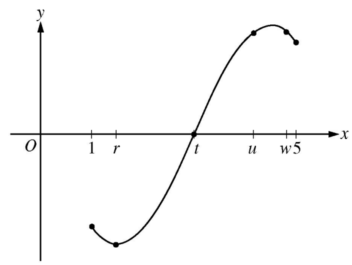
1 point is earned for the correct constant and limits

0/3 for shell around y-axis

$$V = \pi \int_0^{\pi/2} \left[(3\cos x)^2 - \left(-\frac{6}{\pi}x + 3 \right)^2 \right] dx$$

5-1-1

26.



Graph of f

The figure above shows the graph of the differentiable function f for $1 \leq x \leq 5$. Which of the following could be the x-coordinate of a point at which the line tangent to the graph of f is parallel to the secant line through the points $(1, f(1))$ and $(5, f(5))$?

(A) r

(B) t

(C) u ✓

(D) w

(E) There is no such point.

27. The function f is continuous for $-2 \leq x \leq 2$ and $f(-2) = f(2) = 0$. If there is no c, where $-2 < c < 2$ for which $f'(c) = 0$, which of the following statements must be true?

(A) For $-2 < k < 2$, $f'(k) > 0$

(B) For $-2 < k < 2$, $f'(k) < 0$

(C) For $-2 < k < 2$, $f'(k)$ exists.

(D) For $-2 < k < 2$, $f'(k)$ exists, but f' is not continuous.

(E) For some k, where $-2 < k < 2$, $f'(k)$ does not exist. ✓

Answer E

28.

x	$f(x)$
1	2.4
3	3.6
5	5.4

The table above gives selected values of a function f. The function is twice differentiable with $f''(x) > 0$. Which of the following could be the value of $f'(3)$?

(A) 0.6

(B) 0.7 ✓

(C) 0.9

(D) 1.2

(E) 1.5

5-1-1

29. A differentiable function f has the property that $f'(x) \leq 3$ for $1 \leq x \leq 8$ and $f(5) = 6$. Which of the following could be true?

I. $f(2) = 0$

II. $f(6) = -2$

III. $f(7) = 13$

(A) I only

(B) II only

(C) I and II only ✓

(D) I and III only

(E) II and III only

30.

x	$f(x)$
–1	–30
0	–2
3	10
5	18

The table above gives selected values for a twice-differentiable function f. Which of the following must be true?

(A) f has no critical points in the interval $-1 < x < 5$.

(B) $f'(x) = 8$ for some value of x in the interval $-1 < x < 5$. ✓

(C) $f'(x) > 0$ for all values of x in the interval $-1 < x < 5$

(D) $f''(x) < 0$ for for all values of x in the interval $-1 < x < 5$

(E) The graph of f has no points of inflection in the interval $-1 < x < 5$

5-1-1

31. Let f be the function given by $f(x)=x^3-7x+6$.

Find the number c that satisfies the conclusion of the Mean Value Theorem for f on the closed interval $[1,3]$.

Part C

1 point is earned for the answer

1 point is earned for solving $f'(c)$

1 point is earned for solving c

$$\frac{f(3)-f(1)}{3-1}=\frac{12-0}{2}=6$$
$$3c^2-7=f\prime(c)=6$$
$$c=\sqrt{\frac{13}{3}}$$

0	1	2	3

The student response earns three of the following points:

1 point is earned for the answer

1 point is earned for solving $f'(c)$

1 point is earned for solving c

$$\frac{f(3)-f(1)}{3-1}=\frac{12-0}{2}=6$$
$$3c^2-7=f\prime(c)=6$$
$$c=\sqrt{\frac{13}{3}}$$

32. The function f is defined by f(x) = 3x - 4cos (2x + 1), and its derivative is f'(x) = 3 + 8 sin(2x + 1). What are all values of x that satisfy the conclusion of the Mean Value Theorem applied to f on the interval [-1, 2] ?

(A) -0.692 and 1.263
(B) -0.479 and 1.049
(C) 0.285
(D) 0.517
(E) 1.578

33. Let f be the function given by $f(x)=x^3-2x^2+5x-16$. For what value of x in the closed interval [0,5] does the instantaneous rate of change of f equal the average rate of change of f over that interval?

(A) 0
(B) 5/3
(C) 5/2
(D) 3
(E) 5

5-1-1

34. The function f is defined by $f(x) = 2x^3 - 4x^2 + 1$. The application of the Mean Value Theorem to f on the interval $1 \leq x \leq 3$ guarantees the existence of a value c, where $1 < c < 3$, such that $f'(c) =$

(A) 0

(B) 9

(C) 10 ✓

(D) 14

(E) 16

35. Let f be the function given by $f(x) = x^3 + 5x$. For what value of x in the closed interval [1,3] does the instantaneous rate of change of f equal the average rate of change of f on that interval?

(A) $\sqrt{\frac{7}{3}}$

(B) $\sqrt{\frac{13}{3}}$ ✓

(C) $\sqrt{5}$

(D) $\sqrt{6}$

(E) $\sqrt{\frac{19}{3}}$

36. The function g is continuous on the closed interval [1, 4] with $g(1) = 5$ and $g(4) = 8$. Of the following conditions, which would guarantee that there is a number c in the open interval (1, 4) where $g'(c) = 1$?

(A) g is increasing on the closed interval [1, 4].

(B) g is differentiable on the open interval (1, 4). ✓

(C) g has a maximum value on the closed interval [1, 4].

(D) The graph of g has at least one horizontal tangent in the open interval (1, 4).

37. Let f be a polynomial function with degree greater than 2. If $a \neq b$ and $f(a) = f(b) = 1$, which of the following must be true for at least one value of x *between a and b*?

I. $f(x) = 0$

II. $f'(x) = 0$

III. $f''(x) = 0$

(A) None

(B) I only

(C) II only ✓

(D) I and II only

(E) I, II, and III

38. Let f be the function given by $f(x) = x^3 - 3x^2$. What are all values of c that satisfy the conclusion of the Mean Value Theorem of differential calculus on the closed interval [0,3]?

(A) 0 only

(B) 2 only ✓

(C) 3 only

(D) 0 and 3

(E) 2 and 3

5-1-1

39. If $f(x)=\sin(x/2)$ then there exists a number c in the interval $\pi/2<x<3\pi/2$ that satisfies the conclusion of the Mean Value Theorem. Which of the following could be c?

(A) $2\pi/3$

(B) $3\pi/4$

(C) $5\pi/6$

(D) π ✓

(E) $3\pi/2$

40. If c is the number that satisfies the conclusion of the Mean Value Theorem for $f(x) = x^3 - 2x^2$ on the interval $0\le x\le 2$, then c =

(A) 0

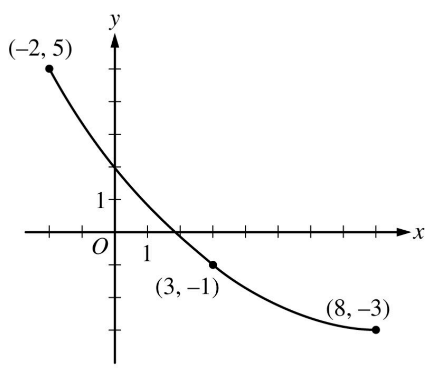
(B) 1/2

(C) 1

(D) 4/3 ✓

(E) 2

41.



Graph of f

A portion of the graph of a differentiable function f is shown above. If the value $c = 3$ satisfies the conclusion of the Mean Value Theorem applied to f on the open interval $-2 < x < 8$, what is the slope of the line tangent to the graph of f at on $x = 3$

(A) $-\frac{7}{5}$

(B) $-\frac{5}{4}$

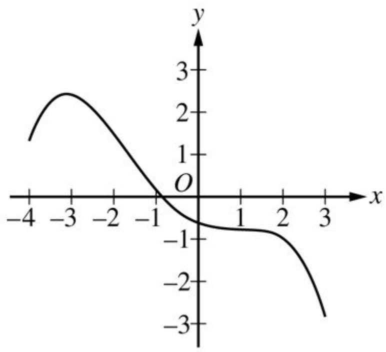
(C) $-\frac{4}{5}$ ✓

(D) $-\frac{5}{7}$

(E) $-\frac{1}{5}$

5-1-1

42.



Graph of f

The graph of a differentiable function f is shown above on the closed interval $[-4, 3]$. How many values of x in the open interval $(-4, 3)$ satisfy the conclusion of the Mean Value Theorem for f on $[-4, 3]$?

(A) zero

(B) one

(C) two

(D) three

Answer D

This option is correct. Since f is continuous and differentiable, the conditions of the Mean Value Theorem are satisfied on the interval [-4, 3]. As shown in the figure to the right, there are three points in the open interval (-4, 3) where the line tangent to the graph of f is parallel to the secant line through the endpoints of the graph on the interval [-4, 3]. At each of these three points the slopes will be the same and will satisfy $f'(c)$

5-1-1

43.

Distance x (cm)	0	1	5	6	8
Temperature $T(x)$ (°C)	100	93	70	62	55

A metal wire of length 8 centimeters (cm) is heated at one end. The table above gives selected values of the temperature $T(x)$, in degrees Celsius (° C), of the wire x cm from the heated end. The function T is decreasing and twice differentiable.

Are the data in the table consistent with the assertion that $T''(x) > 0$ for every x in the interval $0 < x < 8$? Explain your answer.

Part D

1 point is earned for the two slopes of secant lines

1 point is earned for the answer with explanation

Average rate of change of temperature on [1, 5] is $\frac{70-93}{5-1}$ = -5.75.

Average rate of change of temperature on [5, 6] is $\frac{62-70}{6-5}$ = -8.

No. By the MVT, $T'(c1)$ = -5.75 for some c1 in the interval (1, 5) and $T'(c2)$ = -8 for some c2 in the interval (5, 6). It follows that T' must decrease somewhere in the interval (c1, c2). Therefore T'' is not positive for every x in [0, 8].

0	1	2

The student response earns all of the following points:

1 point is earned for the two slopes of secant lines

1 point is earned for the answer with explanation

Average rate of change of temperature on [1, 5] is $\frac{70-93}{5-1}$ = -5.75.

Average rate of change of temperature on [5, 6] is $\frac{62-70}{6-5}$ = -8.

No. By the MVT, $T'(c1)$ = -5.75 for some c1 in the interval (1, 5) and $T'(c2)$ = -8 for some c2 in the interval (5, 6). It follows that T' must decrease somewhere in the interval (c1, c2). Therefore T'' is not positive for every x in [0, 8].

Units

1 point is earned for: units in (a), (b), and (c)

Units of ° C/cm in (a), and ° C in (b) and (c)

0	1

The student response earns all of the following points:

5-1-1

1 point is earned for: units in (a), (b), and (c)

Units of ˚ C/cm in (a), and ˚ C in (b) and (c)

44.

t (hours)	$R(t)$ (gallons per hour)
0	9.6
3	10.4
6	10.8
9	11.2
12	11.4
15	11.3
18	10.7
21	10.2
24	9.6

The rate at which water flows out of a pipe, gallons per hour, is given by a differentiable function R of time t. The table above shows the rate as measured every 3 hours for a 24-hour period.

Is there some time t, $0 < t < 24$, such that $R'(t) = 0$? Justify your answer.

Part B

1 point is earned for the answer

1 point is earned for MVT or equivalent

Yes;

Since $R(0) = R(24) = 9.6$, the Mean Value Theorem guarantees that there is a t, $0 < t < 24$, such that $R'(t) = 0$.

0	1	2

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for MVT or equivalent

Yes;

Since $R(0) = R(24) = 9.6$, the Mean Value Theorem guarantees that there is a t, $0 < t < 24$, such that $R'(t) = 0$.

5-1-1

45.

x	3	4	5	6	7
$f(x)$	20	17	12	16	20

The function f is continuous and differentiable on the closed interval [3, 7]. The table above gives selected values of f on this interval. Which of the following statements must be true?

I. The minimum value off on [3, 7] is 12.

II. There exists c, for $3 < c < 7$, such that $f'(c) = 0$.

III. $f'(x) > 0$ for $5 < x < 7$.

(A) I only

(B) II only

(C) III only

(D) I and III only

(E) I, II, and III

46.

t (min)	0	5	10	15	20	25	30	35	40
$v(t)$ (mpm)	7.0	9.2	9.5	7.0	4.5	2.4	2.4	4.3	7.3

A test plane flies in a straight line with positive velocity $v(t)$, in miles per minute at time t minutes, where v is a differentiable function of t. Selected values of $v(t)$ for $0 \leq t \leq 40$ are shown in the table above.

Based on the values in the table, what is the smallest number of instances at which the acceleration of the plane could equal zero on the open interval $0 < t < 40$? Justify your answer.

Part B

1 point is earned for the two instances

1 point is earned for the justification

By the Mean Value Theorem, $v'(t)=0$ somewhere in the interval (0,15) and somewhere in the interval (25, 30). Therefore the acceleration will equal 0 for at least two values of t.

0	1	2

The student response earns all of the following points:

1 point is earned for the two instances

1 point is earned for the justification

By the Mean Value Theorem, $v'(t)=0$ somewhere in the interval (0,15) and somewhere in the interval (25, 30). Therefore the acceleration will equal 0 for at least two values of t.

5-1-1

47.

x	-2	$-2 < x < -1$	-1	$-1 < x < 1$	1	$1 < x < 3$	3
$f(x)$	12	Positive	8	Positive	2	Positive	7
$f'(x)$	-5	Negative	0	Negative	0	Positive	$\frac{1}{2}$
$g(x)$	-1	Negative	0	Positive	3	Positive	1
$g'(x)$	2	Positive	$\frac{3}{2}$	Positive	0	Negative	-2

The twice-differentiable functions f and g are defined for all real numbers x. Values of f, f', g, and g' for various values of x are given in the table above.

Explain why there must be a value c, for $-1 < c < 1$, such that $f''(c)=0$.

Part B

1 point is earned for $f'(1) - f'(-1) = 0$

1 point is earned for the explanation, using Mean Value Theorem

f' is differentiable ⇒ f' is continuous on the interval $-1 \leq x \leq 1$

$\frac{f'(1)-f'(-1)}{1-(-1)} = \frac{0-0}{2} = 0$ Therefore, by the Mean Value Theorem, there is at least one value c, $-1 < c < 1$, such that $f''(c)=0$.

0	1	2

The student response earns all of the following points:

1 point is earned for $f'(1) - f'(-1) = 0$

1 point is earned for the explanation, using Mean Value Theorem

f' is differentiable ⇒ f' is continuous on the interval $-1 \leq x \leq 1$

$$\frac{f'(1)-f'(-1)}{1-(-1)} = \frac{0-0}{2} = 0$$

Therefore, by the Mean Value Theorem, there is at least one value c, $-1 < c < 1$, such that $f''(c)=0$.

48.

x	0	1	2	3	4
$f(x)$	2	3	4	3	2

The function f is continuous and differentiable on the closed interval [0,4]. The table above gives selected values of f on this interval. Which of the following statements must be true?

(A) The minimum value of f on [0,4] is 2.

(B) The maximum value of f on [0,4] is 4.

(C) $f(x) > 0$ for $0 < x < 4$

(D) $f'(x) < 0$ for $2 < x < 4$

(E) There exists c, with $0 < c < 4$, for which $f'(c) = 0$.

5-1-1

49. Let g be a twice-differentiable, increasing function of t. If $g(0) = 20$ and $g(10) = 220$, which of the following must be true on the interval $0 < t < 10$?

(A) $g\prime(t) = 0$ for some t in the interval.

(B) $g\prime(t) = 20$ for some t in the interval. ✓

(C) $g\prime\prime(t) = 0$ for some t in the interval.

(D) $g\prime\prime(t) > 0$ for all t in the interval.

50. The function f is continuous for $-2 \le x \le 1$ and differentiable for $-2 < x < 1$. If $f(-2) = -5$ and $f(1) = 4$, which of the following statements could be false?

(A) There exists c, where $-2 < c < 1$, such that $f(c) = 0$.

(B) There exists c, where $-2 < c < 1$, such that $f'(c) = 0$. ✓

(C) There exists c, where $-2 < c < 1$, such that $f(c) = 3$.

(D) There exists c, where $-2 < c < 1$, such that $f'(c) = 3$.

(E) There exists c, where $-2 \le c \le 1$, such that $f(c) \ge f(x)$ for all x on the closed interval $-2 \le x \le 1$.

51. The function f is continuous for all real numbers, and the average rate of change of f on the closed interval $[6, 9]$ is $-\frac{3}{2}$. For $6 < c < 9$, there is no value of c such that $f'(c) = -\frac{3}{2}$. Of the following, which must be true?

(A) $\frac{1}{3}\int_6^9 f(x)\,dx = -\frac{3}{2}$

(B) $\int_6^9 f(x)\,dx$ does not exist.

(C) $\frac{f'(6)+f'(9)}{2} = -\frac{3}{2}$

(D) $f'(x) < 0$ for all x in the open interval (6,9).

(E) f is not differentiable on the open interval (6,9). ✓

5-1-1

52. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Consider the curve given by the equation $2(x - y) = 3 + \cos y$. For all points on the curve, $\frac{2}{3} \leq \frac{dy}{dx} \leq 2$.

(a) Show that $\frac{dy}{dx} = \frac{2}{2 - \sin y}$.

(b) For $-\frac{\pi}{2} < y < \frac{\pi}{2}$, there is a point P on the curve through which the line tangent to the curve has slope 1. Find the coordinates of the point P.

(c) Determine the concavity of the curve at points for which $-\frac{\pi}{2} < y < \frac{\pi}{2}$. Give a reason for your answer.

(d) Let $y = f(x)$ be a function, defined implicitly by $2(x - y) = 3 + \cos y$, that is continuous on the closed interval $[2,\ 2.1]$ and differentiable on the open interval $(2,\ 2.1)$. Use the Mean Value Theorem on the interval $[2,\ 2.1]$ to show that $\frac{1}{15} \leq f(2.1) - f(2) \leq \frac{1}{5}$.

Part A

1 point is earned for: implicit differentiation

1 point is earned for: verification

$$\frac{d}{dx}(2(x - y)) = \frac{d}{dx}(3 + \cos y)$$
$$2 - 2\frac{dy}{dx} = (-\sin y)\frac{dy}{dx}$$
$$2 = (2 - \sin y)\frac{dy}{dx}$$
$$\frac{dy}{dx} = \frac{2}{2 - \sin y}$$

0	1	2

The student response earns all of the following points:

1 point is earned for: implicit differentiation

1 point is earned for: verification

$$\frac{d}{dx}(2(x - y)) = \frac{d}{dx}(3 + \cos y)$$
$$2 - 2\frac{dy}{dx} = (-\sin y)\frac{dy}{dx}$$
$$2 = (2 - \sin y)\frac{dy}{dx}$$
$$\frac{dy}{dx} = \frac{2}{2 - \sin y}$$

Part B

1 point is earned for: $\frac{dy}{dx} = 1$

1 point is earned for: answer

5-1-1

$$\frac{dy}{dx} = \frac{2}{2 - \sin y} = 1 \Rightarrow \sin y = 0 \Rightarrow y = 0$$
$$2(x - 0) = 3 + \cos 0 \Rightarrow 2x = 4 \Rightarrow x = 2$$

Point P has coordinates $(2,\ 0)$.

0	1	2

The student response earns all of the following points:

1 point is earned for: $\frac{dy}{dx} = 1$

1 point is earned for: answer

$$\frac{dy}{dx} = \frac{2}{2 - \sin y} = 1 \Rightarrow \sin y = 0 \Rightarrow y = 0$$
$$2(x - 0) = 3 + \cos 0 \Rightarrow 2x = 4 \Rightarrow x = 2$$

Point P has coordinates $(2,\ 0)$.

Part C

2 points are earned for: $\frac{d^2y}{dx^2}$

1 point is earned for: answer with reason

$$\frac{d^2y}{dx^2} = \frac{-2}{(2 - \sin y)^2}(-\cos y)\frac{dy}{dx} = \frac{4\cos y}{(2 - \sin y)^3}$$

$\frac{d^2y}{dx^2} > 0$ for all y in the interval $-\frac{\pi}{2} < y < \frac{\pi}{2}$.

Therefore, the curve is concave up for $-\frac{\pi}{2} < y < \frac{\pi}{2}$.

0	1	2	3

The student response earns all of the following points:

2 points are earned for: $\frac{d^2y}{dx^2}$

1 point is earned for: answer with reason

$$\frac{d^2y}{dx^2} = \frac{-2}{(2 - \sin y)^2}(-\cos y)\frac{dy}{dx} = \frac{4\cos y}{(2 - \sin y)^3}$$

$\frac{d^2y}{dx^2} > 0$ for all y in the interval $-\frac{\pi}{2} < y < \frac{\pi}{2}$.

Therefore, the curve is concave up for $-\frac{\pi}{2} < y < \frac{\pi}{2}$.

5-1-1

Part D

1 point is earned for: applies Mean Value Theorem

1 point is earned for: verification

By the Mean Value Theorem, for some value c in the interval $(2,\ 2.1)$, $f'(c) = \frac{f(2.1)-f(2)}{0.1}$.

For all points on the curve, $\frac{2}{3} \leq f'(x) \leq 2$.

Thus, $\frac{2}{3} \leq \frac{f(2.1)-f(2)}{0.1} \leq 2 \Rightarrow \frac{1}{15} \leq f(2.1) - f(2) \leq \frac{1}{5}$.

0	1	2

The student response earns all of the following points:

1 point is earned for: applies Mean Value Theorem

1 point is earned for: verification

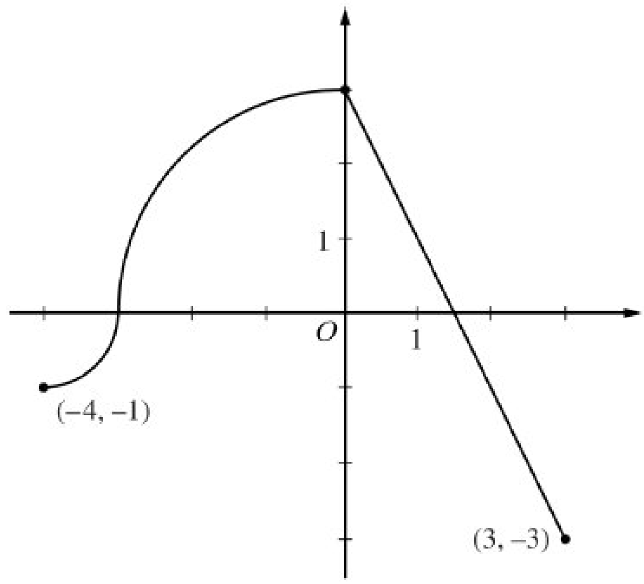
By the Mean Value Theorem, for some value c in the interval $(2,\ 2.1)$, $f'(c) = \frac{f(2.1)-f(2)}{0.1}$.

For all points on the curve, $\frac{2}{3} \leq f'(x) \leq 2$.

Thus, $\frac{2}{3} \leq \frac{f(2.1)-f(2)}{0.1} \leq 2 \Rightarrow \frac{1}{15} \leq f(2.1) - f(2) \leq \frac{1}{5}$.

5-1-1

53.



Graph of f

The continuous function f is defined on the interval $-4 \le x \le 3$. The graph of f consists of two quarter circles and one line segment, as shown in the figure above. Let $g(x) = 2x + \int_0^x f(t)dt$.

(a) Find $g(-3)$. Find $g'(x)$ and evaluate $g'(-3)$.

(b) *Determine the x-coordinate of the point at which g has an absolute maximum on the interval* $-4 \le x \le 3$*. Justify your answer.*

(c) Find all values of x on the interval $-4 < x < 3$ for which the graph of g has a point of inflection. Give a reason for your answer.

(d) *Find the average rate of change of f on the interval* $-4 \le x \le 3$*. There is no point c, -4 < c < 3, for which f′(c)* is equal to that average rate of change. Explain why this statement does not contradict the Mean Value Theorem.

Part A

1 point is earned for $g(-3)$

1 point is earned for $g'(x)$

1 point is earned for $g'(-3)$

$$g(-3) = 2(-3) + \int_0^{-3} f(t)dt = -6 - \frac{9\pi}{4}$$

$g'(x) = 2 + f(x)$

$g'(-3) = 2 + f(-3) = 2$

0	1	2	3

The student response earns all of the following points:

5-1-1

1 point is earned for $g(-3)$

1 point is earned for $g'(x)$

1 point is earned for $g'(-3)$

$$g(-3) = 2(-3) + \int_0^{-3} f(t)dt = -6 - \frac{9\pi}{4}$$

$g'(x) = 2 + f(x)$

$g'(-3) = 2 + f(-3) = 2$

Part B

1 point is earned for considering $g'(x) = 0$

1 point is earned for identifying interior candidate

1 point is earned for answer with justification

$g'(x) = 0$ when $f(x) = -2$. This occurs at $x = \frac{5}{2}$.

$g'(x) > 0$ for $-4 < x < \frac{5}{2}$ and $g'(x) < 0$ for $\frac{5}{2} < x < 3$.

Therefore g has an absolute maximum at $x = \frac{5}{2}$.

0	1	2	3

The student response earns all of the following points:

1 point is earned for considering $g'(x) = 0$

1 point is earned for identifying interior candidate

1 point is earned for answer with justification

$g'(x) = 0$ when $f(x) = -2$. This occurs at $x = \frac{5}{2}$.

$g'(x) > 0$ for $-4 < x < \frac{5}{2}$ and $g'(x) < 0$ for $\frac{5}{2} < x < 3$.

Therefore g has an absolute maximum at $x = \frac{5}{2}$.

Part C

1 point is earned for answer with reason

$g''(x) = f'(x)$ changes sign only at $x = 0$. Thus the graph of g has a point of inflection at $x = 0$.

5-1-1

0	1

The student response earns all of the following points:

1 point is earned for answer with reason

$g''(x) = f'(x)$ changes sign only at $x = 0$. Thus the graph of g has a point of inflection at $x = 0$.

Part D

1 point is earned for average rate of change

1 point is earned for explanation

The average rate of change of f on the interval $-4 \leq x \leq 3$ is $\dfrac{f(3) - f(-4)}{3 - (-4)} = -\dfrac{2}{7}$.

To apply the Mean Value Theorem, f must be differentiable at each point in the interval $-4 < x < 3$. However, f is not differentiable at $x = -3$ and $x = 0$.

0	1	2

The student response earns all of the following points:

1 point is earned for average rate of change

1 point is earned for explanation

The average rate of change of f on the interval $-4 \leq x \leq 3$ is $\dfrac{f(3) - f(-4)}{3 - (-4)} = -\dfrac{2}{7}$.

To apply the Mean Value Theorem, f must be differentiable at each point in the interval $-4 < x < 3$. However, f is not differentiable at $x = -3$ and $x = 0$.

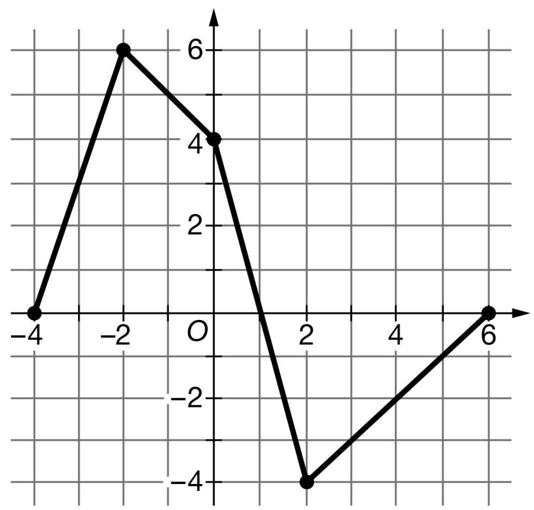
5-1-1

54. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f

Let f be a continuous function defined on the closed interval $-4 \leq x \leq 6$. The graph of f, consisting of four line segments, is shown above. Let G be the function defined by $G(x) = \int_0^x f(t)\,dt$.

(a) On what open intervals is the graph of G concave up? Give a reason for your answer.

(b) Let P be the function defined by $P(x) = G(x) \cdot f(x)$. Find $P'(3)$.

(c) Find $\lim_{x \to 2} \frac{G(x)}{x^2 - 2x}$.

(d) Find the average rate of change of G on the interval $[-4, 2]$. Does the Mean Value Theorem guarantee a value c, $-4 < c < 2$, for which $G'(c)$ is equal to this average rate of change? Justify your answer.

Global

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This "global point" can be earned in any one part. Expressions that show this connection and therefore earn this point include: $G' = f$, $G'(x) = f(x)$, $G''(x) = f'(x)$ in part (a), $G'(3) = f(3)$ in part (b), or $G'(2) = f(2)$ in part (c).

0	1

The student response accurately includes the criteria below.

5-1-1

- [ ] $G'(x) = f(x)$

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Intervals may also include one or both endpoints.

0	1 ✓

The student response accurately includes the criteria below.

- [ ] Answer with reason

Solution:

$G'(x) = f(x)$

The graph of G is concave up for $-4 < x < -2$ and $2 < x < 6$, because $G' = f$ is increasing on these intervals.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for the correct application of the product rule in terms of x or in the evaluation of $P'(3)$. Once earned, this point cannot be lost.

The second point is earned by correctly evaluating $G(3) = -3.5$, $G'(3) = -3$, or $f(3) = -3$.

To be eligible to earn the third point a response must have earned the first two points.

Simplification of the numerical value is not required to earn the third point.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- [ ] Product rule
- [ ] $G(3)$ or $G'(3)$
- [ ] Answer

Solution:

$P'(x) = G(x) \cdot f'(x) + f(x) \cdot G'(x)$

$P'(3) = G(3) \cdot f'(3) + f(3) \cdot G'(3)$

5-1-1

Substituting $G(3) = \int_0^3 f(t)\,dt = -3.5$ and $G'(3) = f(3) = -3$ into the above expression for $P'(3)$ gives the following:

$P'(3) = -3.5 \cdot 1 + (-3) \cdot (-3) = 5.5$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point the response must show $\lim\limits_{x\to2}\left(x^2-2x\right)=0$ and $\lim\limits_{x\to2}G(x)=0$ and must show a ratio of the two derivatives, $G'(x)$ and $2x-2$. The ratio may be shown as evaluations of the derivatives at $x=2$, such as $\frac{G'(2)}{2}$.

To earn the second point the response must evaluate correctly with appropriate limit notation. In particular the response must include $\lim\limits_{x\to2}\frac{G'(x)}{2x-2}$ or $\lim\limits_{x\to2}\frac{f(x)}{2x-2}$.

With any linkage errors (such as $\frac{G'(x)}{2x-2}=\frac{f(2)}{2}$), the response does not earn the second point.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Uses L'Hospital's Rule
- ☐ Answer with justification

Solution:

$\lim\limits_{x\to2}\left(x^2-2x\right)=0$

Because G is continuous for $-4 \le x \le 6$, $\lim\limits_{x\to2}G(x) = \int_0^2 f(t)\,dt = 0$.

Therefore, the limit $\lim\limits_{x\to2}\frac{G(x)}{x^2-2x}$ is an indeterminate form of type $\frac{0}{0}$.

Using L'Hospital's Rule,

$$\lim_{x\to2}\frac{G(x)}{x^2-2x} = \lim_{x\to2}\frac{G'(x)}{2x-2}$$
$$= \lim_{x\to2}\frac{f(x)}{2x-2} = \frac{f(2)}{2} = \frac{-4}{2} = -2$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point a response must present at least a difference and a quotient and a correct evaluation. For example, $\frac{0+16}{6}$ or $\frac{G(2)-G(-4)}{6}=\frac{16}{6}$.

Simplification of the numerical value is not required to earn the first point, but any simplification must be correct.

5-1-1

The second point can be earned without the first point if the response has the correct setup but an incorrect or no evaluation of the average rate of change. The Mean Value Theorem need not be explicitly stated provided the conditions and conclusion are stated.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Average rate of change
- ☐ Answer with justification

Solution:

$G(2) = \int_0^2 f(t)\, dt = 0$ and $G(-4) = \int_0^{-4} f(t)\, dt = -16$

Average rate of change $= \frac{G(2)-G(-4)}{2-(-4)} = \frac{0-(-16)}{6} = \frac{8}{3}$

Yes, $G'(x) = f(x)$ so G is differentiable on $(-4,\ 2)$ and continuous on $[-4,\ 2]$. Therefore, the Mean Value Theorem applies and guarantees a value c, $-4 < c < 2$, such that $G'(c) = \frac{8}{3}$.

5-1-1

55. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

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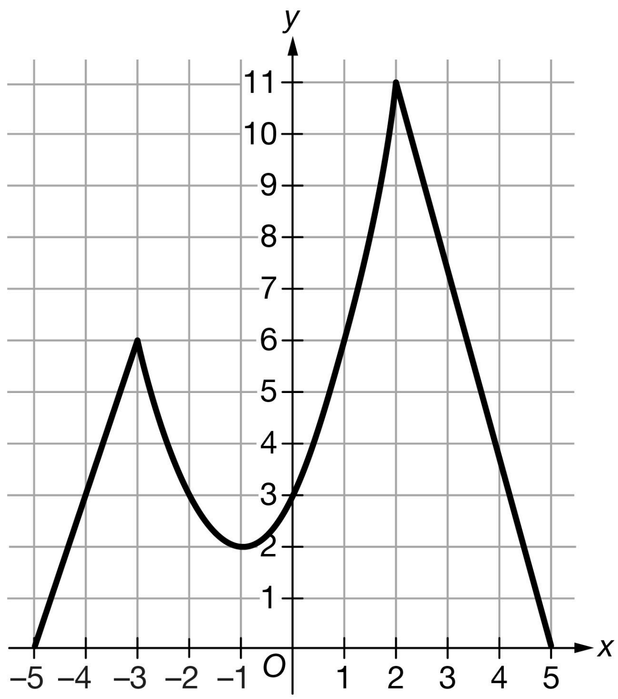
Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f

The continuous function f is defined on the closed interval $[-5, 5]$. The graph of f consists of a parabola and two line segments, as shown in the figure above. Let g be a function such that $g'(x) = f(x)$.

(a) Fill in the missing entries in the table below to describe the behavior of f' and f''. Indicate Positive, Negative, or 0. Give reasons for your answers.

x	$-5 < x < -3$	$-3 < x < -1$	$-1 < x < 2$	$2 < x < 5$
$f(x)$	Positive	Positive	Positive	Positive
$f'(x)$				
$f''(x)$				

(b) There is no value of x in the open interval $(-1, 5)$ at which $f'(x) = \frac{f(5)-f(-1)}{5-(-1)}$. Explain why this does not violate the Mean Value Theorem.

(c) Find all values of x in the open interval $(-5, 5)$ at which the graph of g has a point of inflection. Explain your reasoning.

(d) At what value of x does g attain its absolute maximum on the closed interval $[-5, 5]$? Give a reason for your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

5-1-1

0	1	2	3

The student response accurately includes all three of the criteria below.

- [ ] behavior for $f'(x)$
- [ ] behavior for $f''(x)$
- [ ] reasons

Solution:

x	$-5<x<-3$	$-3<x<-1$	$-1<x<2$	$2<x<5$
$f(x)$	Positive	Positive	Positive	Positive
$f'(x)$	**Positive**	**Negative**	**Positive**	**Negative**
$f''(x)$	**0**	**Positive**	**Positive**	**0**

$f'(x)$ is positive for $-5 < x < -3$ and $-1 < x < 2$ because f is increasing there.

$f'(x)$ is negative for $-3 < x < -1$ and $2 < x < 5$ because f is decreasing there.

$f''(x)$ is positive for $-3 < x < -1$ and $-1 < x < 2$ because the graph of f is concave up there.

$f''(x) = 0$ for $-5 < x < -3$ and $2 < x < 5$ because the graph of f is linear there.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1

The student response accurately includes a correct explanation.

Solution:

This does not violate the Mean Value Theorem because f is not differentiable at $x = 2$ due to a sharp corner. The Mean Value Theorem cannot be applied on the interval $-1 < x < 5$.

Part C

The response is not eligible for any points if incorrect values of x are included.

A response may earn the first point for two correct values of x without explanations or with incomplete, incorrect explanations.

A maximum of 1 point may be earned for three correct values of x without explanations.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

5-1-1

0	1	2	3

The student response accurately includes all three of the criteria below.

- [ ] One correct value of x with explanation OR Two correct values of x without explanations
- [ ] A second correct value of x with explanation
- [ ] A third correct value of x with explanation

Solution:

The graph of g has a point of inflection at $x = -3$, $x = -1$, and $x = 2$.

$g'(x) = f(x)$ changes from increasing to decreasing at $x = -3$ and $x = 2$, and $g'(x) = f(x)$ changes from decreasing to increasing at $x = -1$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] answer
- [ ] reason

Solution:

Because $g'(x) = f(x) > 0$ on the interval $(-5,\ 5)$, g is increasing on the interval $(-5,\ 5)$. Therefore, the absolute maximum value of g on the closed interval $[-5,\ 5]$ occurs at the right endpoint $x = 5$.

5-1-1

56. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

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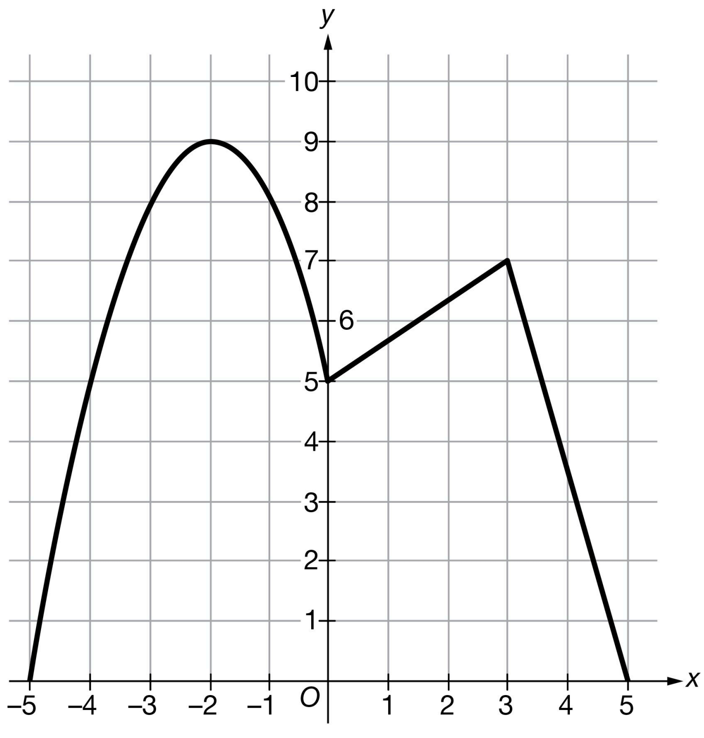

Graph of f

The continuous function f is defined on the closed interval $[-5,5]$. The graph of f consists of a parabola and two line segments, as shown in the figure above. Let g be a function such that $g'(x) = f(x)$.

(a) Fill in the missing entries in the table below to describe the behavior of f' and f''. Indicate Positive, Negative, or 0. Give reasons for your answers.

x	$-5 < x < -2$	$-2 < x < 0$	$0 < x < 3$	$3 < x < 5$
$f(x)$	Positive	Positive	Positive	Positive
$f'(x)$				
$f''(x)$				

(b) There is no value of x in the open interval $(-1,3)$ at which $f'(x) = \frac{f(3)-f(-1)}{3-(-1)}$. Explain why this does not violate the Mean Value Theorem.

(c) Find all values of x in the open interval $(-5,5)$ at which the graph of g has a point of inflection. Explain your reasoning.

(d) At what value of x does g attain its absolute maximum on the closed interval $[-5,5]$? Give a reason for your answer.

Part A

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Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- [ ] behavior for $f'(x)$
- [ ] behavior for $f''(x)$
- [ ] reasons

Solution:

x	$-5<x<-2$	$-2<x<0$	$0<x<3$	$3<x<5$
$f(x)$	Positive	Positive	Positive	Positive
$f'(x)$	**Positive**	**Negative**	**Positive**	**Negative**
$f''(x)$	**Negative**	**Negative**	**0**	**0**

$f'(x)$ is positive for $-5 < x < -2$ and $0 < x < 3$ because f is increasing there.

$f'(x)$ is negative for $-2 < x < 0$ and $3 < x < 5$ because f is decreasing there.

$f''(x)$ is negative for $-5 < x < -2$ and $-2 < x < 0$ because the graph of f is concave down there.

$f''(x) = 0$ for $0 < x < 3$ and $3 < x < 5$ because the graph of f is linear there.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1

The student response accurately includes a correct explanation.

Solution:

This does not violate the Mean Value Theorem because f is not differentiable at $x = 0$ due to a sharp corner. The Mean Value Theorem cannot be applied on the interval $-1 < x < 3$.

Part C

Two values of x must be correct without explanation to earn one point, and no incorrect values included.

A maximum of 1 point may be earned for one correct value with explanation.

A maximum of 2 points may be earned for two correct values with explanation.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

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0	1	2	3

The student response accurately includes all three of the criteria below.

- [ ] values of x (2 points full credit)
- [ ] explanation

Solution:

The graph of g has a point of inflection at $x = -2$, $x = 0$, and $x = 3$.

$g'(x) = f(x)$ changes from increasing to decreasing at $x = -2$ and $x = 3$, and $g'(x) = f(x)$ changes from decreasing to increasing at $x = 0$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] answer
- [ ] reason

Solution:

Because $g'(x) = f(x) > 0$ on the interval $(-5,\ 5)$, g is increasing on the interval $(-5,\ 5)$. Therefore, the absolute maximum value of g on the closed interval $[-5,\ 5]$ occurs at the right endpoint $x = 5$.

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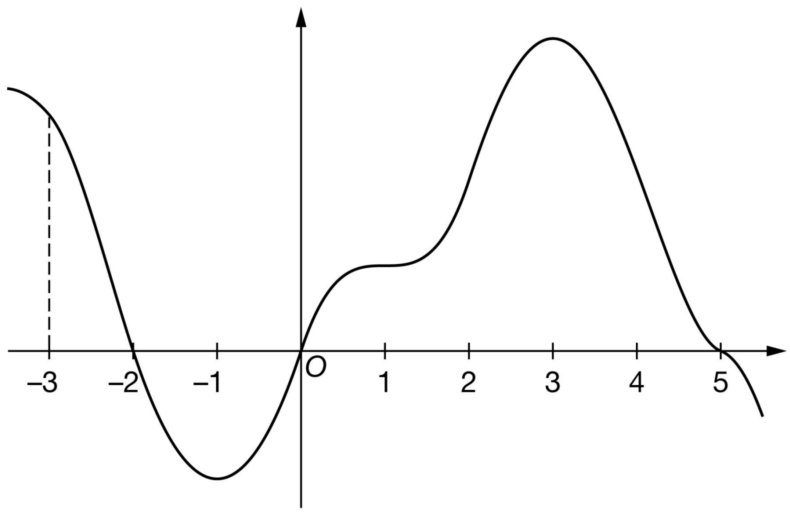
57. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of g

The graph of the differentiable function g with domain $-3.5 \leq x \leq 5.5$ is shown in the figure above. The areas of the regions bounded by the x-axis and the graph of g on the intervals $[-3, -2]$, $[-2, 0]$, and $[0, 5]$ are 10, 13, and 59, respectively. The graph of g has horizontal tangents at $x = -1$, $x = 1$, and $x = 3$. Let h be the function defined by $h(x) = \int_0^x g(t)\,dt$ for $-3.5 \leq x \leq 5.5$.

(a) Find the x-coordinate of each critical point of h on the interval $-3.5 < x < 5.5$.

(b) Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h. Justify each of your answers.

(c) Find the x-coordinate of each point of inflection for the graph of h on the interval $-3.5 < x < 5.5$. Justify your answer.

(d) Find the value of $h(5)$.

(e) Find the value of $h(-3)$. Show the computations that lead to your answer.

(f) Must there exist a value of p, for $-3 < p < 2$, such that $h'(p)$ is equal to the average rate of change of h over the interval $-3 \leq x \leq 2$? Justify your answer.

(g) Find $\lim_{x \to -3} \frac{h(x)}{2x}$. Show the computations that lead to your answer.

(h) The function f is defined by $f(x) = x \cdot h(x)$. It is known that $g(-3) = 18$. Find $f'(-3)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -2$, $x = 0$, and $x = 5$.

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No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3.5,\ 5.5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3.5,\ 5.5)$. The point is still earned provided the three critical points, $x = -2$, $x = 0$, and $x = 5$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-2,\ 0)$, $(0, 0)$, $(5, 0)$, read the x-coordinates, and disregard the y-coordinates.

0	1 ✓

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$h'(x) = g\ (x) = 0$ at $x = -2$, $x = 0$, and $x = 5$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on g, then:

· the response is eligible for all three points if the response presents the connection between $h'\ (x)$ and $g\ (x)$ (that is, $h'(x) = g\ (x)$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of g or h' as $g\ (x) = h'(x)$, throughout all parts of this question.

· the response is eligible for at most two of the three points if the response does not present the connection $h'(x) = g\ (x)$. Any additional error (for example, not justifying that at $x = -2$, h has a relative maximum) is eligible for at most one of the three points.

Read references to "the graph" as references to $g\ (x)$ as indicated by the stem.

A response referencing "the function" or "it" or "the slope" is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of "the function" or "it" or "the slope" with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., " $h'(-2)$ changes from positive to negative") when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h\ (x)$ value given in the table or list of values must be correct to earn any points. In order to earn these points, all three critical points must be included in the table or list and the relevant

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endpoint must be considered. The correct values of relevant inputs include:

· $h(-3) = 3$

· $h(-2) = 13$

· $h(0) = 0$

· $h(5) = 59$

· $h(5.5) < 59$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a "global" rather than a "local" argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

· At $x = -2$, h has a relative maximum because $h' = g$ is positive for $x < -2$ and $h' = g$ is negative for $x > -2$.

· At $x = 0$, h has a relative minimum because $h' = g$ is negative for $x < 0$ and $h' = g$ is positive for $x > 0$.

· At $x = 5$, h has a relative maximum because $h' = g$ is positive for $x < 5$ and $h' = g$ is negative for $x > 5$.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Relative maximum with justification
- [ ] Relative minimum with justification
- [ ] Relative maximum with justification

Solution:

At $x = -2$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

At $x = 0$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

At $x = 5$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each point of inflection must be accompanied by a correct justification to earn the point.

A response giving justification based only h' is eligible for all three points.

If a response gives justifications based only on g, then (in parallel with the approach followed in part (b)):

· the response is eligible for all three points if the response presents the connection between $h'(x)$ and $g(x)$ (that is, $h'(x) = g(x)$). If you do not see this in part (c), look for it in part (a) and part (b). If you find this connection in either part, read all subsequent occurrences of g or h' as $g(x) = h'(x)$, throughout all parts of this question.

· the response is eligible for at most two of the three points if the response does not present the connection $h'(x) = g(x)$.

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Justification for the first two points should reference one of the following:

· $h'(x)$ changing from increasing to decreasing, or vice versa

· relative extrema of $h'(x)$

· $h''(x)$ changes sign

· $g'(x)$ changes sign

"$h'(x) = g(x)$ changes direction" earns neither of the first two points.

Read references to "the graph" as references to g as indicated by the stem.

A response referencing "the function" or "it" or "the slope" is vague. A response of this type can earn a maximum of one of the first two points. This type of response earns one point if replacing every occurrence of "the function" or "it" or "the slope" with h' produces a response that would earn both points.

A response that discusses h' at a particular point (i.e., $h'(-1)$) when the discussion should have been about $h'(x)$ can earn only one of the first two points, with one exception. This communication error should not be penalized twice. If a response failed to earn a point for this error in part (b), it will not lose a point in this part.

If $x = -1$ and $x = 3$ are both listed as points of inflection and no other numbers are listed as points of inflection, the third point is earned. A response does not need to say that there are only two inflection points if only two points are listed.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ $x = -1$ with justification
- ☐ $x = 3$ with justification
- ☐ Response includes no other points of inflection

Solution:

At $x = -1$, the graph of h has a point of inflection because $h'(x) = g(x)$ changes from decreasing to increasing there.

At $x = 3$, the graph of h has a point of inflection because $h'(x) = g(x)$ changes from increasing to decreasing there.

There are no other points of inflection.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of 59 is all that is needed to earn the point. No supporting work is required.

0	1 ✓

The student response accurately includes the criteria below.

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☐ Answer

Solution:

$$h\left(5\right)=\int_0^5 g\left(t\right)\ dt=59$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned when the response identifies ± 10 and ± 13 either linked to their corresponding definite integrals or combined as a sum or difference, often expressed by the sum (or difference) $(\pm 10)\pm(\pm 13)$. The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 10 and ± 13 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

· The expression $-10-(-13)$ earns both points because numerical expressions need not be simplified.

· The expression $(-10)-13$ earns the first point, but not the second point.

· The expression $-10-(-13)-3$ earns neither point.

· The expression $\int_0^{-3} g\left(t\right)\ dt=3$ earns neither point.

· The expression $-\int_{-3}^{-2} g\left(t\right)\ dt-\int_{-2}^{0} g\left(t\right)\ dt=3$ or any other sum or difference of correct symbolic integrals does not earn the first point because there is no representation of ± 10 and ± 13, but it does earn the second point.

· $-10, 13$ earns neither point

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

☐ $\left|\int_{-2}^{-3} g\left(t\right)\ dt\right|=10$ and $\left|\int_{-2}^{0} g\left(t\right)\ dt\right|=13$

☐ Answer

Solution:

$$h\left(-3\right)=\int_0^{-3} g\left(t\right)\ dt=-\int_{-3}^{0} g\left(t\right)\ dt$$

$$=-\int_{-3}^{-2} g\left(t\right)\ dt-\int_{-2}^{0} g\left(t\right)\ dt=-10-(-13)=3$$

Part F

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Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that h is continuous and differentiable. An ambiguous reference to h (e.g., "the function is continuous and differentiable") does not earn the first point. The interval on which h is continuous and differentiable does not need to be stated. If an interval is given, it must contain the interval $(-3,\ 2)$ and be a subset of the domain referenced in the stem of the problem with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., by either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

· "Yes, $h\ (x)$ is continuous and differentiable so MVT applies" earns both points.

· "Yes, $h\ (x)$ is continuous and differentiable, therefore there must be a value where $h'\ (p)$ equals the average rate of change of $h\ (x)$ " earns both points.

· "Yes, $h'\ (x)$ (the derivative of $h\ (x)$) is continuous and differentiable, therefore there must be a value where $h'\ (p)$ equals the average rate of change of $h\ (x)$ " does not earn the first point, but does the earn second point.

· "Yes because of MVT" does not earn the first point, but does earn the second point.

· "Yes, because $h\ (x)$ is continuous and differentiable" earns the first point, but not the second point.

· "Yes, because $h\ (x)$ is continuous" earns neither point.

· "Yes, because $h\ (x)$ is differentiable" earns neither point.

· "Yes" earns neither point.

A response of "No" earns neither point, regardless of any other explanation including any reference to MVT.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$h' = g \Rightarrow h$ is differentiable on $-3 \leq x \leq 2. \Rightarrow h$ is continuous on $-3 \leq x \leq 2.$

Therefore, the Mean Value Theorem can be applied to h on the interval $-3 \leq x \leq 2$ to guarantee that there exists a value of p, for $-3 < p < 2$, such that $h'\ (p)$ equals the average rate of change of h over the interval $-3 \leq x \leq 2$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for the correct answer supported by some work that must include a fraction. Responses that earn the point include:

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- $\frac{h(-3)}{-6} = \frac{3}{-6}$
- $\frac{3}{-6}$

Responses that do not earn the point are:

- $\frac{h(-3)}{-6}$
- $\frac{h(-3)}{2 \cdot (-3)}$

Note: an incorrect value for $h\ (-3)$ can be brought in from part (e) for the numerator of the fraction to earn the point.

The response does not need to state that h is continuous. If the response does state that h is continuous, an explanation is not required.

The response does not need to use limit notation.

0	1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h' = g \Rightarrow h$ is continuous. $\Rightarrow \lim_{x \to -3} h\ (x) = h\ (-3)$

$$\lim_{x \to -3} \frac{h(x)}{2x} = \frac{h(-3)}{2 \cdot (-3)} = \frac{3}{-6} = -\frac{1}{2}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first (product rule) point is awarded for sufficient evidence of an attempt at the product rule. The evidence required is a derivative of f that contains a sum of two terms, with at least one term written as a product, and one term involving h and the other term involving h' (or g).

Fundamental Theorem of Calculus is evidenced by linking h' and g either directly or indirectly in the presented derivative of $f\ (x)$ or in $f'\ (-3)$. Since $g\ (-3) = 18$ is given in the question, replacing $h'\ (-3)$ directly with 18 without an intermediate step earns the second point.

It is possible to earn the second point without earning the first point.

A response that contains an incorrect value for $h\ (-3)$ that is brought in from part (e) can earn the third point.

$-3\ \cdot\ 18 + 3$ is the minimal answer that earns all three points. If there is no supporting work, an answer of the form $a\ \cdot\ b\ +\ c$ where any of $a,\ \ b$, or c is incorrect, will earn zero points.

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0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Product rule
- [ ] Fundamental Theorem of Calculus
- [ ] Answer

Solution:

$f'(x) = x \cdot h'(x) + 1 \cdot h(x)$

$= x \cdot g(x) + h(x)$

$f'(-3) = -3 \cdot g(-3) + h(-3) = -3 \cdot (18) + 3 = -51$

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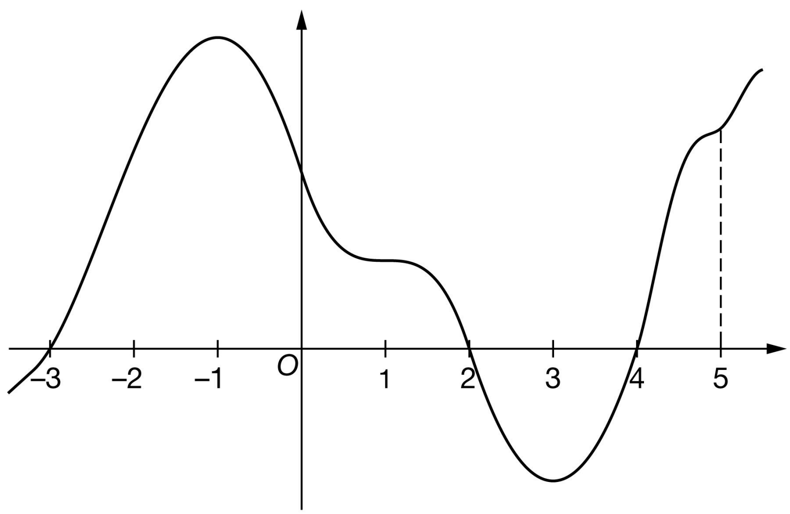
58. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of g

The graph of the differentiable function g with domain $-3.5 \leq x \leq 5.5$ is shown in the figure above. The areas of the regions bounded by the x-axis and the graph of g on the intervals $[-3, 2]$, $[2, 4]$, and $[4, 5]$ are 55, 12, and 10, respectively. The graph of g has horizontal tangents at $x = -1$, $x = 1$, and $x = 3$. Let h be the function defined by $h(x) = \int_4^x g(t)\, dt$ for $-3.5 \leq x \leq 5.5$.

(a) Find the x-coordinate of each critical point of h on the interval $-3.5 < x < 5.5$.

(b) Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h. Justify each of your answers.

(c) Find the x-coordinate of each point of inflection for the graph of h on the interval $-3.5 < x < 5.5$. Justify your answer.

(d) Find the value of $h(5)$.

(e) Find the value of $h(-3)$. Show the computations that lead to your answer.

(f) Must there exist a value of r, for $-3 < r < 2$, such that $h'(r)$ is equal to the average rate of change of h over the interval $-3 \leq x \leq 2$? Justify your answer.

(g) Find $\lim_{x \to 5} \frac{h(x)}{2x}$. Show the computations that lead to your answer.

(h) The function k is defined by $k(x) = x \cdot h(x)$. It is known that $g(5) = 15$. Find $k'(5)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -3$, $x = 2$, and $x = 4$.

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No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3.5, 5.5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3.5,\ 5.5)$. The point is still earned provided the three critical points $x = -3$, $x = 2$, and $x = 4$ are correctly listed.

If the critical points are given as ordered pairs, e.g.,$(-3, 0)$, $(2, 0)$, $(4, 0)$, read the x -coordinates, and disregard the y-coordinates.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$h'(x) = g(x) = 0$ at $x = -3$, $x = 2$, and $x = 4$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on g, then:

· the response is eligible for all three points if the response presents the connection between $h'(x)$ and $g(x)$ (that is, $h'(x) = g(x)$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of g or h' as $g(x) = h'(x)$, throughout all parts of this question.

· the response is eligible for at most two of the three points if the response does not present the connection $h'(x) = g(x)$. Any additional error (for example, not justifying that at $x = -3$, h has a relative minimum) is eligible for at most one of the three points.

Read references to "the graph" as references to $g(x)$ as indicated by the stem.

A response referencing "the function" or "it" or "the slope" is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of "the function" or "it" or "the slope" with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., "$h'(3)$ changes from negative to positive") when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn these points, all three critical points must be included in the table or list and the relevant

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endpoint must be considered. The correct values of relevant inputs include:

· $h(-3.5) > -43$

· $h(-3) = -43$

· $h(2) = 12$

· $h(4) = 0$

· $h(5) = 10$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

· At $x = -3$, h has a relative minimum because $h' = g$ is negative for $x < -3$ and $h' = g$ is positive for $x > -3$.

· At $x = 2$, h has a relative maximum because $h' = g$ h prime equals g is positive for $x < 2$ and $h' = g$ is negative for $x > 2$.

· At $x = 4$, h has a relative minimum because $h' = g$ is negative for $x < 4$ and $h' = g$ is positive for $x > 4$.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- [ ] Relative minimum with justification
- [ ] Relative maximum with justification
- [ ] Relative minimum with justification

Solution:

At $x = -3$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

At $x = 2$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

At $x = 4$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each point of inflection must be accompanied by a correct justification to earn the point.

A response giving justification based only h' is eligible for all three points.

If a response gives justifications based only on g, then (in parallel with the approach followed in part (b)):

· the response is eligible for all three points if the response presents the connection between $h'(x)$ and $g(x)$ (that is, $h'(x) = g(x)$). If you do not see this in part (c), look for it in part (a) and part (b). If you find this connection in either part, read all subsequent occurrences of g or h' as $g(x) = h'(x)$, throughout all parts of this question.

· the response is eligible for at most two of the three points if the response does not present the connection $h'(x) = g(x)$.

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Justification for the first two points should reference one of the following:

· $h'(x)$ changing from increasing to decreasing, or vice versa

· relative extrema of $h'(x) = g(x)$

· $h''(x)$ changes sign

· $g'(x)$ changes sign

"$h'(x) = g(x)$ changes direction" earns neither of the first two points.

Read references to "the graph" as references to g as indicated by the stem.

A response referencing "the function" or "it" or "the slope" is vague. A response of this type can earn a maximum of one of the first two points. This type of response earns one point if replacing every occurrence of "the function" or "it" or "the slope" with h' produces a response that would earn both points.

A response that discusses h' at a particular point (i.e., $h'(-3)$) when the discussion should have been about $h'(x)$ can earn only one of the first two points, with one exception. This communication error should not be penalized twice. If a response failed to earn a point for this error in part (b), it will not lose a point in this part.

If $x = -1$ and $x = 3$ are both listed as points of inflection and no other numbers are listed as points of inflection, the third point is earned. A response does not need to say that there are only two inflection points if only two points are listed.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ $x = -1$ with justification
- ☐ $x = 3$ with justification
- ☐ Response includes no other points of inflection

Solution:

At $x = -1$, the graph of h has a point of inflection because $h'(x) = g(x)$ changes from increasing to decreasing there.

At $x = 3$, the graph of h has a point of inflection because $h'(x) = g(x)$ changes from decreasing to increasing there.

There are no other points of inflection.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of 10 is all that is needed to earn the point. No supporting work is required.

0	1 ✓

The student response accurately includes the criteria below.

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☐ Answer

Solution:

$$h(5) = \int_4^5 g(t)\,dt = 10$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned when the response identifies ± 55 and ± 12 either linked to their corresponding definite integrals or combined as a sum or difference, often expressed by the sum (or difference) $(\pm 55) \pm (\pm 12)$. The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 55 and ± 12 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

· The expression $-55-(-12)$ earns both points because numerical expressions need not be simplified.

· The expression $(-55)-12$ earns the first point, but not the second point.

· The expression $-55-(-12)-8$ earns neither point.

· The expression $\int_4^{-3} g(t)\,dt = -43$ earns neither point.

· The expression $-\int_{-3}^{2} g(t)\,dt - \int_2^4 g(t)\,dt = -43$ or any other sum or difference of correct symbolic integrals does not earn the first point because there is no representation of ± 55 and ± 12, but it does earn the second point.

· $-55, 12$ earns neither point

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

☐ $\left|\int_2^{-3} g(t)\,dt\right| = 55$ and $\left|\int_2^{4} g(t)\,dt\right| = 12$

☐ Answer

Solution:

$$h(-3) = \int_4^{-3} g(t)\,dt = -\int_{-3}^{4} g(t)\,dt$$

$$= -\int_{-3}^{2} g(t)\,dt - \int_2^4 g(t)\,dt = -55-(-12) = -43$$

Part F

5-1-1

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that h is continuous and differentiable. An ambiguous reference to h (e.g., "the function is continuous and differentiable") does not earn the first point. The interval on which h is continuous and differentiable does not need to be stated. If an interval is given, it must contain the interval $(-3, 2)$ and be a subset of the domain referenced in the stem of the problem with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., by either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

· "Yes, $h(x)$ is continuous and differentiable so MVT applies" earns both points.

· "Yes, $h(x)$ is continuous and differentiable, therefore there must be a value where $h'(r)$ equals the average rate of change of $h(x)$ " earns both points.

· "Yes, $h'(x)$ (the derivative of $h(x)$) is continuous and differentiable, therefore there must be a value where $h'(r)$ equals the average rate of change of $h(x)$ " does not earn the first point, but does the earn second point.

· "Yes because of MVT" does not earn the first point, but does earn the second point.

· "Yes, because $h(x)$ is continuous and differentiable" earns the first point, but not the second point.

· "Yes, because $h(x)$ is continuous" earns neither point.

· "Yes, because $h(x)$ is differentiable" earns neither point.

· "Yes" earns neither point.

A response of "No" earns neither point, regardless of any other explanation including any reference to MVT.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$h' = g \Rightarrow h$ is differentiable on $-3 \leq x \leq 2. \Rightarrow h$ is continuous on $-3 \leq x \leq 2.$

Therefore, the Mean Value Theorem can be applied to h on the interval $-3 \leq x \leq 2$ to guarantee that there exists a value of r, for $-3 < r < 2$, such that $h'(r)$ equals the average rate of change of h over the interval $-3 \leq x \leq 2$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for the correct answer supported by some work that must include a fraction. Responses that earn the point include:

5-1-1

- $\dfrac{h(5)}{10} = 1$
- $\dfrac{10}{10}$

Responses that do not earn the point include:

- $\dfrac{h(5)}{10}$
- 1

Note: an incorrect value for $h(5)$ can be brought in from part (d) for the numerator of the fraction to earn the point.

The response does not need to state that h is continuous. If the response does state that h is continuous, an explanation is not required.

The response does not need to use limit notation.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$h' = g \Rightarrow h$, is continuous. $\Rightarrow \lim\limits_{x \to 5} h(x) = h(5)$

$$\lim_{x \to 5} \frac{h(x)}{2x} = \frac{h(5)}{2 \cdot 5} = \frac{10}{10} = 1$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first (product rule) point is awarded for sufficient evidence of an attempt at the product rule. The evidence required is a derivative of k that contains a sum of two terms, with at least one term written as a product, and one term involving h and the other term involving h' (or g).

Fundamental Theorem of Calculus is evidenced by linking h' and g either directly or indirectly in the presented derivative of $k(x)$ or in $k'(5)$. Since $g(5) = 15$ is given in the question, replacing $h'(5)$ directly with 15 without an intermediate step earns the second point.

It is possible to earn the second point without earning the first point.

A response that contains an incorrect value for $h(5)$ that is brought in from part (d) can earn the third point.

$5 \cdot 15 + 10$ is the minimal answer that earns all three points. If there is no supporting work, an answer of the form $a \cdot b + c$ where any of a, b, or c is incorrect, will earn zero points.

5-1-1

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$k'(x) = x \cdot h'(x) + 1 \cdot h(x)$

$= x \cdot g(x) + h(x)$

$k'(5) = 5g(5) + h(5) = 5 \cdot 15 + 10 = 85$

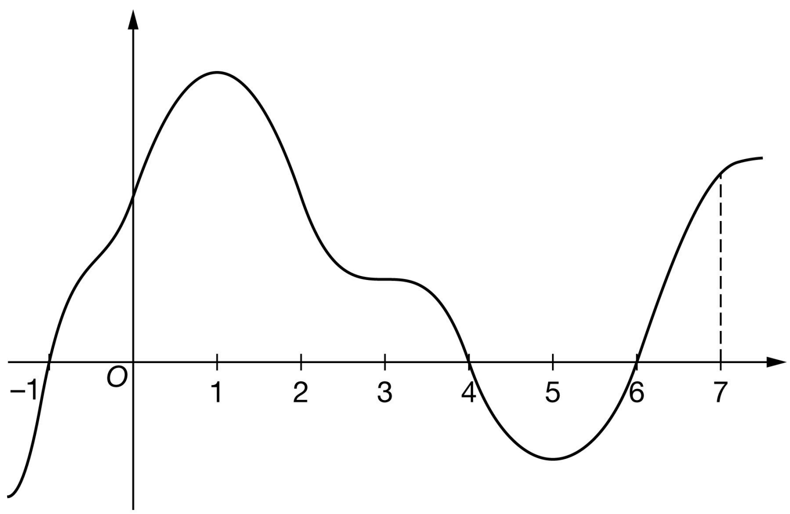
5-1-1

59. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f

The graph of the differentiable function f with domain $-1.5 \le x \le 7.5$ is shown in the figure above. The areas of the regions bounded by the x-axis and the graph of f on the intervals $[-1, 4]$, $[4, 6]$, and $[6, 7]$ are 56, 10, and 8, respectively. The graph of f has horizontal tangents at $x = 1$, $x = 3$, and $x = 5$. Let h be the function defined by $h(x) = \int_6^x f(t)\,dt$ for $-1.5 \le x \le 7.5$.

(a) Find the x-coordinate of each critical point of h on the interval $-1.5 < x < 7.5$.

(b) Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h. Justify each of your answers.

(c) Find the x-coordinate of each point of inflection for the graph of h on the interval $-1.5 < x < 7.5$. Justify your answer.

(d) Find the value of $h(7)$.

(e) Find the value of $h(-1)$. Show the computations that lead to your answer.

(f) Must there exist a value of d, for $-1 < d < 4$, such that $h'(d)$ is equal to the average rate of change of h over the interval $-1 \le x \le 4$? Justify your answer.

(g) Find $\lim_{x \to 7} \frac{h(x)}{4x}$. Show the computations that lead to your answer.

(h) The function q is defined by $q(x) = 3x \cdot h(x)$. It is known that $f(7) = 14$. Find $q'(7)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -1$, $x = 4$, and $x = 6$.

5-1-1

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-1.5, 7.5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-1.5, 7.5)$. The point is still earned provided the three critical points, $x = -1$, $x = 4$, and$x = 6$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-1, 0)$, $(4, 0)$, $(6, 0)$, read the x-coordinates and disregard the y-coordinates.

0	1 ✓

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h\prime(x) = f(x) = 0$ at $x = -1$, $x = 4$, and $x = 6$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on f, then:

· the response is eligible for all three points if the response presents the connection between $h'(x)$ and $f(x)$ (that is, $h'(x) = f(x)$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of f or h' as $f(x) = h'(x)$, throughout all parts of this question. m

· the response is eligible for at most two of the three points if the response does not present the connection $h'(x) = f(x)$. Any Any additional error (for example, not justifying that at $x = -1$, h has a relative minimum) is eligible for at most one of the three points.

Read references to “the graph” as references to $f(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., "$h'(-1)$ changes from negative to positive”) when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

5-1-1

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn these points, all three critical points must be included in the table or list and the relevant endpoint must be considered. The correct values of relevant inputs include:

· $h(-1.5) > -46$

· $h(-1) > -46$

· $h(4) = 10$

· $h(6) = 0$

· $h(7) = 8$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a "global" rather than a "local" argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

· At $x = -1$, h has a relative minimum because $h' = f$ is negative for $x < -1$ and $h' = f$ is positive for $x > -1$.

· At $x = 4$, h has a relative maximum because $h' = f$ is positive for $x < 4$ and $h' = f$ is negative for $x > 4$.

· At $x = 6$, h has a relative minimum because $h' = f$ is negative for $x < 6$ and $h' = f$ is positive for $x > 6$.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Relative minimum with justification
- [ ] Relative maximum with justification
- [ ] Relative minimum with justification

Solution:

At $x = -1$, h has a relative minimum because $h'(x) = f(x)$ changes from negative to positive there.

At $x = 4$, h has a relative maximum because $h'(x) = f(x)$ changes from positive to negative there.

At $x = 6$, h has a relative minimum because $h'(x) = f(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each point of inflection must be accompanied by a correct justification to earn the point.

A response giving justification based only h' is eligible for all three points.

If a response gives justifications based only on f, then (in parallel with the approach followed in part (b)):

· the response is eligible for all three points if the response presents the connection between $h'(x)$ and $f(x)$ (that is, $h'(x) = f(x)$). If you do not see this in part (c), look for it in part (a) and part (b). If you find this connection in either part, read all subsequent occurrences of f or h' as $f(x) = h'(x)$, throughout all parts of this question.

5-1-1

· the response is eligible for at most two of the three points if the response does not present the connection $h'(x) = f(x)$.

Justification for the first two points should reference one of the following:

· $h'(x)$ changing from increasing to decreasing, or vice versa

· relative extrema of $h'(x) = f(x)$

· h'' changes sign

· $f'(x)$ changes sign

" $h'(x) = f(x)$ changes direction" earns neither of the first two points.

Read references to "the graph" as references to f as indicated by the stem.

A response referencing "the function" or "it" or "the slope" is vague. A response of this type can earn a maximum of one of the first two points. This type of response earns one point if replacing every occurrence of "the function" or "it" or "the slope" with h' produces a response that would earn both points.

A response that discusses h' at a particular point (i.e., $h'(-1)$) when the discussion should have been about $h'(x)$ can earn only one of the first two points, with one exception. This communication error should not be penalized twice. If a response failed to earn a point for this error in part (b), it will not lose a point in this part.

If $x = 1$ and $x = 5$ are both listed as points of inflection and no other numbers are listed as points of inflection, the third point is earned. A response does not need to say that there are only two inflection points if only two points are listed.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ $x = 1$ with justification
- ☐ $x = 5$ with justification
- ☐ Response includes no other points of inflection

Solution:

At $x = 1$, the graph of h has a point of inflection because $h'(x) = f(x)$ changes from increasing to decreasing there.

At $x = 5$, the graph of h has a point of inflection because $h'(x) = f(x)$ changes from decreasing to increasing there.

There are no other points of inflection.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of 8 is all that is needed to earn the point. No supporting work is required.

0	1 ✓

5-1-1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$$h(7) = \int_6^7 f(t)\,dt = 8$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned when the response identifies ± 56 and ± 10 either linked to their corresponding definite integrals or combined as a sum or difference, often expressed by the sum (or difference) $(\pm 56) \pm (\pm 10)$. The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 56 and ± 10 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

· The expression $-56 - (-10)$ earns both points because numerical expressions need not be simplified.

· The expression $(-56) - 10$ earns the first point, but not the second point.

· The expression $-56 - (-10) - 8$ earns neither point.

· The expression $\int_6^{-1} g(t)\,dt = -46$ earns neither point.

· The expression $-\int_{-1}^{4} f(t)\,dt - \int_4^6 f(t)\,dt = -46$ or any other sum or difference of correct symbolic integrals does not earn the first point because there is no representation of ± 56 and ± 10, but it does earn the second point.

· $-56,\ 10$ earns neither point.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- [ ] $\left|\int_4^{-1} f(t)\,dt\right| = 56$ and $\left|\int_4^{6} f(t)\,dt\right| = 10$
- [ ] Answer

Solution:

$$h(-1) = \int_6^{-1} f(t)\,dt = -\int_{-1}^{6} f(t)\,dt$$

$$= -\int_{-1}^{4} f(t)\,dt - \int_4^6 f(t)\,dt = -56 - (-10) = -46$$

5-1-1

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that h is continuous and differentiable. An ambiguous reference to h (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which h is continuous and differentiable does not need to be stated. If an interval is given, it must contain the interval $(-1,\ 4)$ and be a subset of the domain referenced in the stem of the problem with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., by either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

· “Yes, $h\ (x)$ is continuous and differentiable so MVT applies” earns both points.

· “Yes, $h\ (x)$ is continuous and differentiable, therefore there must be a value where $h'\ (d)$ equals the average rate of change of $h\ (x)$“ earns both points.

· “Yes, $h'\ (x)$ (the derivative of $h\ (x)$) is continuous and differentiable, therefore there must be a value where $h'\ (d)$ equals the average rate of change of $h\ (x)$“ does not earn the first point, but does the earn second point.

· “Yes because of MVT” does not earn the first point but does earn the second point.

· “Yes, because $h\ (x)$ is continuous and differentiable” earns the first point, but not the second point.

· “Yes, because $h\ (x)$ is continuous” earns neither point.

· “Yes, because $h\ (x)$ is differentiable” earns neither point.

· “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$h' = f \Rightarrow h$ is differentiable on $-1 \le x \le 4$. $\Rightarrow h$ is continuous on $-1 \le x \le 4$.

Therefore, the Mean Value Theorem can be applied to h on the interval $-1 \le x \le 4$ to guarantee that there exists a value of d, for $-1 < d < 4$, such that $h'\ (d)$ equals the average rate of change of h over the interval $-1 \le x \le 4$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

5-1-1

The point is earned for the correct answer supported by some work that must include a fraction. Responses that earn the point include:

- $\frac{h(7)}{28} = \frac{2}{7}$
- $\frac{8}{28}$

Responses that do not earn the point include:

- $\frac{h(7)}{4 \cdot 7}$
- $\frac{h(7)}{28}$

Note: an incorrect value for $h\,(7)$ can be brought in from part (d) for the numerator of the fraction to earn the point.

The response does not need to state that h is continuous. If the response does state that h is continuous, an explanation is not required.

The response does not need to use limit notation.

0	1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h' = f \Rightarrow h$ is continuous. $\Rightarrow \lim_{x \to 7} h\,(x) = h\,(7)$

$$\lim_{x \to 7} \frac{h(x)}{4x} = \frac{h(7)}{4 \cdot (7)} = \frac{8}{28} = \frac{2}{7}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first (product rule) point is awarded for sufficient evidence of an attempt at the product rule. The evidence required is a derivative of q that contains a sum of two terms, with at least one term written as a product, and one term involving h and the other term involving h' (or f).

Fundamental Theorem of Calculus is evidenced by linking h' and f either directly or indirectly in the presented derivative of $q\,(x)$ or in $q'\,(7)$. Since $f\,(7) = 15$ is given in the question, replacing $h'\,(7)$ directly with 14 without an intermediate step earns the second point.

It is possible to earn the second point without earning the first point.

A response that contains an incorrect value for $h\,(7)$ that is brought in from part (d) can earn the third point.

$21 \cdot 14 + 3 \cdot 8$ is the minimal answer that earns all three points. If there is no supporting work, an answer of the form $a \cdot b + c \cdot d$ where any of a, b, c, or d is incorrect, will earn zero points.

5-1-1

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Product rule
- [ ] Fundamental Theorem of Calculus
- [ ] Answer

Solution:

$q'(x) = 3x \cdot h'(x) + 3 \cdot h(x)$

$= 3x \cdot f(x) + 3 \cdot h(x)$

$q'(7) = 3 \cdot 7 \cdot f(7) + 3h(7) = 21 \cdot (14) + 3 \cdot (8) = 318$

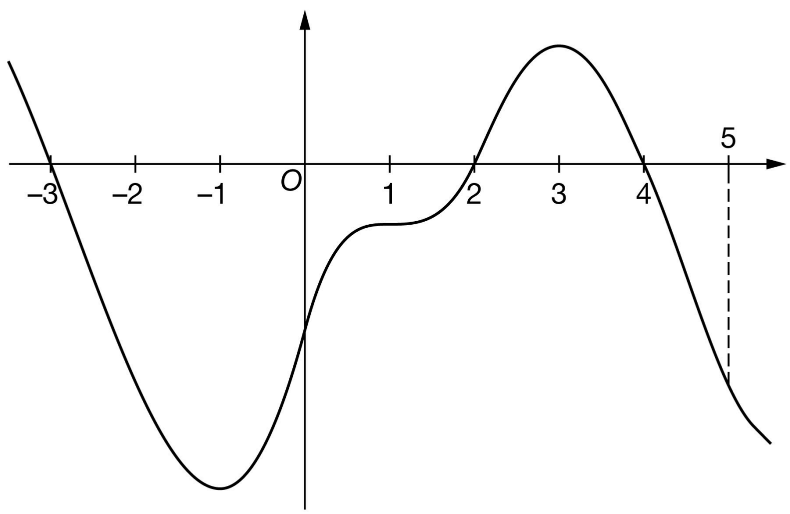
5-1-1

60. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of h

The graph of the differentiable function h with domain $-3.5 \leq x \leq 5.5$ is shown in the figure above. The areas of the regions bounded by the x-axis and the graph of h on the intervals $[-3, 2]$, $[2, 4]$, and $[4, 5]$ are 58, 11, and 8, respectively. The graph of h has horizontal tangents at $x = -1$, $x = 1$, and $x = 3$. Let g be the function defined by $g(x) = \int_4^x h(t)\,dt$ for $-3.5 \leq x \leq 5.5$.

(a) Find the x-coordinate of each critical point of g on the interval $-3.5 < x < 5.5$.

(b) Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for g. Justify each of your answers.

(c) Find the x-coordinate of each point of inflection for the graph of g on the interval $-3.5 < x < 5.5$. Justify your answer.

(d) Find the value of $g(5)$.

(e) Find the value of $g(-3)$. Show the computations that lead to your answer.

(f) Must there exist a value of b, for $-3 < b < 2$, such that $g'(b)$ is equal to the average rate of change of g over the interval $-3 \leq x \leq 2$? Justify your answer.

(g) Find $\lim_{x \to 5} \frac{g(x)}{3x}$. Show the computations that lead to your answer.

(h) The function a is defined by $a(x) = x^2 \cdot g(x)$. It is known that $h(5) = -16$. Find $a'(5)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -3$, $x = 2$, and $x = 4$.

5-1-1

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3.5,\ \ 5.5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3.5,\ \ 5.5)$. The point is still earned provided the three critical points, $x = -3,\ \ x = 2$, and $x = 4$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3,\ \ 0),\ \ (2,\ \ 0),\ \ (4,\ \ 0)$, read the x-coordinates, and disregard the y-coordinates.

0	1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$g'(x) = h(x) = 0$ at $x = -3$, $x = 2$, and $x = 4$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on g' is eligible for all three points.

If a response gives justifications based only on h, then:

· the response is eligible for all three points if the response presents the connection between $g'(x)$ and $h(x)$ (that is, $g'(x) = h(x)$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of h or g' as $h(x) = g'(x)$, throughout all parts of this question.

· the response is eligible for at most two of the three points if the response does not present the connection $g'(x) = h(x)$ Any additional error (for example, not justifying that at $x = 2$, h has a relative minimum) is eligible for at most one of the three points.

Read references to “the graph” as references to $h(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn all three points.

A response that discusses g' at a particular point (e.g., “ $g'(-3)$ changes from negative to positive”) when the discussion should have been about $g'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

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Responses using a Candidates Test are eligible for all three points. Every $g(x)$ value given in the table or list of values must be correct to earn any points. In order to earn these points, all three critical points must be included in the table or list and the relevant endpoint must be considered. The correct values of relevant inputs include:

· $g(-3.5) < 47$

· $g(-3) = 47$

· $g(2) = -11$

· $g(4) = 0$

· $g(5) = -8$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a "global" rather than a "local" argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

· At $x = -3$, g has a relative maximum because $g' = h$ is positive for $x < -3$ and $g' = h$ is negative for $x > -3$.

· At $x = 2$, g has a relative minimum because $g' = h$ is negative for $x < 2$ and $g' = h$ is positive for $x > 2$.

· At $x = 4$, g has a relative maximum because $g' = h$ is positive for $x < 4$ and $g' = h$ is negative for $x > 4$.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Relative maximum with justification
- ☐ Relative minimum with justification
- ☐ Relative maximum with justification

Solution:

At $x = -3$, g has a relative maximum because $g'(x) = h(x)$ changes from positive to negative there.

At $x = 2$, g has a relative minimum because $g'(x) = h(x)$ changes from negative to positive there.

At $x = 4$, g has a relative maximum because $g'(x) = h(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each point of inflection must be accompanied by a correct justification to earn the point.

A response giving justification based only g is eligible for all three points.

If a response gives justifications based only on h, then (in parallel with the approach followed in part (b)):

· the response is eligible for all three points if the response presents the connection between $g'(x)$ and $h(x)$ (that is, $g'(x) = h(x)$). If you do not see this in part (c), look for it in part (a) and part (b). If you find this connection in either part, read all subsequent occurrences of h or g' as $h(x) = g'(x)$, throughout all parts of this question.

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· the response is eligible for at most two of the three points if the response does not present the connection $g'(x) = h(x)$.

Justification for the first two points should reference one of the following:

· $g'(x)$ changing from increasing to decreasing, or vice versa

· relative extrema of $g'(x)$

· $g''(x)$ changes sign

· $h'(x)$ changes sign

“ $g'(x) = h(x)$ changes direction” earns neither of the first two points.

Read references to “the graph” as references to h as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one of the first two points. This type of response earns one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn both points.

A response that discusses g' at a particular point (i.e., $g'(-3)$) when the discussion should have been about $g'(x)$ can earn only one of the first two points, with one exception. This communication error should not be penalized twice. If a response failed to earn a point for this error in part (b), it will not lose a point in this part.

If $x = -1$ and $x = 3$ are both listed as points of inflection and no other numbers are listed as points of inflection, the third point is earned. A response does not need to say that there are only two inflection points if only two points are listed.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ $x = -1$ with justification
- ☐ $x = 3$ with justification
- ☐ Response includes no other points of inflection

Solution:

At $x = -1$, the graph of g has a point of inflection because $g'(x) = h(x)$ changes from decreasing to increasing there.

At $x = 3$, the graph of g has a point of inflection because $g'(x) = h(x)$ changes from increasing to decreasing there.

There are no other points of inflection.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of -8 is all that is needed to earn the point. No supporting work is required.

0	1 ✓

5-1-1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g\left(5\right)=\int_{4}^{5}h\left(t\right)\,dt=-8$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned when the response identifies ± 58 and ± 11 either linked to their corresponding definite integrals or combined as a sum or difference, often expressed by the sum (or difference) $\left(\pm 58\right)\pm\left(\pm 11\right)$. The values do not need to be combined, but if they are, they cannot be combined with any additional values. The ± 58 and ± 11 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

· The expression $-\left(-58\right)-11$ earns both points because numerical expressions need not be simplified.

· The expression $\left(-58\right)-11$ earns the first point, but not the second point.

· The expression $-58-11-8$ earns neither point.

· The expression $\int_{4}^{-3}h\left(t\right)\,dt=47$ earns neither point.

· The expression $-\int_{-3}^{2}h\left(t\right)\,dt-\int_{2}^{4}h\left(t\right)\,dt=47$ or any other sum or difference of correct symbolic integrals does not earn the first point because there is no representation of ± 58 and ± 11, but it does earn the second point.

· $-58,\quad 11$ earns neither point

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

☐ $\left|\int_{-3}^{2}h\left(t\right)\,dt\right|=58$ and $\left|\int_{4}^{2}h\left(t\right)dt\right|=11$

☐ Answer

Solution:

$$g\left(-3\right)=\int_{4}^{-3}h\left(t\right)\,dt=-\int_{-3}^{4}h\left(t\right)\,dt$$

$$=-\int_{-3}^{2}h\left(t\right)\,dt-\int_{2}^{4}h\left(t\right)\,dt=-\left(-58\right)-11=47$$

Part F

5-1-1

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that g is continuous and differentiable. An ambiguous reference to g (e.g., "the function is continuous and differentiable") does not earn the first point. The interval on which g is continuous and differentiable does not need to be stated. If an interval is given, it must contain the interval $(-3,\ 2)$ and be a subset of the domain referenced in the stem of the problem with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., by either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

· "Yes, $g\ (x)$ is continuous and differentiable so MVT applies" earns both points.

· "Yes, $g\ (x)$ is continuous and differentiable, therefore there must be a value where $g'\ (b)$ equals the average rate of change of $g\ (x)$" earns both points.

· "Yes, $g'\ (x)$ (the derivative of $g\ (x)$) is continuous and differentiable, therefore there must be a value where $g'\ (b)$ equals the average rate of change of $g\ (x)$ " does not earn the first point, but does the earn second point.

· "Yes because of MVT" does not earn the first point, but does earn the second point.

· "Yes, because $g\ (x)$ is continuous and differentiable" earns the first point, but not the second point.

· "Yes, because $g\ (x)$ is continuous" earns neither point.

· "Yes, because $g\ (x)$ is differentiable" earns neither point.

· "Yes" earns neither point.

A response of "No" earns neither point, regardless of any other explanation including any reference to MVT.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- [ ] Conditions
- [ ] Conclusion using Mean Value Theorem

Solution:

$g' = h \Rightarrow g$ is differentiable on $-3 \leq x \leq 2. \Rightarrow g$ is continuous on $-3 \leq x \leq 2$.

Therefore, the Mean Value Theorem can be applied to g on the interval $-3 \leq x \leq 2$ to guarantee that there exists a value of b, for $-3 < b < 2$, such that $g'\ (b)$ equals the average rate of change of g over the interval $-3 \leq x \leq 2$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for the correct answer supported by some work that must include a fraction. Responses that earn the point include:

5-1-1

- $\frac{g(5)}{15} = \frac{-8}{15}$
- $\frac{-8}{15}$

Responses that do not earn the point are:

- $\frac{g(5)}{15}$
- $\frac{g(5)}{3 \cdot 5}$

Note: an incorrect value for $g\ (5)$ can be brought in from part (d) for the numerator of the fraction to earn the point.

The response does not need to state that g is continuous. If the response does state that g is continuous, an explanation is not required.

The response does not need to use limit notation.

0	1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$g' = h \Rightarrow g$ is continuous. $\Rightarrow \lim_{x \to 5} g\ (x) = g\ (5)$

$\lim_{x \to 5} \frac{g(x)}{3x} = \frac{g(5)}{3 \cdot (5)} = \frac{-8}{15}$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first (product rule) point is awarded for sufficient evidence of an attempt at the product rule. The evidence required is a derivative of a that contains a sum of two terms, with at least one term written as a product, and one term involving g and the other term involving g' (or h).

Fundamental Theorem of Calculus is evidenced by linking g' and h either directly or indirectly in the presented derivative of $g\ (x)$ or in $g'\ (5)$. Since $h\ (5) = -16$ is given in the question, replacing $g'\ (5)$ directly with -16 without an intermediate step earns the second point.

It is possible to earn the second point without earning the first point.

A response that contains an incorrect value for $g\ (5)$ that is brought in from part (d) can earn the third point.

$25 \cdot (-16) + 10 \cdot (-8)$ is the minimal answer that earns all three points. If there is no supporting work, an answer of the form $a \cdot b + c \cdot d$ where any of $a,\ b, c,$ or d is incorrect, will earn zero points.

5-1-1

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Product rule
- [ ] Fundamental Theorem of Calculus
- [ ] Answer

Solution:

$a'\left(x\right)=x^{2}\cdot g'\left(x\right)+2x\cdot g\left(x\right)$

$=x^{2}\cdot h\left(x\right)+2x\cdot g\left(x\right)$

$a'\left(5\right)=25\cdot h\left(5\right)+2\cdot 5\cdot g\left(5\right)=25\cdot\left(-16\right)+10\cdot\left(-8\right)=-480$

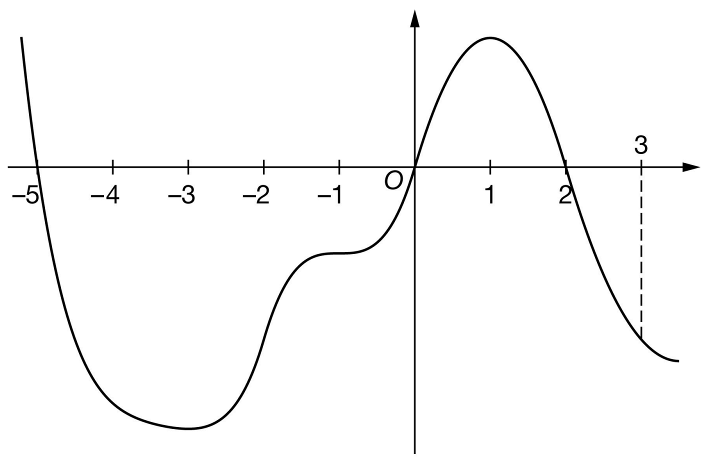
5-1-1

61. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f

The graph of the differentiable function f with domain $-5.5 \le x \le 3.5$ is shown in the figure above. The areas of the regions bounded by the x-axis and the graph of f on the intervals $[-5, 0]$, $[0, 2]$, and $[2, 3]$ are 57, 12, and 7, respectively. The graph of f has horizontal tangents at $x = -3$, $x = -1$, and $x = 1$. Let g be the function defined by $g(x) = \int_2^x f(t)\,dt$ for $-5.5 \le x \le 3.5$.

(a) Find the x-coordinate of each critical point of g on the interval $-5.5 < x < 3.5$.

(b) Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for g. Justify each of your answers.

(c) Find the x-coordinate of each point of inflection for the graph of g on the interval $-5.5 < x < 3.5$. Justify your answer.

(d) Find the value of $g(3)$.

(e) Find the value of $g(-5)$. Show the computations that lead to your answer.

(f) Must there exist a value of k, for $-5 < k < 0$, such that $g'(k)$ is equal to the average rate of change of g over the interval $-5 \le x \le 0$? Justify your answer.

(g) Find $\lim_{x \to 3} \frac{5g(x)}{2x}$. Show the computations that lead to your answer.

(h) The function r is defined by $r(x) = x^2 \cdot g(x)$. It is known that $f(3) = -12$. Find $r'(3)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -5$, $x = 0$, and $x = 2$.

No supporting work is required, although any presented work must be correct.

5-1-1

The point is not earned if the response presents any extra numbers in the open interval $(-5.5,\ 3.5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-5.5, 3.5)$. The point is still earned provided the three critical points, $x = -5,\ \ x = 0$, and $x = 2$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-5,\ 0),\ \ (0,\ 0),\ \ (2,\ 0)$, read the x-coordinates, and disregard the y-coordinates.

0	1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$g'(x) = f(x) = 0$ at $x = -5$, $x = 0$, and $x = 2$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on g' is eligible for all three points.

If a response gives justifications based only on f, then:

· the response is eligible for all three points if the response presents the connection between $g'(x)$ and $f(x)$ (that is, $g'(x) = f(x)$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of f or g' as $f(x) = g'(x)$, throughout all parts of this question.

· the response is eligible for at most two of the three points if the response does not present the connection $g'(x) = f(x)$. Any additional error (for example, not justifying that at $x = -5$, g has a relative maximum) is eligible for at most one of the three points.

Read references to “the graph” as references to $f(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn all three points.

A response that discusses g' at a particular point (e.g., “ $g'(-5)$ changes from positive to negative”) when the discussion should have been about $g'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn these points, all three critical points must be included in the table or list and the relevant endpoint must be considered. The correct values of relevant inputs include:

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· $h(-5.5) < 45$

· $h(-5) = 45$

· $h(0) = -12$

· $h(2) = 0$

· $h(3) = -7$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a "global" rather than a "local" argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

· At $x = -5$, g has a relative maximum because $g' = f$ is positive for $x < -5$ and $g' = f$ is negative for $x > -5$.

· At $x = 0$, g has a relative minimum because $g' = f$ is negative for $x < 0$ and $g'(x) = f(x)$ is positive for $x > 0$.

· At $x = 2$, g has a relative maximum because $g' = f$ is positive for $x < 2$ and $g' = f$ is negative for $x > 2$.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Relative maximum with justification
- ☐ Relative minimum with justification
- ☐ Relative maximum with justification

Solution:

At $x = -5$, g Alt text: g has a relative maximum because $g'(x) = f(x)$ changes from positive to negative there.

At $x = 0$, g Alt text: g has a relative minimum because $g'(x) = f(x)$ changes from negative to positive there.

At $x = 2$, g Alt text: g has a relative maximum because $g'(x) = f(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each point of inflection must be accompanied by a correct justification to earn the point.

A response giving justification based only g' is eligible for all three points.

If a response gives justifications based only on f, then (in parallel with the approach followed in part (b)):

· the response is eligible for all three points if the response presents the connection between $g'(x)$ and $f(x)$ (that is, $g'(x) = f(x)$). If you do not see this in part (c), look for it in part (a) and part (b). If you find this connection in either part, read all subsequent occurrences of f or g' as $f(x) = g'(x)$, throughout all parts of this question.

· the response is eligible for at most two of the three points if the response does not present the connection $g'(x) = f(x)$.

Justification for the first two points should reference one of the following:

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· $g'(x)$ changing from increasing to decreasing, or vice versa

· relative extrema of $g'(x)$

· $g''(x)$ changes sign

· $f'(x)$ changes sign

" $g'(x) = f(x)$ changes direction" earns neither of the first two points.

Read references to "the graph" as references to f as indicated by the stem.

A response referencing "the function" or "it" or "the slope" is vague. A response of this type can earn a maximum of one of the first two points. This type of response earns one point if replacing every occurrence of "the function" or "it" or "the slope" with g' produces a response that would earn both points.

A response that discusses g' at a particular point (i.e., $g'(-3)$) when the discussion should have been about $g'(x)$ can earn only one of the first two points, with one exception. This communication error should not be penalized twice. If a response failed to earn a point for this error in part (b), it will not lose a point in this part.

If $x = -3$ and $x = 1$ are both listed as points of inflection and no other numbers are listed as points of inflection, the third point is earned. A response does not need to say that there are only two inflection points if only two points are listed.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ $x = -3$ with justification
- ☐ $x = 1$ with justification
- ☐ Response includes no other points of inflection

Solution:

At $x = -3$, the graph of g has a point of inflection because $g'(x) = f(x)$ changes from decreasing to increasing there.

At $x = 1$, the graph of g has a point of inflection because $g'(x) = f(x)$ changes from increasing to decreasing there.

There are no other points of inflection.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of -7 is all that is needed to earn the point. No supporting work is required.

0	1 ✓

The student response accurately includes the criteria below.

·Answer

5-1-1

☐

Solution:

$$g(3) = \int_{2}^{3} f(t)\ dt = -7$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned when the response identifies ± 57 and ± 12 either linked to their corresponding definite integrals or combined as a sum or difference, often expressed by the sum (or difference) $(\pm 57) \pm (\pm 12)$. The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 57 and ± 12 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

· The expression $-(-57) - (-12)$ earns both points because numerical expressions need not be simplified.

· The expression $(-57) - 12$ earns the first point, but not the second point.

· The expression $-57 - (-12) - 7$ earns neither point.

· The expression $\int_{2}^{-5} f(t)\ dt = 45$ earns neither point.

· The expression $-\int_{-5}^{0} f(t)dt - \int_{0}^{2} f(t)dt = 45$ or any other sum or difference of correct symbolic integrals does not earn the first point because there is no representation of ± 57 and ± 12, but it does earn the second point.

· $-57, -12$ earns neither point

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

☐ $\left| \int_{-5}^{0} f(t)\ dt \right| = 57$ and $\left| \int_{2}^{0} f(t)\ dt \right| = 12$

☐ Answer

Solution:

$$g(-5) = \int_{2}^{-5} f(t)\ dt = -\int_{-5}^{2} f(t)\ dt$$

$$= -\int_{-5}^{0} f(t)\ dt - \int_{0}^{2} f(t)\ dt = -(-57) - 12 = 45$$

Part F

5-1-1

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that g is continuous and differentiable. An ambiguous reference to g (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which g is continuous and differentiable does not need to be stated. If an interval is given, it must contain the interval $(-5,\ 0)$ and be a subset of the domain referenced in the stem of the problem with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., by either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

· “Yes, $g\ (x)$ is continuous and differentiable so MVT applies” earns both points.

· “Yes, $g\ (x)$ is continuous and differentiable, therefore there must be a value where $g'\ (k)$ equals the average rate of change of $g\ (x)$ “ earns both points.

· “Yes, $g'\ (x)$ (the derivative of $g\ (x)$) is continuous and differentiable, therefore there must be a value where $g'\ (k)$ equals the average rate of change of $g\ (x)$ “ does not earn the first point, but does the earn second point.

· “Yes because of MVT” does not earn the first point, but does earn the second point.

· “Yes, because $g\ (x)$ is continuous and differentiable” earns the first point, but not the second point.

· “Yes, because $g\ (x)$ is continuous” earns neither point.

· “Yes, because $g\ (x)$ is differentiable” earns neither point.

· “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- [ ] Conditions
- [ ] Conclusion using Mean Value Theorem

Solution:

$g' = f \Rightarrow g$ is differentiable on $-5 \leq x \leq 0. \Rightarrow g$ is continuous on $-5 \leq x \leq 0$.

Therefore, the Mean Value Theorem can be applied to g on the interval $-5 \leq x \leq 0$ to guarantee that there exists a value of k , for $-5 < k < 0$, such that $g'\ (k)$ equals the average rate of change of g over the interval $-5 \leq x \leq 0$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for the correct answer supported by some work that must include a fraction. Responses that earn the first point include:

5-1-1

- $\frac{5(-7)}{2\cdot 3}$
- $-\frac{35}{6}$

Responses that do not earn the first point include:

- $\frac{5\cdot g(3)}{6}$
- $\frac{5\cdot g(3)}{2\cdot 3}$

Note: an incorrect value for $g\left(3\right)$ can be brought in from part (d) for the numerator of the fraction to earn the point.

The response does not need to state that g is continuous. If the response does state that g is continuous, an explanation is not required.

The response does not need to use limit notation.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$g' = f \Rightarrow g$ is continuous. $\Rightarrow \lim\limits_{x\to 3} g\left(x\right) = g\left(3\right)$

$$\lim_{x\to 3}\frac{5g(x)}{2x} = \frac{5g(3)}{2\cdot(3)} = \frac{5(-7)}{6} = -\frac{35}{6}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first (product rule) point is awarded for sufficient evidence of an attempt at the product rule. The evidence required is a derivative of r that contains a sum of two terms, with at least one term written as a product, and one term involving g and the other term involving g' (or f).

Fundamental Theorem of Calculus is evidenced by linking g' and f either directly or indirectly in the presented derivative of $r\left(x\right)$ or in $r'\left(3\right)$. Since $f\left(3\right) = -12$ is given in the question, replacing $g'\left(3\right)$ directly with -12 without an intermediate step earns the second point.

It is possible to earn the second point without earning the first point.

A response that contains an incorrect value for $g\left(3\right)$ that is brought in from part (d) can earn the third point.

$9\left(-12\right) + 6\left(-7\right)$ is the minimal answer that earns all three points. If there is no supporting work, an answer of the form $a\cdot b + c\cdot d$ where any of a, b, c, or d is incorrect, will earn zero points.

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0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$r'(x) = x^2 \cdot g'(x) + 2x \cdot g(x)$

$= x^2 \cdot f(x) + 2x \cdot g(x)$

$r'(3) = 3^2 \cdot f(3) + 2 \cdot 3 \cdot g(3) = 9 \cdot (-12) + 6 \cdot (-7) = -150$

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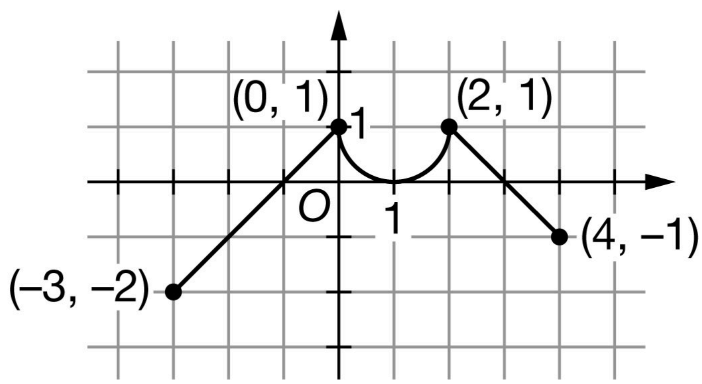
62. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of g

The continuous function g has domain $-3 \leq x \leq 4$. The graph of g, consisting of two line segments and a semicircle, is shown in the figure above. The graph of g has a horizontal tangent at $x = 1$. Let h be the function defined by $h(x) = \int_0^x g(t)\, dt$ for $-3 \leq x \leq 4$.

(a) Find the x-coordinate of each critical point of h on the interval $-3 < x < 4$.

(b) Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h. Justify each of your answers.

(c) For $-3 < x < 4$, on what open intervals is h decreasing and concave down? Give a reason for your answer.

(d) Find the value of $h(2)$. Show the computations that lead to your answer.

(e) Find the value of $h(-3)$. Show the computations that lead to your answer.

(f) Find the value of $h''(-1)$, or explain why it does not exist.

(g) Must there exist a value of r, for $-3 < r < 1$, such that $h'(r)$ is equal to the average rate of change of h over the interval $-3 \leq x \leq 1$? Justify your answer.

(h) Find $\lim_{x \to 2} \frac{h'(x)}{5x}$. Show the computations that lead to your answer.

(i) The function k is defined by $k(x) = h\left(x^2\right)$. Find $k'(-2)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = 1$, $x = 1$, and $x = 3$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3,\ \ 4)$, or if any critical values are missing.

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Disregard presentations of the endpoints or any other numbers outside the open interval $(-3,\ \ 4)$. The point is still earned provided the three critical points $x=-1,\ \ x=1$, and $x=3$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-1,\ \ 0)$, $(1,\ \ 0)$, and $(3,\ \ 0)$, read the x -coordinates, and disregard the y -coordinates.

0	1 ✓

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$h'\ (x)=g\ (x)=0$ at $x=-1$, $x=1$, and $x=3$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on g, then:

· the response is eligible for all three points, if the response presents the connection between $h'\ (x)$ and $g\ (x)$, (that is, $h'=g$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of g or h' as $h'=g$, throughout all parts of this question.

· the response is eligible for at most two of the three points, if the response does not present the connection $h'\ (x)=g\ (x)$. Any additional error (for example, not justifying that at $x=-1$, h has a relative minimum) is eligible for at most one of the three points.

Read references to “the graph” as references to $g\ (x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'\ (3)$ changes from negative to positive”) when the discussion should have been about $h'\ (x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. $h\ (x)$ values for $x=-1,\ \ 1,\ \ 3$ must be included in the table or list of values. The correct values of relevant inputs include:

· $h\ (-3)=\frac{3}{2}$

· $h\ (-1)=-\frac{1}{2}$

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· $h(1) = 1 - \frac{\pi}{4}$

· $h(3) = \frac{5}{2} - \frac{\pi}{2}$

· $h(4) = 2 - \frac{\pi}{2}$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

· At $x = -1$, h has a relative minimum because $h' = g$ is negative for $x < -1$ and $h' = g$ is positive for $x > -1$.

· At $x = 1$, h has neither because $h' = g$ is positive for $x < 1$ and $h' = g$ is positive for $x > 1$.

· At $x = 3$, h has a relative maximum because $h' = g$ is positive for $x < 3$ and $h' = g$ is negative for $x > 3$.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Relative minimum with justification
- [ ] Neither with justification
- [ ] Relative maximum with justification

Solution:

At $x = -1$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

At $x = 1$, h has neither because $h'(x) = g(x)$ does not change sign there.

At $x = 3$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[3,\ 4]$, $(3,\ 4]$, $[3,\ 4)$, and $(3,\ 4)$ all earn the first point.

In order to be eligible for the second point, the first point must have been earned.

A response referencing only h' is eligible to earn the second point.

A response referencing only g is not eligible to earn the second point, unless the response has established the link between $g(x)$ and $h'(x)$ previously in part (a) or part (b), in which case the response is eligible to earn the second point.

The reason “ h' is decreasing” can also be expressed by either “ h'' is negative” or “the slope of h' is negative.”

A response referencing just “the function” or “it” or “the slope” is vague and does not earn the second point. Note that responses of this type are eligible for a maximum of one point in part (b) and one point in part (c); that is, a response of this type fails to earn points in both parts (b) and (c).

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0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

h is decreasing where $h' = g$ is negative.

h is concave down where $h' = g$ is decreasing.

h is decreasing and concave down on the open interval $(3,\ 4)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Correct supporting work is needed to earn this point.

The correct decimal value is $h\ (2) =\ 0.429$. If this decimal value is given, it must be correct, rounded or truncated, to three decimal places.

0	1

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$h\ (2) = 2\ \cdot\ 1 - \frac{\pi}{2}\ \cdot\ 1^2 = 2 - \frac{\pi}{2}$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $-\frac{1}{2} + 2$ earns both points. This is the minimal amount of work that earns both points.

As indicated by this minimal answer, the value of an integral (that is, -2 or $\frac{1}{2}$) does not need to be explicitly linked with an integral to earn the first point.

A response of $h\ (-3) = \frac{3}{2}$ with no supporting work earns the second point only.

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0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] $\int_{-3}^{-1} g(t)\,dt = -2$ – OR – $\int_{-1}^{0} g(t)\,dt = \frac{1}{2}$
- [ ] Answer

Solution:

$$h(-3) = \int_{0}^{-3} g(t)\,dt = -\int_{-3}^{0} g(t)\,dt$$

$$= -\int_{-3}^{-1} g(t)\,dt - \int_{-1}^{0} g(t)\,dt = -(-2) - \frac{1}{2} = \frac{3}{2}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $h''(-1) = 1$ or $g'(-1) = 1$ or even simply stating a numerical value of 1 clearly marked as the answer earns the point.

No supporting work is required because the answer of 1 can easily be read from the graph.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$$h''(x) = g'(x)$$

$$h''(-1) = g'(-1) = \frac{1-(-2)}{0-(-3)} = 1$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that h is continuous and differentiable. An ambiguous reference to h (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which h is continuous and differentiable does not need to be stated. If an interval is given, it must be correct with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., either by referencing the Mean Value Theorem or by restating the statement in the stem of the

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problem). Examples include:

· "Yes, $h\left(x\right)$ is continuous and differentiable so MVT applies" earns both points.

· "Yes, $h\left(x\right)$ is continuous and differentiable, therefore there must be a value where $h'\left(r\right)$ equals the average rate of change of $h'\left(x\right)$ " earns both points.

· "Yes, $h'\left(x\right)$ (the derivative of $h\left(x\right)$) is continuous and differentiable, therefore there must be a value where $h'\left(r\right)$ equals the average rate of change of $h\left(x\right)$ " does not earn the first point, but does earn the second point.

· "Yes because of MVT" does not earn the first point, but does earn the second point.

· "Yes, because $h\left(x\right)$ is continuous and differentiable" earns the first point, but not the second point.

· "Yes, because $h\left(x\right)$ is continuous" earns neither point.

· "Yes, because $h\left(x\right)$ is differentiable" earns neither point.

· "Yes" earns neither point.

A response of "No" earns neither point, regardless of any other explanation including any reference to MVT.

Special case: a response that claims "h is not differentiable, so therefore the MVT does not apply" earns one of the two points.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Conditions
- [ ] Conclusion using Mean Value Theorem

Solution:

$h' = g \Rightarrow h$ is differentiable on $-3 \leq x \leq 1 \Rightarrow h$ is continuous on $-3 \leq x \leq 1$.

Therefore, the Mean Value Theorem can be applied to h on the interval $-3 \leq x \leq 1$ to guarantee that there exists a value of r, for $-3 < r < 1$, such that $h'\left(r\right)$ equals the average rate of change of h over the interval $-3 \leq x \leq 1$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\lim\limits_{x\to 2}\frac{h'(x)}{5x} = \frac{1}{10}$ earns the point, as does just stating the answer of $\frac{1}{10}$.

No additional supporting work is required.

The response does not need to state that h' is continuous. If the response does state that h' is continuous, an explanation is not required.

0	1

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The student response accurately includes the criteria below.

☐ Answer

Solution:

$h' = g \Rightarrow\ h'$ is continuous. $\Rightarrow \lim\limits_{x\to 2} h'(x) = h'(2)$

$\lim\limits_{x\to 2} \dfrac{h'(x)}{5x} = \dfrac{h'(2)}{5\cdot 2} = \dfrac{g(2)}{10} = \dfrac{1}{10}$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x=-2$ is substituted for x, then $x=-2$ must appear unsimplified in at least one factor in order to earn the first point. Responses that earn the point include:

- $h'\left(x^2\right)\cdot 2x$
- $g\left(x^2\right)\cdot 2x$
- $h'\left((-2)^2\right)\cdot 2\cdot(-2)$
- $g\left((-2)^2\right)\cdot 2\cdot(-2)$
- $h'(4)\cdot 2\cdot(-2)$
- $g(4)\cdot 2\cdot(-2)$
- $h'\left((-2)^2\right)\cdot(-4)$
- $g\left((-2)^2\right)\cdot(-4)$

Responses that do not earn the point include:

- $h'\left(x^2\right)\cdot 2$
- $g\left(x^2\right)\cdot 2$
- $h'(x)\cdot 2x$
- $g(x)\cdot 2x$
- $h'(4)\cdot(-4)$
- $g(4)\cdot(-4)$
- $h\left(x^2\right)\cdot 2x$

The second point is earned for evidence of attempting to apply the Fundamental Theorem of Calculus as evidenced by linking h' and g either directly or indirectly in the presented derivative of $k(x)$ or in $k'(-2)$. For example, the second point is earned by $g\left(x^2\right)$ or by $g\left((-2)^2\right)$.

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The second point may be earned without earning the first point. For example, $h'\left(x^2\right)\cdot 2=g\left(x^2\right)\cdot 2$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because h' and Ag are linked.

The second point may be earned by going directly from $h'(4)$ to -1. The function g does not have to be stated explicitly. Responses that earn the first two points include:

$\cdot\, h'\left((-2)^2\right)\cdot 2\cdot(-2)=-1\cdot 2\cdot(-2)$

$\cdot\, h'\left((-2)^2\right)\cdot(-4)=-1\cdot(-4)$

Recall, if $x=-2$ is substituted for x, then $x=-2$ must appear unsimplified in at least one factor to earn the first point.

The first two points may be earned by a single expression. For example, both $g\left(x^2\right)\cdot 2x$ and $g\left((-2)^2\right)\cdot 2\cdot(-2)$ earn the first two points.

The third point is earned for the correct answer of 4 simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

Be careful of parenthesis errors in part (i). Responses of the form $h'(x)^2\cdot 2x$ or $\left(h'(x)\right)^2\cdot 2x$ are not uncommon. These types of responses fail to earn the first point, are eligible for the second point, and fail to earn the third point.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$k'(x)=h'\left(x^2\right)\cdot 2x$

$=g\left(x^2\right)\cdot 2x$

$k'(-2)=g\left((-2)^2\right)\cdot 2\cdot(-2)=g(4)\cdot(-4)=-1\cdot(-4)=4$

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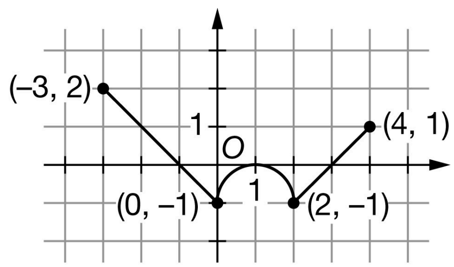
63. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f

The continuous function f has domain $-3 \le x \le 4$. The graph of f, consisting of two line segments and a semicircle, is shown in the figure above. The graph of f has a horizontal tangent at $x = 1$. Let g be the function defined by $g(x) = \int_0^x f(t)\,dt$ for $-3 \le x \le 4$.

(a) Find the x-coordinate of each critical point of g on the interval $-3 < x < 4$.

(b) Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for g. Justify each of your answers.

(c) For $-3 < x < 4$, on what open intervals is g decreasing and concave up? Give a reason for your answer.

(d) Find the value of $g(2)$. Show the computations that lead to your answer.

(e) Find the value of $g(-3)$. Show the computations that lead to your answer.

(f) Find the value of $g''(-2)$, or explain why it does not exist.

(g) Must there exist a value of b, for $-3 < b < 1$, such that $g'(b)$ is equal to the average rate of change of g over the interval $-3 \le x \le 1$? Justify your answer.

(h) Find $\lim_{x \to 2} \frac{g'(x)}{4x}$. Show the computations that lead to your answer.

(i) The function z is defined by $z(x) = g(x^2)$. Find $z'(-2)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -1$, $x = 1$, and $x = 3$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3, 4)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3, 4)$. The point is still earned provided the three critical points $x = -1$, $x = 1$, and $x = 3$ are correctly listed.

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If the critical points are given as ordered pairs, e.g., $(-1,0)$, $(1,0)$, and $(3,0)$, read the x-coordinates, and disregard the y-coordinates.

0	1 ✓

The student response accurately includes the criteria below.

☐ Answer

Solution:

$g'(x) = f(x) = 0$ at $x = -1$, $x = 1$, and $x = 3$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on g' is eligible for all three points.

If a response gives justifications based only on f, then:

· the response is eligible for all three points, if the response presents the connection between $g'(x)$ and $f(x)$, (that is, $g' = f$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of f or g' as $g\prime = f$, throughout all parts of this question.

· the response is eligible for at most two of the three points, if the response does not present the connection $g'(x) = f(x)$. Any additional error (for example, not justifying that at $x = -1$, f has a relative maximum) is eligible for at most one of the three points.

Read references to "the graph" as references to $f(x)$ as indicated by the stem.

A response referencing "the function" or "it" or "the slope" is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of "the function" or "it" or "the slope" with g' produces a response that would earn all three points.

A response that discusses g' at a particular point (e.g., " $g'(3)$ changes from negative to positive") when the discussion should have been about $g'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. $g(x)$ values for $x = -1,\ \ 1,\ \ 3$ must be included in the table or list of values. The correct values of relevant inputs include:

· $g(-3) = -\frac{3}{2}$

· $g(-1) = \frac{1}{2}$

· $g(1) = \frac{\pi}{4} - 1$

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· $g(3) = \frac{\pi}{2} - \frac{5}{2}$

· $g(4) = \frac{\pi}{2} - 2$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a "global" rather than a "local" argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

· At $x = -1$, g has a relative maximum because $g' = f$ is positive for $x < -1$ and $g' = f$ is negative for $x > -1$.

· At $x = 1$, g has neither because $g' = f$ is negative for $x < 1$ and $g' = f$ is negative for $x > 1$.

· At $x = 3$, g has a relative minimum because $g' = f$ is negative for $x < 3$ and $g' = f$ is positive for $x > 3$.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Relative maximum with justification
- ☐ Neither with justification
- ☐ Relative minimum with justification

Solution:

At $x = -1$, g has a relative maximum because $g'(x) = f(x)$ changes from positive to negative there.

At $x = 1$, g has neither because $g'(x) = f(x)$ does not change sign there.

At $x = 3$, g has a relative minimum because $g'(x) = f(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[0,\ 1]$ and $(2,\ 3]$ or $[0,\ 1]$ and $(2,\ 3)$ both earn the first point.

In order to be eligible for the second point, the first point must have been earned.

A response referencing only g' is eligible to earn the second point.

A response referencing only f is not eligible to earn the second point, unless the response has established the link between $f(x)$ and $g'(x)$ previously in part (a) or part (b), in which case the response is eligible to earn the second point.

A response referencing just "the function" or "it" or "the slope" is vague and does not earn the second point. Note that responses of this type are eligible for a maximum of one point in part (b) and one point in part (c); that is, a response of this type fails to earn points in both parts (b) and (c).

The reason " g' is increasing" can also be expressed by either " g'' is positive" or "the slope of g' is positive."

Responses giving a completely correct justification for only one of the two subintervals earn one of the two points.

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0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- [ ] Answer
- [ ] Reason

Solution:

g is decreasing where $g' = f$ is negative.

g is concave up where $g' = f$ is increasing.

g is decreasing and concave up on the open intervals $(0,\ 1)$ and $(2,\ 3)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Correct supporting work is needed to earn this point.

The correct decimal value is $g\,(2) = -0.429$. If this decimal value is given, it must be correct, rounded or truncated, to three decimal places.

0	1 ✓

The student response accurately includes the criteria below.

- [ ] Answer

$g\,(2) = -\left(2 \cdot 1 - \frac{\pi}{2} \cdot 1^2\right) = -\left(2 - \frac{\pi}{2}\right) = \frac{\pi}{2} - 2$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\frac{1}{2} - 2$ earns both points. This is the minimal amount of work that earns both points.

As indicated by this minimal answer, the value of an integral (that is, 2 or $-\frac{1}{2}$) does not need to be explicitly linked with an integral to earn the first point.

A response of $g\,(-3) = -\frac{3}{2}$ with no supporting work earns the second point only.

0	1	2 ✓

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The student response accurately includes **both** of the criteria below.

- ☐ $\int_{-3}^{-1} f(t)\,dt = 2$ – OR – $\int_{-1}^{0} f(t)\,dt = -\frac{1}{2}$
- ☐ Answer

Solution:

$$g(-3) = \int_{0}^{-3} f(t)\,dt = -\int_{-3}^{0} f(t)\,dt$$
$$= -\int_{-3}^{-1} f(t)\,dt - \int_{-1}^{0} f(t)\,dt = -2 - \left(-\tfrac{1}{2}\right) = -\tfrac{3}{2}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $g''(-2) = -1$ or $f'(-2) = -1$ or even simply stating a numerical value of -1 clearly marked as the answer earns the point.

No supporting work is required because the answer of -1 can easily be read from the graph.

0	1 ✓

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$g''(x) = f'(x)$$

$$g''(-2) = f'(-2) = \frac{-1-2}{0-(-3)} = -1$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that g is continuous and differentiable. An ambiguous reference to g (e.g., "the function is continuous and differentiable") does not earn the first point. The interval on which g is continuous and differentiable does not need to be stated. If an interval is given, it must be correct with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

· "Yes, $g(x)$ is continuous and differentiable so MVT applies" earns both points.

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· "Yes, $g\left(x\right)$ is continuous and differentiable, therefore there must be a value where $g'\left(b\right)$ equals the average rate of change of $g\left(x\right)$ " earns both points.

· "Yes, $g'\left(x\right)$ (the derivative of $g\left(x\right)$) is continuous and differentiable, therefore there must be a value where $g'\left(b\right)$ equals the average rate of change of $g\left(x\right)$ " does not earn the first point, but does earn the second point.

· "Yes because of MVT" does not earn the first point, but does earn the second point.

· "Yes, because $g\left(x\right)$ is continuous and differentiable" earns the first point, but not the second point.

· "Yes, because $g\left(x\right)$ is continuous" earns neither point.

· "Yes, because $g\left(x\right)$ is differentiable" earns neither point.

· "Yes" earns neither point.

A response of "No" earns neither point, regardless of any other explanation including any reference to MVT.

Special case: a response that claims " g is not differentiable, so therefore the MVT does not apply" earns one of the two points.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- [ ] Conditions
- [ ] Conclusion using Mean Value Theorem

Solution:

$g' = f \Rightarrow g$ is differentiable on $-3 \le x \le 1. \Rightarrow g$ is continuous on $-3 \le x \le 1.$

Therefore, the Mean Value Theorem can be applied to g on the interval $-3 \le x \le 1$ to guarantee that there exists a value of b, for $-3 < b < 1$, such that $g'\left(b\right)$ equals the average rate of change of g over the interval $-3 \le x \le 1$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\lim\limits_{x\to 2}\frac{g'(x)}{4x} = \frac{-1}{8}$ earns the point, as does just stating the answer of $\frac{-1}{8}$.

No additional supporting work is required.

The response does not need to state that g' is continuous. If the response does state that g' is continuous, an explanation is not required.

0	1 ✓

The student response accurately includes the criteria below.

Answer

5-1-1

☐

Solution:

$g' = f \Rightarrow g'$ is continuous. $\Rightarrow \lim\limits_{x \to 2} g'(x) = g'(2)$

$$\lim_{x \to 2} \frac{g'(x)}{4x} = \frac{g'(2)}{4 \cdot 2} = \frac{f(2)}{8} = \frac{-1}{8}$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = -2$ is substituted for x, then $x = -2$ must appear unsimplified in at least one factor to earn the first point. Responses that earn the point include:

- $g'\left(x^2\right) \cdot 2x$
- $f\left(x^2\right) \cdot 2x$
- $g'\left((-2)^2\right) \cdot 2 \cdot (-2)$
- $f\left((-2)^2\right) \cdot 2 \cdot (-2)$
- $g'(4) \cdot 2 \cdot (-2)$
- $f(4) \cdot 2 \cdot (-2)$
- $g'\left((-2)^2\right) \cdot (-4)$
- $f\left((-2)^2\right) \cdot (-4)$

Responses that do not earn the point include:

- $g'\left(x^2\right) \cdot 2$
- $f\left(x^2\right) \cdot 2$
- $g'(x) \cdot 2x$
- $f(x) \cdot 2x$
- $g'(4) \cdot (-4)$
- $f(4) \cdot (-4)$
- $g\left(x^2\right) \cdot 2x$

The second point is earned for evidence of attempting to apply the Fundamental Theorem of Calculus as evidenced by linking g' and f either directly or indirectly in the presented derivative of $z(x)$ or in $z'(-2)$. For example, the second point is earned by $f\left(x^2\right)$ or by $f\left((-2)^2\right)$.

The second point may be earned without earning the first point. For example, $g'\left(x^2\right) \cdot 2 = f\left(x^2\right) \cdot 2$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because g' and f are linked.

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The second point may be earned by going directly from $g'(4)$ to 1. The function f does not have to be stated explicitly. Responses that earn the first two points include:

- $g'\left((-2)^2\right) \cdot 2 \cdot (-2) = 1 \cdot 2 \cdot (-2)$
- $g'\left((-2)^2\right) \cdot (-4) = 1 \cdot (-4)$

Recall, if $x = -2$ is substituted for x, then $x = -2$ must appear unsimplified in at least one factor to earn the first point.

The first two points may be earned by a single expression. For example, both $f\left(x^2\right) \cdot 2x$ and $f\left((-2)^2\right) \cdot 2 \cdot (-2)$ earn the first two points.

The third point is earned for the correct answer of -4, simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

Be careful of parenthesis errors in part (i). Responses of the form $g'(x)^2 \cdot 2x$ or $(g'(x))^2 \cdot 2x$ are not uncommon. These type of responses fail to earn the first point, are eligible for the second point, and fail to earn the third point.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$z'(x) = g'\left(x^2\right) \cdot 2x$

$= f\left(x^2\right) \cdot 2x$

$z'(-2) = f\left((-2)^2\right) \cdot 2 \cdot (-2) = f(4) \cdot (-4) = 1 \cdot (-4) = -4$

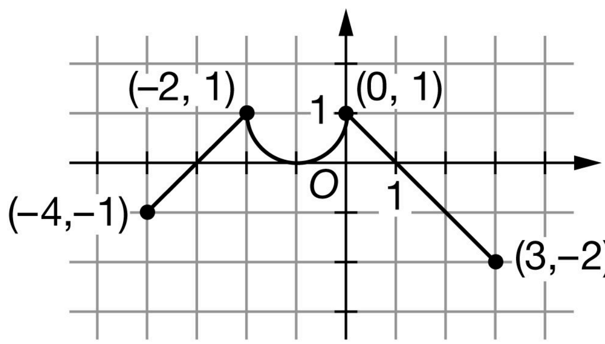
5-1-1

64. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f

The continuous function f has domain $-4 \leq x \leq 3$. The graph of f, consisting of two line segments and a semicircle, is shown in the figure above. The graph of f has a horizontal tangent at $x=-1$. Let h be the function defined by $h(x)=\int_0^x f(t)\,dt$ for $-4 \leq x \leq 3$.

(a) Find the x-coordinate of each critical point of h on the interval $-4 < x < 3$.

(b) Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h. Justify each of your answers.

(c) For $-4 < x < 3$, on what open intervals is h increasing and concave down? Give a reason for your answer.

(d) Find the value of $h(-2)$. Show the computations that lead to your answer.

(e) Find the value of $h(3)$. Show the computations that lead to your answer.

(f) Find the value of $h''(2)$, or explain why it does not exist.

(g) Must there exist a value of k, for $-4 < k < 0$, such that $h'(k)$ is equal to the average rate of change of h over the interval $-4 \leq x \leq 0$? Justify your answer.

(h) Find $\lim\limits_{x \to -2} \frac{h'(x)}{3x}$. Show the computations that lead to your answer.

(i) The function r is defined by $r(x)=h(-3x)$. Find $r'(-1)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x=-3,\ \ x=-1$, and $x=1$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-4,\ 3)$, or if any critical values are missing.

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Disregard presentations of the endpoints or any other numbers outside the open interval $(-4,\ 3)$. The point is still earned provided the three critical points $x = -3,\ \ x = -1$, and $x = 1$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3,\ \ 0)$, $(-1,\ \ 0)$, and $(1,\ \ 0)$, read the x-coordinates, and disregard the y-coordinates.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$h'(x) = f(x) = 0$ at $x = -3$, $x = -1$, and $x = 1$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on f, then:

· the response is eligible for all three points, if the response presents the connection between $h'(x)$ and $f(x)$, (that is, $h' = f$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of f or h' as $h' = f$, throughout all parts of this question.

· the response is eligible for at most two of the three points, if the response does not present the connection $h'(x) = f(x)$. Any additional error (for example, not justifying that at $x = -3$, h has a relative minimum) is eligible for at most one of the three points.

Read references to “the graph” as references to $f(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'(-3)$ changes from negative to positive”) when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. $h(x)$ values for $x = -3,\ \ -1,\ 1$ must be included in the table or list of values. The correct values of relevant inputs include:

· $h(-4) = \frac{\pi}{2} - 2$

· $h(-3) = \frac{\pi}{2} - \frac{5}{2}$

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· $h(-1) = \frac{\pi}{4} - 1$

· $h(1) = \frac{1}{2}$

· $h(3) = -\frac{3}{2}$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a "global" rather than a "local" argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

· At $x = -3$, h has a relative minimum because $h' = f$ is negative for $x < -3$ and $h' = f$ is positive for $x > -3$.

· At $x = -1$, h has neither because $h' = f$ is positive for $x < -1$ and $h' = f$ is positive for $x > -1$.

· At $x = 1$, h has a relative maximum because $h' = f$ is positive for $x < 1$ and $h' = f$ is negative for $x > 1$.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Relative minimum with justification
- ☐ Neither with justification
- ☐ Relative maximum with justification

Solution:

At $x = -3$, h has a relative minimum because $h'(x) = f(x)$ changes from negative to positive there.

At $x = -1$, h has neither because $h'(x) = f(x)$ does not change sign there.

At $x = 1$, h has a relative maximum because $h'(x) = f(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[-2, -1]$ and $(0, 1]$ or $[-2, -1)$ and $(0, 1)$ both earn the first point.

In order to be eligible for the second point, the first point must have been earned.

A response referencing only h' is eligible to earn the second point.

A response referencing only f is not eligible to earn the second point, unless the response has established the link between $f(x)$ and $h'(x)$ previously in part (a) or part (b), in which case the response is eligible to earn the second point.

The reason " h' is decreasing" can also be expressed by either " h'' is negative" or "the slope of h' is negative."

A response referencing just "the function" or "it" or "the slope" is vague and does not earn the second point. Note that responses of this type are eligible for a maximum of one point in part (b) and one point in part (c); that is, a response of this type fails to earn points in both parts (b) and (c).

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Responses giving a completely correct justification for only one of the two subintervals earn one of the two points.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

h is increasing where $h' = f$ is positive.

h is concave down where $h' = f$ is decreasing.

h is increasing and concave down on the open intervals $(-2, \ -1)$ and $(0, \ 1)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Correct supporting work is needed to earn this point.

The correct decimal value is $h\,(-2) = \ -0.429$. If this decimal value is given, it must be correct, rounded or truncated, to three decimal places.

0	1 ✓

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$h\,(-2) = -\left(2 \cdot 1 - \frac{\pi}{2} \cdot 1^2\right) = -\left(2 - \frac{\pi}{2}\right) = \frac{\pi}{2} - 2$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\frac{1}{2} - 2$ earns both points. This is the minimal amount of work that earns both points.

As indicated by this minimal answer, the value of an integral (that is, -2 or $\frac{1}{2}$) does not need to be explicitly linked with an integral to earn the first point.

A response of $h\,(3) = -\frac{3}{2}$ with no supporting work earns the second point only.

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0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ $\int_0^1 f(t)\, dt = \frac{1}{2}$ – OR – $\int_1^3 f(t)\, dt = -2$
- ☐ Answer

Solution:

$$h(3) = \int_0^3 f(t)\, dt$$

$$= \int_0^1 f(t)\, dt + \int_1^3 f(t)\, dt = \frac{1}{2} + (-2) = -\frac{3}{2}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $h''(2) = -1$ or $f'(2) = -1$ or even simply stating a numerical value of -1 clearly marked as the answer earns the point.

No supporting work is required because the answer of -1 can easily be read from the graph.

0	1

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$h''(x) = f'(x)$$

$$h''(2) = f'(2) = \frac{1-(-2)}{0-3} = -1$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that h is continuous and differentiable. An ambiguous reference to h (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which h is continuous and differentiable does not need to be stated. If an interval is given, it must be correct with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., either by referencing the Mean Value Theorem or by restating the statement in the stem of the

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problem). Examples include:

· “Yes, $h\ (x)$ is continuous and differentiable so MVT applies” earns both points.

· “Yes, $h\ (x)$ is continuous and differentiable, therefore there must be a value where $h'\ (k)$ equals the average rate of change of $h\ (x)$ ” earns both points.

· “Yes, $h'\ (x)$ (the derivative of $h\ (x)$) is continuous and differentiable, therefore there must be a value where $h'\ (k)$ equals the average rate of change of $h\ (x)$” does not earn the first point, but does earn the second point.

· “Yes because of MVT” does not earn the first point, but does earn the second point.

· “Yes, because $h\ (x)$ is continuous and differentiable” earns the first point, but not the second point.

· “Yes, because $h\ (x)$ is continuous” earns neither point.

· “Yes, because $h\ (x)$ is differentiable” earns neither point.

· “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.

Special case: a response that claims “ h is not differentiable, so therefore the MVT does not apply” earns one of the two points.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- [ ] Conditions
- [ ] Conclusion using Mean Value Theorem

Solution:

$h' = f \Rightarrow h$ is differentiable on $-4 \leq x \leq 0$. $\Rightarrow h$ is continuous on $-4 \leq x \leq 0$.

Therefore, the Mean Value Theorem can be applied to h on the interval $-4 \leq x \leq 0$ to guarantee that there exists a value of k, for $-4 < k < 0$, such that $h'\ (k)$ equals the average rate of change of h over the interval $-4 \leq x \leq 0$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\lim\limits_{x \to -2} \dfrac{h'(x)}{3x} = \dfrac{-1}{6}$ earns the point, as does just stating the answer of $\dfrac{-1}{6}$.

No additional supporting work is required.

The response does not need to state that h' is continuous. If the response does state that h' is continuous, an explanation is not required.

0	1 ✓

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The student response accurately includes the criteria below.

☐ Answer

Solution:

$h' = f \Rightarrow h'$ is continuous. $\Rightarrow \lim\limits_{x \to -2} h'(x) = h'(-2)$

$$\lim_{x \to -2} \frac{h'(x)}{3x} = \frac{h'(-2)}{3 \cdot (-2)} = \frac{f(-2)}{-6} = \frac{1}{-6}$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = -1$ is substituted for x, then $x = -1$ must appear unsimplified to earn the first point. Responses that earn the point include:

· $h'(-3x) \cdot (-3)$

· $f(-3x) \cdot (-3)$

· $h'(-3 \cdot -1) \cdot (-3)$

· $f(-3 \cdot -1) \cdot (-3)$

Responses that do not earn the point include:

· $h'(x) \cdot (-3)$

· $f(x) \cdot (-3)$

· $h'(3) \cdot (-3)$

· $f(3) \cdot (-3)$

· $h(-3x) \cdot (-3)$

The second point is earned for evidence of attempting to apply the Fundamental Theorem of Calculus as evidenced by linking h' and f either directly or indirectly in the presented derivative of $r(x)$ or in $r'(-1)$. For example, the second point is earned by $f(-3x)$ or by $f(-3 \cdot -1)$.

The second point may be earned without earning the first point. For example, $h'(-3x) = f(-3x)$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because h' and f are linked.

The second point may be earned by going directly from $h'(3)$ to -2. The function f does not have to be stated explicitly. For example, $h'(-3 \cdot -1) \cdot (-3) = -2 \cdot (-3)$ earns the first two points. Recall, if $x = -1$ is substituted for x, then $x = -1$ must appear unsimplified to earn the first point.

The first two points may be earned by a single expression. For example, both $f(-3x) \cdot (-3)$ and $f(-3 \cdot -1) \cdot (-3)$ earn the first two points.

The third point is earned for the correct answer of 6, simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

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0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$r'(x) = h'(-3x) \cdot (-3)$

$= f(-3x) \cdot (-3)$

$r'(-1) = f((-3) \cdot (-1)) \cdot (-3) = f(3) \cdot (-3) = -2 \cdot (-3) = 6$

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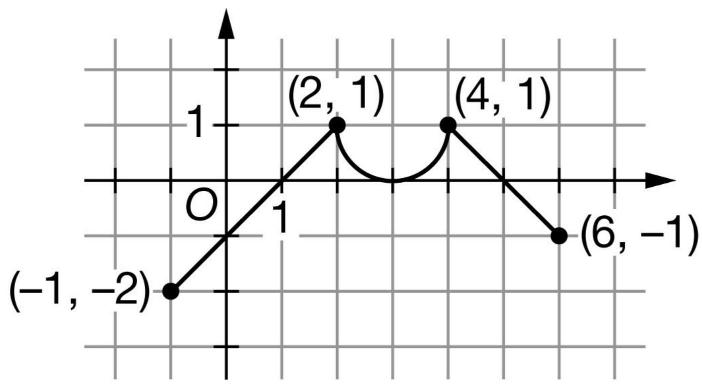
65. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of h

The continuous function h has domain $-1 \leq x \leq 6$. The graph of h, consisting of two line segments and a semicircle, is shown in the figure above. The graph of h has a horizontal tangent at $x = 3$. Let g be the function defined by $g(x) = \int_2^x h(t)\,dt$ for $-1 \leq x \leq 6$.

(a) Find the x-coordinate of each critical point of g on the interval $-1 \leq x \leq 6$.

(b) Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for g. Justify each of your answers.

(c) For $-1 \leq x \leq 6$, on what open intervals is g decreasing and concave up? Give a reason for your answer.

(d) Find the value of $g(4)$. Show the computations that lead to your answer.

(e) Find the value of $g(-1)$. Show the computations that lead to your answer.

(f) Find the value of $g''(0)$, or explain why it does not exist.

(g) Must there exist a value of d, for $-1 < d < 3$, such that $g'(d)$ is equal to the average rate of change of g over the interval $-1 \leq x \leq 3$? Justify your answer.

(h) Find $\lim_{x \to 4} \frac{g'(x)}{3x}$. Show the computations that lead to your answer.

(i) The function a is defined by $a(x) = g(2x)$. Find $a'(0)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = 1$, $x = 3$, and $x = 5$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-1, 6)$, or if any critical values are missing.

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Disregard presentations of the endpoints or any other numbers outside the open interval $(-1, 6)$. The point is still earned provided the three critical points $x = 1$, $x = 3$, and $x = 5$ are correctly listed.

If the critical points are given as ordered pairs, e.g.,$(1, 0)$, $(3, 0)$, and $(5, 0)$, read the x -coordinates, and disregard the y -coordinates.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$g'(x) = h(x) = 0$ at $x = 1$, $x = 3$, and $x = 5$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on g' is eligible for all three points.

If a response gives justifications based only on h , then:

· the response is eligible for all three points, if the response presents the connection between $g'(x)$ and $h(x)$, (that is, $g' = h$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of h or g' as $g' = h$, throughout all parts of this question.

· the response is eligible for at most two of the three points, if the response does not present the connection $g'(x) = h(x)$. Any additional error (for example, not justifying that at $x = 1$, g has a relative minimum) is eligible for at most one of the three points.

Read references to “the graph” as references to $h(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn all three points.

A response that discusses g' at a particular point (e.g., “$g'(1)$ changes from negative to positive”) when the discussion should have been about $g'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. $g(x)$ value for $x = 1, 3, 5$ must be included in the table or list of values. The correct values of relevant inputs include:

· $g(-1) = \frac{3}{2}$

· $g(1) = -\frac{1}{2}$

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· $g(3) = 1 - \frac{\pi}{4}$

· $g(5) = \frac{5}{2} - \frac{\pi}{4}$

· $g(6) = 2 - \frac{\pi}{4}$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a "global" rather than a "local" argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

· At $x = 1$, g has a relative minimum because $g' = h$ is negative for $x < 1$ and $g' = h$ is positive for $x > 1$.

· At $x = 3$, g has neither because $g' = h$ is positive for $x < 3$ and $g' = h$ is positive for $x > 3$.

· At $x = 5$, g has a relative maximum because $g' = h$ is positive for $x < 5$ and $g' = h$ is negative for $x > 5$.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- [ ] Relative minimum with justification
- [ ] Neither with justification
- [ ] Relative maximum with justification

Solution:

At $x = 1$, g has a relative minimum because $g'(x) = h(x)$ changes from negative to positive there.

At $x = 3$, g has neither because $g'(x) = h(x)$ does not change sign there.

At $x = 5$, g has a relative maximum because $g'(x) = h(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[-1, 1]$, $(-1, 1]$, $[-1, 1)$, and $(-1, 1)$ all earn the first point.

In order to be eligible for the second point, the first point must have been earned.

A response referencing only g' is eligible to earn the second point.

A response referencing only h is not eligible to earn the second point, unless the response has established the link between $h(x)$ and $g'(x)$ previously in part (a) or part (b), in which case the response is eligible to earn the second point.

The reason "g' is increasing" can also be expressed by either "g'' is positive" or "the slope of g' is positive."

A response referencing just "the function" or "it" or "the slope" is vague and does not earn the second point. Note that responses of this type are eligible for a maximum of one point in part (b) and one point in part (c); that is, a response of this type fails to earn points in both parts (b) and (c).

5-1-1

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

g is decreasing where $g' = h$ is negative.

g is concave up where $g' = h$ is increasing.

g is decreasing and concave up on the open interval $(-1, 1)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Correct supporting work is needed to earn this point.

The correct decimal value is $g(4) = \ 0.429$. If this decimal value is given, it must be correct, rounded or truncated, to three decimal places.

0	1

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$g(4) = 2 \cdot 1 - \frac{\pi}{2} \cdot 1^2 = 2 - \frac{\pi}{2}$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $-\frac{1}{2} + 2$ earns both points. This is the minimal amount of work that earns both points.

As indicated by this minimal answer, the value of an integral (that is, -2 or $\frac{1}{2}$) does not need to be explicitly linked with an integral to earn the first point.

A response of $g\,(-1) = \frac{3}{2}$ with no supporting work earns the second point only.

5-1-1

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] $\int_{-1}^{1} h(t)\ dt = -2$ – OR – $\int_{1}^{2} h(t)\ dt = \frac{1}{2}$
- [ ] Answer

Solution:

$$g(-1) = \int_{2}^{-1} h(t)\ dt = -\int_{-1}^{2} h(t)\ dt$$
$$= -\int_{-1}^{1} h(t)\ dt - \int_{1}^{2} h(t)\ dt = -(-2) - \frac{1}{2} = \frac{3}{2}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $g''(0) = 1$ or $h'(0) = 1$ or even simply stating a numerical value of 1 clearly marked as the answer earns the point.

No supporting work is required because the answer of 1 can easily be read from the graph.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$$g''(x) = h'(x)$$

$$g''(0) = h'(0) = \frac{1}{2} = \frac{1-(-2)}{2-(-1)}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that g is continuous and differentiable. An ambiguous reference to g (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which g is continuous and differentiable does not need to be stated. If an interval is given, it must be correct with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

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· "Yes, $g(x)$ is continuous and differentiable so MVT applies" earns both points.

· "Yes, $g(x)$ is continuous and differentiable, therefore there must be a value where $g'(d)$ equals the average rate of change of $g(x)$ " earns both points.

· "Yes, $g'(x)$ (the derivative of $g(x)$) is continuous and differentiable, therefore there must be a value where $g'(d)$ equals the average rate of change of $g(x)$" does not earn the first point, but does earn the second point.

· "Yes because of MVT" does not earn the first point, but does earn the second point.

· "Yes, because $g(x)$ is continuous and differentiable" earns the first point, but not the second point.

· "Yes, because $g(x)$ is continuous" earns neither point.

· "Yes, because $g(x)$ is differentiable" earns neither point.

· "Yes" earns neither point.

A response of "No" earns neither point, regardless of any other explanation including any reference to MVT.

Special case: a response that claims "g is not differentiable, so therefore the MVT does not apply" earns one of the two points.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$g' = h \Rightarrow g$ is differentiable on $-1 \leq x \leq 3. \Rightarrow g$ is continuous on $-1 \leq x \leq 3.$

Therefore, the Mean Value Theorem can be applied to g on the interval $-1 \leq x \leq 3$ to guarantee that there exists a value of d, for $-1 < d < 3$, such that $g'(d)$ equals the average rate of change of g over the interval $-1 \leq x \leq 3$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\lim\limits_{x\to 4} \frac{g'(x)}{3x} = \frac{1}{12}$ earns the point, as does just stating the answer of $\frac{1}{12}$.

No additional supporting work is required.

The response does not need to state that g' is continuous. If the response does state that g' is continuous, an explanation is not required.

0	1 ✓

The student response accurately includes the criteria below.

5-1-1

☐ Answer

Solution:

$g' = h \Rightarrow g'$ is continuous. $\Rightarrow \lim_{x \to 4} g'(x) = g'(4)$

$$\lim_{x \to 4} \frac{g'(x)}{3x} = \frac{g'(4)}{3 \cdot 4} = \frac{h(4)}{12} = \frac{1}{12}$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = 0$ is substituted for x, then $x = 0$ must appear unsimplified to earn the first point. Responses that earn the point include:

· $g'(2x) \cdot 2$

· $h(2x) \cdot 2$

· $g'(2 \cdot 0) \cdot 2$

· $h(2 \cdot 0) \cdot 2$

Responses that do not earn the point include:

· $g'(x) \cdot 2$

· $h(x) \cdot 2$

· $g'(0) \cdot 2$

· $h(0) \cdot 2$

· $g(2x) \cdot 2$

The second point is earned for evidence of attempting to apply the Fundamental Theorem of Calculus as evidenced by linking g' and h either directly or indirectly in the presented derivative of $a(x)$ or in $a'(0)$. For example, the second point is earned by $h(2x)$ or by $h(2 \cdot 0)$.

The second point may be earned without earning the first point. For example, $g'(2x) = h(2x)$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because g' and h are linked.

The second point may be earned by going directly from $g'(0)$ to -1. The function h does not have to be stated explicitly. For example, $g'(2 \cdot 0) \cdot 2 = -1 \cdot 2$ earns the first two points. Recall, if $x = 0$ is substituted for x, then $x = 0$ must appear unsimplified to earn the first point.

The first two points may be earned by a single expression. For example, both $h(2x) \cdot 2$ and $h(2 \cdot 0) \cdot 2$ earn the first two points.

The third point is earned for the correct answer of -2, simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

0	1	2	3 ✓

5-1-1

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$a'(x) = g'(2x) \cdot 2$

$= h(2x) \cdot 2$

$a'(0) = g'(2 \cdot 0) \cdot 2 = g'(0) \cdot 2 = h(0) \cdot 2 = -1 \cdot 2 = -2$

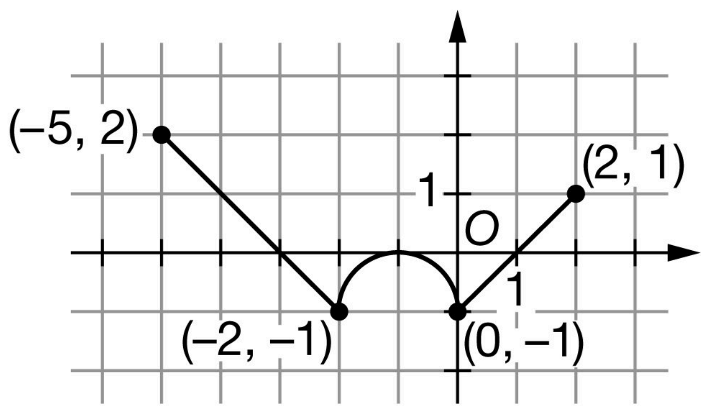
5-1-1

66. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of g

The continuous function g has domain $-5 \leq x \leq 2$. The graph of g, consisting of two line segments and a semicircle, is shown in the figure above. The graph of g has a horizontal tangent at $x = -1$. Let h be the function defined by $h(x) = \int_{-2}^{x} g(t)\,dt$ for $-5 \leq x \leq 2$.

(a) Find the x-coordinate of each critical point of h on the interval $-5 < x < 2$.

(b) Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h. Justify each of your answers.

(c) For $-5 < x < 2$, on what open intervals is h increasing and concave up? Give a reason for your answer.

(d) Find the value of $h(0)$. Show the computations that lead to your answer.

(e) Find the value of $h(-5)$. Show the computations that lead to your answer.

(f) Find the value of $h''(-3)$, or explain why it does not exist.

(g) Must there exist a value of p, for $-3 < p < 1$, such that $h'(p)$ is equal to the average rate of change of h over the interval $-3 \leq x \leq 1$? Justify your answer.

(h) Find $\lim_{x \to -2} \frac{h'(x)}{4x}$. Show the computations that lead to your answer.

(i) The function f is defined by $f(x) = h(-2x)$. Find $f'(1)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -3$, $x = -1$, and $x = 1$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-5,\ 2)$, or if any critical values are missing.

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Disregard presentations of the endpoints or any other numbers outside the open interval $(-5,\ 2)$. The point is still earned provided the three critical points $x = -3,\ \ x = -1$ and $x = 1$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3,\ \ 0)$, $(-1,\ \ 0)$ and $(1,\ \ 0)$, read the x -coordinates, and disregard the y-coordinates.

0	1 ✓

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$h'\ (x) = g\ (x) = 0$ at $x = -3$, $x = -1$, and $x = 1$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on g, then:

· the response is eligible for all three points, if the response presents the connection between $h'\ (x)$ and $g\ (x)$, (that is, $h' = g$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of g or h' as $h' = g$, throughout all parts of this question.

· the response is eligible for at most two of the three points, if the response does not present the connection $h'\ (x) = g\ (x)$. Any additional error (for example, not justifying that at $x = -3$, h has a relative maximum) is eligible for at most one of the three points.

Read references to “the graph” as references to $g\ (x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'\ (1)$ changes from negative to positive”) when the discussion should have been about $h'\ (x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. $h\ (x)$ values for $x = -3,\ \ -1,\ \ 1$ must be included in the table or list of values. The correct values of relevant inputs include:

· $h\ (-5) = -\frac{3}{2}$

· $h\ (-3) = \frac{1}{2}$

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$\cdot\, h(-1) = \frac{\pi}{4} - 1$

$\cdot\, h(1) = \frac{\pi}{2} - \frac{5}{2}$

$\cdot\, h(2) = \frac{\pi}{2} - 2$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a "global" rather than a "local" argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

· At $x = -3$, h has a relative maximum because $h' = g$ is positive for $x < -3$ and $h' = g$ is negative for $x > -3$.

· At $x = -1$, h has neither because $h' = g$ is negative for $x < -1$ and $h' = g$ is negative for $x > -1$.

· At $x = 1$, h has a relative minimum because $h' = g$ is negative for $x < 1$ and $h' = g$ is positive for $x > 1$.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Relative maximum with justification
- ☐ Neither with justification
- ☐ Relative minimum with justification

Solution:

At $x = -3$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

At $x = -1$, h has neither because $h'(x) = g(x)$ does not change sign there.

At $x = 1$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[1,\ 2]$, $(1,\ 2]$, $[1,\ 2)$, and $(1,\ 2)$ all earn the first point.

In order to be eligible for the second point, the first point must have been earned.

A response referencing only h' is eligible to earn the second point.

A response referencing only g is not eligible to earn the second point, unless the response has established the link between $g(x)$ and $h'(x)$ previously in part (a) or part (b), in which case the response is eligible to earn the second point.

The reason " h' is increasing" can also be expressed by either " h'' is positive" or "the slope of h' is positive."

A response referencing just "the function" or "it" or "the slope" is vague and does not earn the second point. Note that responses of this type are eligible for a maximum of one point in part (b) and one point in part (c); that is, a response of this type fails to earn points in both parts (b) and (c).

5-1-1

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

h is increasing where $h' = g$ is positive.

h is concave up where $h' = g$ is increasing.

h is increasing and concave up on the open interval $(1,\ 2)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Correct supporting work is needed to earn this point.

The correct decimal value is $h\,(0) = \ -0.429$. If this decimal value is given, it must be correct, rounded or truncated, to three decimal places.

0	1

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$h\,(0) = -\left(2 \cdot 1 - \frac{\pi}{2} \cdot 1^2\right) = -\left(2 - \frac{\pi}{2}\right) = \frac{\pi}{2} - 2$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\frac{1}{2} - 2$ earns both points. This is the minimal amount of work that earns both points.

As indicated by this minimal answer, the value of an integral (that is, -2 or $-\frac{1}{2}$) does not need to be explicitly linked with an integral to earn the first point.

A response of $h\,(-5) = -\frac{3}{2}$ with no supporting work earns the second point only.

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0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] $\int_{-5}^{-3} g(t)\ dt = 2$ – OR – $\int_{-3}^{-2} g(t)\ dt = -\frac{1}{2}$
- [ ] Answer

Solution:

$$h(-5) = \int_{-2}^{-5} g(t)\ dt = -\int_{-5}^{-2} g(t)\ dt$$
$$= -\int_{-5}^{-3} g(t)\ dt - \int_{-3}^{-2} g(t)\ dt = -(2) - \left(-\tfrac{1}{2}\right) = -\tfrac{3}{2}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $h''(-3) = -1$ or $g'(-3) = -1$ or even simply stating a numerical value of -1 clearly marked as the answer earns the point.

No supporting work is required because the answer of -1 can easily be read from the graph.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$$h''(x) = g'(x)$$

$$h''(-3) = g'(-3) = \frac{2-(-1)}{-5-(-2)} = -1$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that h is continuous and differentiable. An ambiguous reference to h (e.g., "the function is continuous and differentiable") does not earn the first point. The interval on which h is continuous and differentiable does not need to be stated. If an interval is given, it must be correct with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., either by referencing the Mean Value Theorem or by restating the statement in the stem of the

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problem). Examples include:

· "Yes, $h\left(x\right)$ is continuous and differentiable so MVT applies" earns both points.

· "Yes, $h\left(x\right)$ is continuous and differentiable, therefore there must be a value where $h'\left(p\right)$ equals the average rate of change of $h\left(x\right)$ " earns both points.

· "Yes, $h'\left(x\right)$ (the derivative of $h\left(x\right)$) is continuous and differentiable, therefore there must be a value where $h'\left(p\right)$ equals the average rate of change of $h\left(x\right)$ " does not earn the first point, but does earn the second point.

· "Yes because of MVT" does not earn the first point, but does earn the second point.

· "Yes, because $h\left(x\right)$ is continuous and differentiable" earns the first point, but not the second point.

· "Yes, because $h\left(x\right)$ is continuous" earns neither point.

· "Yes, because $h\left(x\right)$ is differentiable" earns neither point.

· "Yes" earns neither point.

A response of "No" earns neither point, regardless of any other explanation including any reference to MVT.

Special case: a response that claims " h is not differentiable, so therefore the MVT does not apply" earns one of the two points.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$h' = g \Rightarrow h$ is differentiable on $-3 \leq x \leq 1. \Rightarrow h$ is continuous on $-3 \leq x \leq 1.$

Therefore, the Mean Value Theorem can be applied to h on the interval $-3 \leq x \leq 1$ to guarantee that there exists a value of p, for $-3 < p < 1$, such that $h'\left(p\right)$ equals the average rate of change of h over the interval $-3 \leq x \leq 1$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\lim\limits_{x \to 2} \frac{h'(x)}{4x} = \frac{1}{8}$ earns the point, as does just stating the answer of $\frac{1}{8}$.

No additional supporting work is required.

The response does not need to state that h' is continuous. If the response does state that h' is continuous, an explanation is not required.

0	1 ✓

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The student response accurately includes the criteria below.

☐ Answer

Solution:

$h' = g \Rightarrow h'$ is continuous. $\Rightarrow \lim\limits_{x\to-2} h'(x) = h'(-2)$

$$\lim_{x\to-2} \frac{h'(x)}{4x} = \frac{h'(-2)}{4\cdot(-2)} = \frac{g(-2)}{-8} = \frac{-1}{-8} = \frac{1}{8}$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = 1$ is substituted for x, then $x = 1$ must appear unsimplified to earn the first point. Responses that earn the point include:

· $h'(-2x) \cdot (-2)$

· $g(-2x) \cdot (-2)$

· $h'(-2 \cdot 1) \cdot (-2)$

· $g(-2 \cdot 1) \cdot (-2)$

Responses that do not earn the point include:

· $h'(x) \cdot (-2)$

· $g(x) \cdot (-2)$

· $h'(-2) \cdot (-2)$

· $g(-2) \cdot (-2)$

· $h(-2x) \cdot (-2)$

The second point is earned for evidence of attempting to apply the Fundamental Theorem of Calculus as evidenced by linking h' and g either directly or indirectly in the presented derivative of $f(x)$ or in $f'(1)$. For example, the second point is earned by $g(-2x)$ or by $g(-2 \cdot 1)$.

The second point may be earned without earning the first point. For example, $h'(-2x) = g(-2x)$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because h' and g are linked.

The second point may be earned by going directly from $h'(-2)$ to -1. The function g does not have to be stated explicitly. For example, $h'(-2 \cdot 1) \cdot (-2) = -1 \cdot (-2)$ earns the first two points. Recall, if $x = 1$ is substituted for x, then $x = 1$ must appear unsimplified to earn the first point.

The first two points may be earned by a single expression. For example, both $g(-2x) \cdot (-2)$ and $g(-2 \cdot 1) \cdot (-2)$ earn the first two points.

The third point is earned for the correct answer of 2, simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

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0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

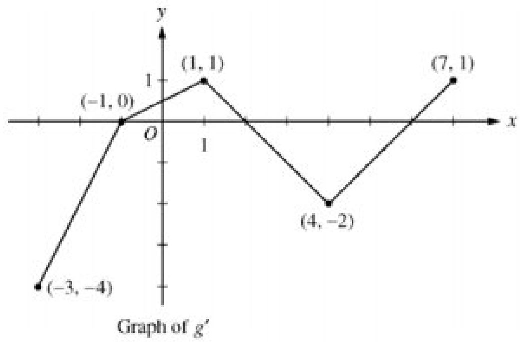
$f'(x) = h'(-2x) \cdot (-2)$

$= g(-2x) \cdot (-2)$

$f'(1) = g(-2 \cdot 1) \cdot (-2) = g(-2) \cdot (-2) = (-1) \cdot (-2) = 2$

5-1-1

67.



Let g be a continuous function with $g(2) = 5$. The graph of the piecewise-linear function g', the derivative of g, is shown above for $-3 \le x \le 7$.

(a) Find the x-coordinate of all points of inflection of the graph of $y = g(x)$ for $-3 < x < 7$. Justify your answer.

(b) Find the absolute maximum value of g on the interval $-3 \le x \le 7$. Justify your answer.

(c) Find the average rate of change of $g(x)$ on the interval $-3 \le x \le 7$.

(d) Find the average rate of change of $g'(x)$ on the interval $-3 \le x \le 7$. Does the Mean Value Theorem applied on the interval $-3 \le x \le 7$ guarantee a value of c, for $-3 < c < 7$, such that $g''(c)$ is equal to this average rate of change? Why or why not?

Part A

1 point is earned for x-values

1 point is earned for the justification

g' changes from increasing to decreasing at $x = 1$;

g' changes from decreasing to increasing at $x = 4$.

Points of inflection for the graph of $y = g(x)$ occur at $x = 1$ and $x = 4$.

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for x-values

1 point is earned for the justification

g' changes from increasing to decreasing at $x = 1$;

g' changes from decreasing to increasing at $x = 4$.

Points of inflection for the graph of $y = g(x)$ occur at $x = 1$ and $x = 4$.

Part B

1 point is earned if identifies $x = 2$ as a candidate

1 point is earned if considers endpoints

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1 point is earned for maximum value and justification

The only sign change of g' from positive to negative in the interval is at $x = 2$.

$$g(-3) = 5 + \int_{2}^{-3} g'(x)dx = 5 + \left(-\frac{3}{2}\right) + 4 = \frac{15}{2}$$
$$g(2) = 5$$
$$g(7) = 5 + \int_{2}^{7} g'(x)dx = 5 + (-4) + \frac{1}{2} = \frac{3}{2}$$

The maximum value of g for $-3 \leq x \leq 7$ is $\frac{15}{2}$.

0	1	2	3

The student response earns all of the following points:

1 point is earned if identifies $x = 2$ as a candidate

1 point is earned if considers endpoints

1 point is earned for maximum value and justification

The only sign change of g' from positive to negative in the interval is at $x = 2$.

$$g(-3) = 5 + \int_{2}^{-3} g'(x)dx = 5 + \left(-\frac{3}{2}\right) + 4 = \frac{15}{2}$$
$$g(2) = 5$$
$$g(7) = 5 + \int_{2}^{7} g'(x)dx = 5 + (-4) + \frac{1}{2} = \frac{3}{2}$$

The maximum value of g for $-3 \leq x \leq 7$ is $\frac{15}{2}$.

Part C

1 point is earned for the difference quotient

1 point is earned for the answer

$$\frac{g(7) - g(-3)}{7 - (-3)} = \frac{\frac{3}{2} - \frac{15}{2}}{10} = -\frac{3}{5}$$

0	1	2

The student response earns all of the following points:

5-1-1

1 point is earned for the difference quotient

1 point is earned for the answer

$$\frac{g(7)-g(-3)}{7-(-3)}=\frac{\frac{3}{2}-\frac{15}{2}}{10}=-\frac{3}{5}$$

Part D

1 point is earned for average value of $g'(x)$

1 point is earned for answer "No" with reason

$$\frac{g'(7)-g'(-3)}{7-(-3)}=\frac{1-(-4)}{10}=\frac{1}{2}$$

No, the MVT does *not* guarantee the existence of a value c with the stated properties because g' is not differentiable for at least one point in $-3 < x < 7$.

0	1	2

The student response earns all of the following points:

1 point is earned for average value of $g'(x)$

1 point is earned for answer "No" with reason

$$\frac{g'(7)-g'(-3)}{7-(-3)}=\frac{1-(-4)}{10}=\frac{1}{2}$$

No, the MVT does *not* guarantee the existence of a value c with the stated properties because g' is not differentiable for at least one point in $-3 < x < 7$.

5-1-1

68.

Distance x (cm)	0	1	5	6	8
Temperature $T(x)$ (°C)	100	93	70	62	55

A metal wire of length 8 centimeters (cm) is heated at one end. The table above gives selected values of the temperature *T(x)*, in degrees Celsius (˚ C), of the wire x cm from the heated end. The function T is decreasing and twice differentiable.

(a) Estimate $T'(7)$. Show the work that leads to your answer. Indicate units of measure.

(b) Write an integral expression in terms of $T(x)$ for the average temperature of the wire. Estimate the average temperature of the wire using a trapezoidal sum with the four subintervals indicated by the data in the table. Indicate units of measure.

(c) Find $\int_0^8 T'(x)dx$, and indicate units of measure. Explain the meaning of $\int_0^8 T'(x)dx$ in terms of the temperature of the wire.

(d) Are the data in the table consistent with the assertion that $T''(x) > 0$ for every x in the interval $0 < x < 8$? Explain your answer.

Part A

1 point is earned for the answer

$$\frac{T(8)-T(6)}{8-6}=\frac{55-62}{2}=-\frac{7}{2}{}^{\circ}\mathrm{C/cm}$$

1 point is earned for: units in (a), (b), and (c)

Units of ˚C/cm in (a), and ˚C in (b) and (c)

0	1	2

The student response earns all of the following points:

1 point is earned for the answer

$$\frac{T(8)-T(6)}{8-6}=\frac{55-62}{2}=-\frac{7}{2}{}^{\circ}\mathrm{C/cm}$$

1 point is earned for: units in (a), (b), and (c)

Units of ˚C/cm in (a), and ˚C in (b) and (c)

Part B

1 point is earned for: $\frac{1}{8}\int_0^8 T(x)dx$

1 point is earned for the trapezoidal sum

5-1-1

1 point is earned for the answer

$\frac{1}{8}\int_0^8 T(x)dx$

Trapezoidal approximation for $\int_0^8 T(x)dx$:

$$A = \frac{100+93}{2}\cdot 1 + \frac{93+70}{2}\cdot 4 + \frac{70+62}{2}\cdot 1 + \frac{62+55}{2}\cdot 2$$

Average temperature $\approx \frac{1}{8}A = 75.6875^{\circ}\mathrm{C}$

1 point is earned for: units in (a), (b), and (c)

Units of $^{\circ}$C/cm in (a), and $^{\circ}$C in (b) and (c)

0	1	2	3	4 ✓

The student response earns all of the following points:

1 point is earned for: $\frac{1}{8}\int_0^8 T(x)dx$

1 point is earned for the trapezoidal sum

1 point is earned for the answer

$\frac{1}{8}\int_0^8 T(x)dx$

Trapezoidal approximation for $\int_0^8 T(x)dx$:

$$A = \frac{100+93}{2}\cdot 1 + \frac{93+70}{2}\cdot 4 + \frac{70+62}{2}\cdot 1 + \frac{62+55}{2}\cdot 2$$

Average temperature $\approx \frac{1}{8}A = 75.6875^{\circ}\mathrm{C}$

1 point is earned for: units in (a), (b), and (c)

Units of $^{\circ}$C/cm in (a), and $^{\circ}$C in (b) and (c)

5-1-1

Part C

1 point is earned for the value

1 point is earned for the meaning

$$\int_0^8 T'(x)dx = T(8) - T(0) = 55 - 100 = -45^\circ\text{C}$$

The temperature drops 45˚C from the heated end of the wire of the other end of the wire.

1 point is earned for: units in (a), (b), and (c)

Units of ˚C/cm in (a), and ˚C in (b) and (c)

0	1	2	3

The student response earns all of the following points:

1 point is earned for the value

1 point is earned for the meaning

$$\int_0^8 T'(x)dx = T(8) - T(0) = 55 - 100 = -45^\circ\text{C}$$

The temperature drops 45˚C from the heated end of the wire of the other end of the wire.

1 point is earned for: units in (a), (b), and (c)

Units of ˚C/cm in (a), and ˚C in (b) and (c)

Part D

1 point is earned for the two slopes of secant lines

1 point is earned for the answer with explanation

Average rate of change of temperature on [1, 5] is $\dfrac{70-93}{5-1}$ = -5.75.

Average rate of change of temperature on [5, 6] is $\frac{62-70}{6-5}$ = -8.

5-1-1

No. By the MVT, $T'(c_1)$ = -5.75 for some c_1 in the interval (1, 5) and $T'(c_2)$ = -8 for some c_2 in the interval (5, 6). It follows that

T' must decrease somewhere in the interval (c_1, c_2). Therefore T'' is not positive for every x in [0, 8].

0	1	2

The student response earns all of the following points:

1 point is earned for the two slopes of secant lines

1 point is earned for the answer with explanation

Average rate of change of temperature on [1, 5] is $\frac{70-93}{5-1}$ = -5.75.

Average rate of change of temperature on [5, 6] is $\frac{62-70}{6-5}$ = -8.

No. By the MVT, $T'(c_1)$ = -5.75 for some c_1 in the interval (1, 5) and $T'(c_2)$ = -8 for some c_2 in the interval (5, 6). It follows that

T' must decrease somewhere in the interval (c_1, c_2). Therefore T'' is not positive for every x in [0, 8].

5-1-1

69. Let f be the function given by $f(x) = x^3 - 7x + 6$.

(a) Find the zeros of f.

(b) Write an equation of the line tangent to the graph of f at $x = -1$.

(c) Find the number c that satisfies the conclusion of the Mean Value Theorem for f on the closed interval $[1\ ,\ 3]$.

Part A

1 point is earned for first correct zero

2 points are earned for zeros of f

1 point is **deducted** for incorrect zero

$$\begin{aligned} f(x) &= x^3 - 7x + 6 \\ &= (x-1)(x-2)(x+3) \end{aligned}$$
$$\boxed{x = 1,\ x = 2,\ x = -3}$$

0	1	2	3 ✓

The student response earns all of the following points:

1 point is earned for first correct zero

2 points are earned for zeros of f

1 point is **deducted** for incorrect zero

$$\begin{aligned} f(x) &= x^3 - 7x + 6 \\ &= (x-1)(x-2)(x+3) \end{aligned}$$
$$\boxed{x = 1,\ x = 2,\ x = -3}$$

Part B

1 point is earned for $f'(x)$

1 point is earned for evaluating $f'(-1)$ and $f(-1)$

1 point is earned for equation of line

5-1-1

$f'(x) = 3x^2 - 7$

$f'(-1) = -4, f(-1) = 12$

$$\boxed{\begin{array}{l} y - 12 = -4(x+1) \\ \quad\text{or} \\ 4x + y = 8 \\ \quad\text{or} \\ y = -4x + 8 \end{array}}$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for $f'(x)$

1 point is earned for evaluating $f'(-1)$ and $f(-1)$

1 point is earned for equation of line

$f'(x) = 3x^2 - 7$

$f'(-1) = -4, f(-1) = 12$

$$\boxed{\begin{array}{l} y - 12 = -4(x+1) \\ \quad\text{or} \\ 4x + y = 8 \\ \quad\text{or} \\ y = -4x + 8 \end{array}}$$

Part C

1 point is earned for the answer

1 point is earned for solving $f'(c)$

1 point is earned for solving c

$$\frac{f(3)-f(1)}{3-1} = \frac{12-0}{2} = 6$$

$$3c^2 - 7 = f'(c) = 6$$

$$c = \sqrt{\frac{13}{3}}$$

0	1	2	3

The student response earns all of the following points:

5-1-1

1 point is earned for the answer

1 point is earned for solving $f'(c)$

1 point is earned for solving c

$\frac{f(3)-f(1)}{3-1} = \frac{12-0}{2} = 6$

$3c^2 - 7 = f'(c) = 6$

$c = \sqrt{\frac{13}{3}}$

70. $f(x) = \begin{cases} -x^2+3 & \text{if } x \le 5 \\ -10x+28 & \text{if } x > 5 \end{cases}$

Let f be the function defined above.

Which of the following statements about f is true?

(A) f is continuous and differentiable at $x = 5$.

(B) f is continuous but not differentiable at $x = 5$.

(C) f is differentiable but not continuous at $x = 5$.

(D) f is defined but neither continuous nor differentiable at $x = 5$.

Answer A

Correct. This statement is true.

$\lim_{x\to5^-} f(x) = \lim_{x\to5^-} \left(-x^2+3\right) = -25+3 = -22$

$\lim_{x\to5^+} f(x) = \lim_{x\to5^+} (-10x+28) = -50+28 = -22$

Therefore, $\lim_{x\to5} f(x)$ exists and $\lim_{x\to5} f(x) = -22 = f(5)$, so f is continuous at $x = 5$.

$\lim_{h\to0^-} \frac{f(5+h)-f(5)}{h} = \lim_{h\to0^-} \frac{-(5+h)^2+3-(-22)}{h} = \lim_{h\to0^-} \frac{-10h-h^2}{h} = -10$

$\lim_{h\to0^+} \frac{f(5+h)-f(5)}{h} = \lim_{h\to0^+} \frac{-10(5+h)+28-(-22)}{h} = \lim_{h\to0^+} \frac{-10h}{h} = -10$

Therefore, f is also differentiable at $x = 5$ and $f'(5) = -10$. An alternative way to see that the piecewise-defined function f is differentiable at $x = 5$ is to observe that $f'(x) = \begin{cases} -2x & \text{for } x < 5 \\ -10 & \text{for } x > 5 \end{cases}$. Since f is continuous at $x = 5$ and the derivatives $-2x$ and -10 are equal at $x = 5$, f is differentiable at $x = 5$.

5-1-1

71. Let f be the function given by $f(x) = \frac{x^2-9}{\sin x}$ on the closed interval $[0, 5]$. Of the following intervals, on which can the Mean Value Theorem be applied to f ?

I. $[1, 3]$, because f is continuous on $[1, 3]$ and differentiable on $(1, 3)$.
II. $[4, 5]$, because f is continuous on $[4, 5]$ and differentiable on $(4, 5)$.
III. $[1, 4]$, because f is continuous on $[1, 4]$ and differentiable on $(1, 4)$.

(A) None

(B) I only

(C) I and II only

(D) I, II, and III

Answer C

Correct. The Mean Value Theorem can be applied on the closed interval $[1, 3]$, since the function f is continuous on the closed interval $[1, 3]$ and differentiable on the open interval $(1, 3)$. The Mean Value Theorem can also be applied on the closed interval $[4, 5]$, since the function f is continuous on the closed interval $[4, 5]$ and differentiable on the open interval $(4, 5)$.

72. Let f be a differentiable function with $f(-3) = 7$ and $f(3) = 8$. Which of the following must be true for some c in the interval $(-3, 3)$?

(A) $f'(c) = 0$, since the Extreme Value Theorem applies.

(B) $f'(c) = \frac{8+7}{3-(-3)}$, since the Mean Value Theorem applies.

(C) $f'(c) = \frac{8-7}{3-(-3)}$, since the Mean Value Theorem applies.

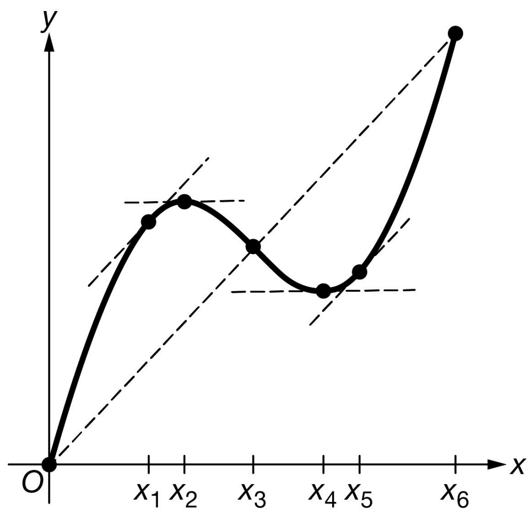
(D) $f'(c) = 7.5$, since the Intermediate Value Theorem applies.

Answer C

Correct. Since f is differentiable, it is continuous on the closed interval $[-3, 3]$ and differentiable on the open interval $(-3, 3)$. Therefore, the Mean Value Theorem guarantees a point in the open interval $(-3, 3)$ where the instantaneous rate of change equals the average rate of change over the interval. The instantaneous rate of change is given by the derivative $f'(x)$, and the average rate of change over the interval $[-3, 3]$ is $\frac{f(3)-f(-3)}{3-(-3)} = \frac{8-7}{3-(-3)}$. Therefore, the equation $f'(c) = \frac{8-7}{3-(-3)}$ is guaranteed to have at least one solution in the interval $(-3, 3)$.

5-1-1

73.



Graph of g

The function g shown in the figure above is continuous on the closed interval $[0, x_6]$ and differentiable on the open interval $(0, x_6)$, where x_1, x_2, x_3, x_4, x_5, and x_6 are points on the x-axis. Based on the graph, what are all values of x that satisfy the conclusion of the Mean Value Theorem applied to g on the closed interval $[0, x_6]$?

(A) x_3 only, because this is the value where $g(x)$ equals the average rate of change of g on $[0, x_6]$.

(B) x_2 and x_4 only, because these are the values where $g'(x) = 0$ on $[0, x_6]$.

(C) x_1 and x_5 only, because these are the values where the instantaneous rate of change of g at those values is equal to the average rate of change of g on $[0, x_6]$.

(D) x_1, x_3, and x_5 only, because these are the values where either the instantaneous rate of change of g at the value is equal to the average rate of change of g on $[0, x_6]$ or the value of $g(x)$ is equal to the average rate of change of g on $[0, x_6]$.

Answer C

Correct. The conclusion of the Mean Value Theorem is that there is at least one point x_k in the interval $(0, x_6)$ where $g'(x_k) = \frac{g(x_6)-g(0)}{x_6-0}$; that is, where the instantaneous rate of change is equal to the average rate of change over the interval $[0, x_6]$. The slopes of the tangent lines shown in the figure at $x = x_1$ and $x = x_5$ represent the instantaneous rate of change at those two points, and the slope of the dotted secant line represents the average rate of change over the interval $[0, x_6]$. Since each tangent line is parallel to the secant line, the slopes are the same. Therefore, $x = x_1$ and $x = x_5$ both satisfy the conclusion of the Mean Value Theorem.

5-1-1

74. The Mean Value Theorem can be applied to which of the following functions on the closed interval $[-5, 5]$?

(A) $f(x) = \frac{1}{\sin x}$

(B) $f(x) = \frac{x-1}{|x-1|}$

(C) $f(x) = \frac{x^2}{x^2-36}$ ✓

(D) $f(x) = \frac{x^2}{x^2-4}$

Answer C

Correct. The Mean Value Theorem can be applied to a function on the closed interval $[-5, 5]$ if the function is continuous on the closed interval $[-5, 5]$ and differentiable on the open interval $(-5, 5)$. This rational function satisfies both of those conditions. The only points at which the function is not continuous and differentiable, $x = -6$ and $x = 6$, are outside of the interval $[-5, 5]$.

75.

x	1	3	5	7	9
$f(x)$	0	6	18	29	42

Selected values of a differentiable function f are given in the table above. What is the fewest possible number of values of c in the interval $[1, 9]$ for which the Mean Value Theorem guarantees that $f'(c) = 6$?

(A) Zero

(B) One

(C) Two ✓

(D) Three

Answer C

Correct. Since f is differentiable, it is continuous on the closed interval $[3, 5]$ and differentiable on the open interval $(3, 5)$. By the Mean Value Theorem, there is point c_1 in the interval $(3, 5)$ where $f'(c_1) = \frac{f(5)-f(3)}{5-3} = \frac{18-6}{2} = 6$. The function f is also continuous on the closed interval $[5, 9]$ and differentiable on the open interval $(5, 9)$. By the Mean Value Theorem, there is a point c_2 in the interval $(5, 9)$ where $f'(c_2) = \frac{f(9)-f(5)}{9-5} = \frac{42-18}{4} = 6$. The Mean Value Theorem also guarantees a point c_3 in the open interval $(3, 9)$ where $f'(c_3) = \frac{f(9)-f(3)}{9-3} = \frac{42-6}{6} = 6$. However, it is possible that $c_3 = c_1$ or $c_3 = c_2$. Therefore, the fewest possible number of values of c for which $f'(c) = 6$ is two.

5-1-1

76.

x	0	2	3	5
$f(x)$	2	3	6	12

The function f is continuous on the closed interval $[0, 5]$ and differentiable on the open interval $(0, 5)$. Selected values of f are given in the table shown. The value $c = 4$ satisfies the conclusion of the Mean Value Theorem for f on the closed interval $[0, 5]$. What is the slope of the line tangent to the graph of f at $x = 4$?

(A) 2 ✓

(B) 2.8

(C) 4.5

(D) 10

Answer A

Correct. Since value $c = 4$ satisfies the conclusion of the Mean Value Theorem for f on the closed interval $[0, 5]$, $f'(4) = f'(c) = \frac{f(5)-f(0)}{5-0} = \frac{12-2}{5} = \frac{10}{5} = 2$.

77. The function f is given by $f(x) = x^3$. The application of the Mean Value Theorem to f on the interval $[1, 3]$ guarantees a point in the interval $(1, 3)$ at which the slope of the line tangent to the graph of f is equal to which of the following?

(A) 0

(B) 12

(C) 13 ✓

(D) 24

Answer C

Correct. Since f is continuous on the closed interval $[1, 3]$ and differentiable on the open interval $(1, 3)$, the Mean Value Theorem guarantees that there is a point c in the interval $(1, 3)$ satisfying $f'(c) = \frac{f(3)-f(1)}{3-1} = \frac{27-1}{2} = 13$.
The line tangent to the graph of f at $x = c$ would therefore have slope 13.

5-1-1

78.

x	0	4	8	12	16
$f(x)$	8	0	2	10	1

The table above gives selected values for the differentiable function f.

In which of the following intervals must there be a number c such that $f'(c) = 2$?

(A) $(0, 4)$

(B) $(4, 8)$

(C) $(8, 12)$

(D) $(12, 16)$

Answer C

Correct. The function f is continuous on the closed interval $[8, 12]$ and differentiable on the open interval $(8, 12)$. By the Mean Value Theorem, there is a number c in the interval $(8, 12)$ such that $f'(c) = \frac{f(12)-f(8)}{12-8} = \frac{10-2}{4} = 2$.

79. Let f be the function given by $f(x) = 5\cos^2\left(\frac{x}{2}\right) + \ln(x+1) - 3$. The derivative of f is given by $f'(x) = -5\cos\left(\frac{x}{2}\right)\sin\left(\frac{x}{2}\right) + \frac{1}{x+1}$. What value of c satisfies the conclusion of the Mean Value Theorem applied to f on the interval $[1, 4]$?

(A) 2.132 because $f(2.132) = \frac{f(4)-f(1)}{3}$

(B) 2.749 because $f'(2.749) = \frac{f(4)-f(1)}{3}$

(C) 3.042 because $f'(3.042) = 0$

(D) 3.252 because $f'(3.252) = \frac{f(1)+f(4)}{2}$

Answer B

Correct. The Mean Value Theorem guarantees that there is a value of c in the interval $[1, 4]$ where $f'(c) = \frac{f(4)-f(1)}{3} = -0.689525$. The calculator is used to find that $c = 2.749$.

5-1-1

80. Let f be the function given by $f(x) = \frac{x}{(x-4)(x+2)}$ on the closed interval $[-7, 7]$. Of the following intervals, on which can the Mean Value Theorem be applied to f ?

I. $[-1, 3]$ because f is continuous on $[-1, 3]$ and differentiable on $(-1, 3)$.
II. $[5, 7]$ because f is continuous on $[5, 7]$ and differentiable on $(5, 7)$.
III. $[1, 5]$ because f is continuous on $[1, 5]$ and differentiable on $(1, 5)$.

(A) None

(B) I only

(C) I and II only

(D) I, II, and III

Answer C

Correct. The Mean Value Theorem can be applied on the closed interval $[-1, 3]$, since the function f is continuous on the closed interval $[-1, 3]$ and differentiable on the open interval $(-1, 3)$. The Mean Value Theorem can also be applied on the closed interval $[5, 7]$, since the function f is continuous on the closed interval $[5, 7]$ and differentiable on the open interval $(5, 7)$.

81. Let f be a differentiable function with $f(0) = -4$ and $f(10) = 11$. Which of the following must be true for some c in the interval $(0, 10)$?

(A) $f'(c) = 0$, since the Extreme Value Theorem applies.

(B) $f'(c) = \frac{11+(-4)}{10-0}$, since the Mean Value Theorem applies.

(C) $f'(c) = \frac{11-(-4)}{10-0}$, since the Mean Value Theorem applies.

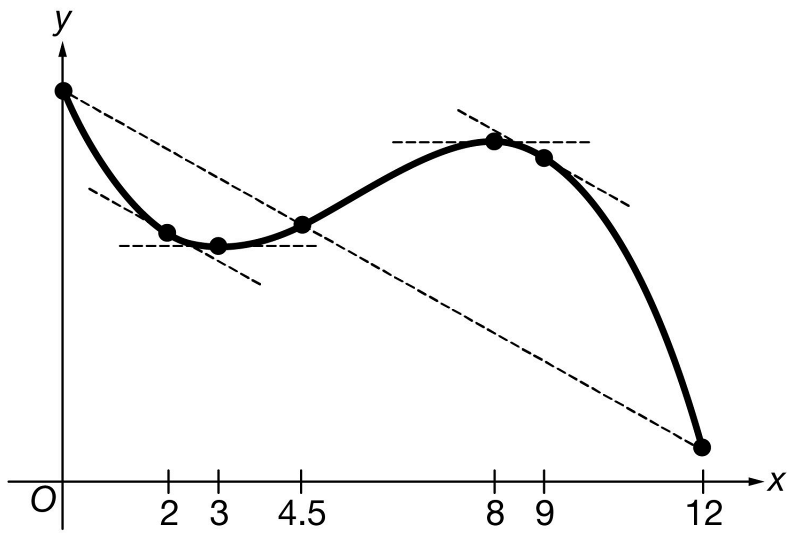
(D) $f'(c) = 1.5$, since the Intermediate Value Theorem applies.

Answer C

Correct. Since f is differentiable, it is continuous on the closed interval $[0, 10]$ and differentiable on the open interval $(0, 10)$. Therefore, the Mean Value Theorem guarantees a point in the open interval $(0, 10)$ where the instantaneous rate of change equals the average rate of change over the interval. The instantaneous rate of change is given by the derivative $f'(x)$, and the average rate of change over the interval $[0, 10]$ is $\frac{f(10)-f(0)}{10-0} = \frac{11-(-4)}{10-0}$. Therefore, the equation $f'(c) = \frac{11-(-4)}{10-0}$ is guaranteed to have at least one solution in the interval $(0, 10)$.

5-1-1

82.



The function f shown in the figure above is continuous on the closed interval $[0, 12]$ and differentiable on the open interval $(0, 12)$. Based on the graph, what are all values of x that satisfy the conclusion of the Mean Value Theorem applied to f on the closed interval $[0, 12]$?

(A) 4.5 only because this is the value where $f(x)$ equals the average rate of change of f on $[0, 12]$.

(B) 3 and 8 because these are the values where $f'(x) = 0$ on $[0, 12]$.

(C) 2 and 9 only because these are the values where the instantaneous rate of change of $f$at those values is equal to the average rate of change of f on $[0, 12]$.

(D) 2, 4.5, and 9 because these are the values where either the instantaneous rate of change of $f$at the value is equal to the average rate of change of f on $[0, 12]$ or the value of $f(x)$ is equal to the average rate of change of f on $[0, 12]$.

Answer C

Correct. The conclusion of the Mean Value Theorem is that there is a point c in the interval $(0, 12)$ where $f'\left(c\right) = \frac{f(12)-f(0)}{12-0}$; that is, where the instantaneous rate of change is equal to the average rate of change over the interval $[0, 12]$. The slopes of the tangent lines shown in the figure at $x = 2$ and $x = 9$ represent the instantaneous rate of change at those two points, and the slope of the dotted secant line represents the average rate of change over the interval $[0, 12]$. Since each tangent line is parallel to the secant line, the slopes are the same. Therefore, $x = 2$ and $x = 9$ both satisfy the conclusion of the Mean Value Theorem.

5-1-1

83. The Mean Value Theorem can be applied to which of the following functions on the closed interval $[-3, 3]$?

(A) $f(x) = x^{\frac{2}{3}}$

(B) $f(x) = |x - 1|$

(C) $f(x) = \frac{x-2}{x-5}$ ✓

(D) $f(x) = \frac{x-5}{x-2}$

Answer C

Correct. The Mean Value Theorem can be applied to a function on the closed interval $[-3, 3]$ if the function is continuous on the closed interval $[-3, 3]$ and differentiable on the open interval $(-3, 3)$. This rational function satisfies both of those conditions, since the only point not in its domain is at $x = 5$.

84.

x	0	2	5	7	11
$f(x)$	13	5	17	28	41

Selected values of a differentiable function f are given in the table above. What is the fewest possible number of values of c in the interval $[0, 11]$ for which the Mean Value Theorem guarantees that $f'(c) = 4$?

(A) Zero

(B) One

(C) Two ✓

(D) Three

Answer C

Correct. Since f is differentiable, it is continuous on the closed interval $[2, 5]$ and differentiable on the open interval $(2, 5)$. By the Mean Value Theorem, there is point c_1 in the interval $(2, 5)$ where $f'(c_1) = \frac{f(5)-f(2)}{5-2} = \frac{17-5}{3} = \frac{12}{3} = 4$. The function f is also continuous on the closed interval $[5, 11]$ and differentiable on the open interval $(5, 11)$. By the Mean Value Theorem, there is a point c_2 in the interval $(5, 11)$ where $f'(c_2) = \frac{f(11)-f(5)}{11-5} = \frac{41-17}{6} = \frac{24}{6} = 4$. The Mean Value Theorem also guarantees a point c_3 in the open interval $(2, 11)$ where $f'(c_3) = \frac{f(11)-f(2)}{11-2} = \frac{41-5}{9} = \frac{36}{9} = 4$. However, it is possible that $c_3 = c_1$ or $c_3 = c_2$. Therefore, the fewest possible number of values of c for which $f'(c) = 4$ is two.

5-1-1

85. Let f be the function given by $f(x) = \cos(x^2 + x) + 2$. The derivative of f is given by $f'(x) = -(2x+1)\sin(x^2 + x)$. What value of c satisfies the conclusion of the Mean Value Theorem applied to f on the interval $[1, 2]$?

(A) 1.079 because $f(1.079) = \frac{f(2)-f(1)}{2-1}$

(B) 1.438 because $f'(1.438) = \frac{f(2)-f(1)}{2-1}$

(C) 1.750 because $f''(1.750) = 0$

(D) 1.932 because $f'(1.932) = f(1.932)$

Answer B

Correct. The Mean Value Theorem guarantees that there is a value of c in the interval $[1, 2]$ where $f'(c) = \frac{f(2)-f(1)}{2-1} = 1.376$. The calculator is used to find that $c = 1.438$.

5-1-1

86. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	5	15	30	35
$A(t)$ (gallons)	10	18	25	16	8

The number of gallons of olive oil in a tank at time t is given by the twice-differentiable function A, where t is measured in hours and $0 \le t \le 35$. Values of $A(t)$ at selected times t are given in the table above.

(a) Use the data in the table to estimate the rate at which the number of gallons of olive oil in the tank is changing at time $t = 10$ hours. Show the computations that lead to your answer. Indicate units of measure.

(b) For $0 \le t \le 30$, is there a time t at which $A'(t) = \frac{1}{5}$? Justify your answer.

(c) The number of gallons of olive oil in the tank at time t is also modeled by the function G defined by $G(t) = 5t - \frac{2}{3}(t+9)^{\frac{3}{2}} + 28$, where t is measured in hours and $0 \le t \le 35$. Based on the model, at what time t, for $0 \le t \le 35$, is the number of gallons of olive oil in the tank an absolute maximum? Justify your answer.

(d) For the function G defined in part (c), the locally linear approximation near $t = 7$ is used to approximate $G(8)$. Is this approximation an overestimate or an underestimate for the value of $G(8)$? Give a reason for your answer.

Part A

Substitution of function values is required. The answer does not need to be simplified.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The student response accurately includes the criteria below.

☐ approximation with units

Solution:

$$A'(10) \approx \frac{A(15)-A(5)}{15-5} = \frac{25-18}{15-5} = \frac{7}{5} \text{ gallons per hour}$$

Part B

The first point requires identification of the difference quotient. The substitution of function values and conclusion impacts the second point.

5-1-1

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] $\frac{A(30)-A(0)}{30-0}$
- [ ] justification using Mean Value Theorem

Solution:

A is twice differentiable. $\Rightarrow A$ is differentiable. $\Rightarrow A$ is continuous.

$$\frac{A(30)-A(0)}{30-0} = \frac{16-10}{30-0} = \frac{6}{30} = \frac{1}{5}$$

Therefore, by the Mean Value Theorem, there is a time t, for $0 \le t \le 30$, at which $A'(t) = \frac{1}{5}$.

Part C

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

The response is eligible for additional points based on consistent answers using an incorrect $G'(t)$ with a maximum of one computational error.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3	4

The student response accurately includes all four of the criteria below.

- [ ] $G'(t)$
- [ ] sets $G'(t) = 0$
- [ ] identifies $t = 16$ as a candidate
- [ ] answer with justification

Solution:

$$G'(t) = 5 - (t+9)^{\frac{1}{2}} = 0 \;\Rightarrow\; (t+9)^{\frac{1}{2}} = 5$$
$$\Rightarrow\; t+9 = 25 \;\Rightarrow t = 16$$

Because $G'(t)$ changes from positive to negative at $t = 16$, G has a relative maximum at $t = 16$.

Since $t = 16$ is the only critical point of G on the interval $0 \le t \le 35$ that is the location of a relative maximum, it is also the location of the absolute maximum of G on the interval $0 \le t \le 35$.

5-1-1

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

- [ ] $G''(t)$
- [ ] overestimate with reason

Solution:

$G''(t) = -\frac{1}{2}(t+9)^{-\frac{1}{2}}$

$G''(t) < 0$ for $7 \le t \le 8$, and so the graph of G is concave down for $7 \le t \le 8$. Therefore, the approximation of $G(8)$ using the locally linear approximation of G near $t = 7$ is an overestimate.

5-1-1

87. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	1	5	8	10
$G(t)$ (thousands of gallons)	22	18.5	19.5	22	24

A gasoline storage tank is filled by a pipeline from a refinery. At the same time, gasoline flows from the tank into trucks that will make deliveries to gasoline stations. The amount of gasoline in the storage tank at time t is given by the twice-differentiable function G, where t is measured in hours and $0 \le t \le 10$. Values of $G(t)$, in thousands of gallons, at selected times t are given in the table above. It is known that $G''(t) > 0$ for $5 \le t < 10$.

(a) Use the data in the table to estimate the rate of change of the amount of gasoline in the storage tank at time $t = 3$ hours. Show the computations that lead to your answer. Indicate units of measure.

(b) For $0 < t < 10$, is there a time t at which $G(t)$ is increasing at a rate of 0.2 thousand gallons per hour? Justify your answer.

(c) It is known that $G'(5) = 0.5$. Use the locally linear approximation for G at time $t = 5$ to approximate the amount of gasoline in the storage tank at time $t = 7$. Is this approximation an overestimate or an underestimate for the actual amount of gasoline in the storage tank at time $t = 7$? Give a reason for your answer.

(d) The rate at which gasoline flows out of the storage tank into trucks at time t can be modeled by the function R defined by $R(t) = \frac{100t}{t^2+4}$, where t is measured in hours, and $R(t)$ is measured in thousands of gallons. Based on the model, at what time t, for $0 \le t \le 10$, is the rate at which gasoline flows out of the storage tank an absolute maximum? Justify your answer.

Part A

Substitution of function values is required. The answer does not need to be simplified.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1

The student response accurately includes a correct approximation with units.

Solution:

$$G'(3) \approx \frac{G(5)-G(1)}{5-1} = \frac{19.5-18.5}{5-1} = \frac{1}{4} = 0.25 \text{ thousand gallons per hour (which is equivalent to 250 gallons per hour)}$$

Part B

The first point requires identification of the difference quotient. The substitution of function values and conclusion impacts the second point.

5-1-1

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] $\frac{G(10)-G(0)}{10-0}$
- [ ] justification using Mean Value Theorem

Solution:

G is twice differentiable. $\Rightarrow$ G is differentiable. $\Rightarrow$ G is continuous.

$\frac{G(10)-G(0)}{10-0} = \frac{24-22}{10-0} = \frac{2}{10} = 0.2$ thousand gallons per hour (which is equivalent to 200 gallons per hour)

Therefore, by the Mean Value Theorem, there is a time t, for $0 < t < 10$, at which $G'(t) = 0.2$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] approximation of $G(7)$
- [ ] underestimate with reason

Solution:

$G(7) \approx G(5) + G'(5)(7-5) = 19.5 + (0.5)(2) = 20.5$ thousand gallons

$G''(t) > 0$ for $5 \le t \le 7$, and so the graph of G is concave up for $5 \le t \le 7$. Therefore, the approximation for $G(7)$ using the locally linear approximation for G at time $t = 5$ is an underestimate for the actual amount.

Part D

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

The response is eligible for additional points based on consistent answers using an incorrect $R'(t)$ with a maximum of one computational error.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

5-1-1

0	1	2	3	4

The student response accurately includes all four of the criteria below.

- [ ] $R'(t)$
- [ ] sets $R'(t) = 0$
- [ ] identifies $t = 2$ as a candidate
- [ ] answer with justification

Solution:

$$R'(t) = \frac{100(t^2+4) - 100t(2t)}{(t^2+4)^2} = \frac{400 - 100t^2}{(t^2+4)^2} = \frac{100(4-t^2)}{(t^2+4)^2} = 0$$

$$\Rightarrow 4 - t^2 = 0 \Rightarrow t = 2$$

Because $R'(t)$ changes from positive to negative at $t = 2$, R has a relative maximum at $t = 2$. R is increasing on $0 \le t \le 2$ and R is decreasing on $2 \le t \le 10$. Since $t = 2$ is the only critical point of R on the interval $0 \le t \le 10$ that is the location of a relative maximum, it is also the location of the absolute maximum of R on the interval $0 \le t \le 10$.

5-1-1

88. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	60	90	120	135	150
$f(t)$ (gallons per second)	0	0.1	0.15	0.1	0.05	0

A customer at a gas station is pumping gasoline into a gas tank. The rate of flow of gasoline is modeled by a differentiable function f, where $f(t)$ is measured in gallons per second and t is measured in seconds since pumping began. Selected values of $f(t)$ are given in the table.

(a) Using correct units, interpret the meaning of $\int_{60}^{135} f(t)\,dt$ in the context of the problem. Use a right Riemann sum with the three subintervals $[60, 90]$, $[90, 120]$, and $[120, 135]$ to approximate the value of $\int_{60}^{135} f(t)\,dt$.

(b) Must there exist a value of c, for $60 < c < 120$, such that $f'(c) = 0$? Justify your answer.

(c) The rate of flow of gasoline, in gallons per second, can also be modeled by $g(t) = \left(\frac{t}{500}\right)\cos\left(\left(\frac{t}{120}\right)^2\right)$ for $0 \le t \le 150$. Using this model, find the average rate of flow of gasoline over the time interval $0 \le t \le 150$. Show the setup for your calculations.

(d) Using the model g defined in part (c), find the value of $g'(140)$. Interpret the meaning of your answer in the context of the problem.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point the response must reference gallons of gasoline added/pumped and the time interval $t = 60$ to $t = 135$.

To earn the second point at least five of the six factors in the Riemann sum must be correct.

If there is any error in the Riemann sum, the response does not earn the third point.

A response of $(0.15)(30) + (0.1)(30) + (0.05)(15)$ earns both the second and third points, unless there is a subsequent error in simplification, in which case the response would earn only the second point.

5-1-1

A response that presents a correct value with accompanying work that shows the three products in the Riemann sum but does not show all six of the factors and/or the sum process, does not earn the second point but does earn the third point. For example, responses of either $4.5 + 3.0 + 0.75$ or $(0.15)(30), 0.1(30), 0.05(15) \rightarrow 8.25$ earn the third point but not the second.

A response of $f(90)(90-60) + f(120)(120-90) + f(135)(135-120) = 8.25$ earns both the second and the third points.

A response that presents an answer of only 8.25 does not earn either the second or third point.

A response that provides a completely correct left Riemann sum with accompanying work, $f(60)(30) + f(90)(30) + f(120)(15) = 9$, or $(0.1)(30) + (0.15)(30) + (0.1)(15)$ earns 1 of the last 2 points. A response with any errors or missing factors in a left Riemann sum earns neither of the last 2 points.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Interpretation with units
- ☐ Form of Riemann sum
- ☐ Answer

Solution:

$\int_{60}^{135} f(t)dt$ represents the total number of gallons of gasoline pumped into the gas tank from time $t = 60$ seconds to time $t = 135$ seconds.

$$\int_{60}^{135} f(t)dt$$
$$\approx f(90)(90-60) + f(120)(120-90) + f(135)(135-120)$$
$$= (0.15)(30) + (0.1)(30) + (0.05)(15) = 8.25$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point a response must present either $f(120) - f(60) = 0$, $0.1 - 0.1 = 0$ (perhaps as the numerator of a quotient), or $f(60) = f(120)$.

To earn the second point a response must:

- have earned the first point,
- state that f is continuous because f is differentiable (or equivalent), and
- answer “yes” in some way.

A response may reference either the Mean Value Theorem or Rolle’s Theorem.

A response that references the Intermediate Value Theorem cannot earn the second point.

5-1-1

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ $f(120) - f(60) = 0$
- ☐ Answer with justification

Solution:

f is differentiable. $\Rightarrow$ f is continuous on $[60, 120]$.

$\frac{f(120)-f(60)}{120-60} = \frac{0.1-0.1}{60} = 0$

By the Mean Value Theorem, there must exist a c, for $60 < c < 120$, such that $f'(c) = 0$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The exact value of $\frac{1}{150}\int_0^{150} g(t)dt$ is $\frac{12}{125}\sin(\frac{25}{16})$.

A response may present the average value formula in single or multiple steps. For example, the following response earns both points: $\int_0^{150} g(t)dt = 14.399504$ so the average rate is 0.0959967.

A response that presents the average value formula in multiple steps but provides incorrect or incomplete communication (e.g., $\int_0^{150} g(t)dt = \frac{14.399504}{150} = 0.0959967$) earns 1 out of 2 points.

A response of $\displaystyle\int_0^{150} g(t)dt = 0.0959967$ does not earn either point.

Degree mode: A response that presents answers obtained by using a calculator in degree mode does not earn the first point it would have otherwise earned. The response is generally eligible for all subsequent points (unless no answer is possible in degree mode or the question is made simpler by using degree mode). In degree mode, $\frac{1}{150}\int_0^{150} g(t)dt = 0.149981$ or 0.002618.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Average value formula
- ☐ Answer

Solution:

5-1-1

$$\frac{1}{150-0}\int_{0}^{150} g(t)dt = 0.0959967$$

The average rate of flow of gasoline, in gallons per second, is 0.096 (or 0.095).

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The exact value of $g'(140)$ is $\frac{1}{500}\cos(\frac{49}{36}) - \frac{49}{9000}\sin(\frac{49}{36})$.

The value of $g'(140)$ may appear only in the interpretation.

To be eligible for the second point a response must present some numerical value for $g'(140)$.

To earn the second point the interpretation must include “the rate of flow of gasoline is changing at a rate of [the declared value of $g'(140)$]” and “at $t = 140$” (or equivalent).

An interpretation of “decreasing at a rate of -0.005” or “increasing at a rate of ” does not earn the second point.

Degree mode: In degree mode, $g'(140) = 0.001997$ or 0.00187.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] $g'(140)$
- [ ] Interpretation

Solution:

$g'(140) \approx -0.004908$

$g'(140) = -0.005$ (or -0.004)

The rate at which gasoline is flowing into the tank is decreasing at a rate of 0.005 (or 0.004) gallon per second per second at time $t = 140$ seconds.

5-1-1

89. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	1	2	3	4
$B(t)$ (miles per hour)	1	8	1.5	-5	11

Brandon and Chloe ride their bikes for 4 hours along a flat, straight road. Brandon's velocity, in miles per hour, at time t hours is given by a differentiable function B for $0 \le t \le 4$. Values of $B(t)$ for selected times t are given in the table above. Chloe's velocity, in miles per hour, at time t hours is given by the piecewise function C defined by

$$C(t) = \begin{cases} te^{4-t^2} & \text{for } 0 \le t \le 2 \\ 12 - 3t - t^2 & \text{for } 2 < t \le 4. \end{cases}$$

(a) How many miles did Chloe travel from time $t = 0$ to time $t = 2$?

(b) At time $t = 3$, is Chloe's speed increasing or decreasing? Give a reason for your answer.

(c) Is there a time t, for $0 \le t \le 4$, at which Brandon's acceleration is equal to 2.5 miles per hour per hour? Justify your answer.

(d) Is there a time t, for $0 \le t \le 2$, at which Brandon's velocity is equal to Chloe's velocity? Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned with a connection to the correct part of the piecewise defined function. $\int_0^2 C(t)\, dt$ by itself is not sufficient to earn the point.

The second point is earned with $-\frac{1}{2}e^{4-t^2}$ or with $-\frac{1}{2}e^u$ for $u = 4 - t^2$. Limits of integration are part of the third point.

The third point does not require units.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ integral
- ☐ antiderivative
- ☐ answer

5-1-1

Solution:

$$\int_0^2 te^{4-t^2}\,dt = -\frac{1}{2}e^{4-t^2}\Big|_{t=0}^{t=2} = -\frac{1}{2}+\frac{1}{2}e^4$$

Chloe traveled $-\frac{1}{2}+\frac{1}{2}e^4$ miles from time $t=0$ to time $t=2$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned by showing both $C\left(3\right)$ and $C'\left(3\right)$ are less than zero or that both have the same sign.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ $C\left(3\right)<0$ and $C'\left(3\right)<0$
- ☐ answer with reason

Solution:

$C\left(3\right)=-6<0$

$C'\left(3\right)=-9<0$

Chloe's speed is increasing at time $t=3$ because her velocity and acceleration have the same sign.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The second point is earned with the justification that B is continuous and a numerical difference quotient equal to 2.5. There is no requirement for the response to indicate that B is differentiable, nor to refer to the Mean Value Theorem by name.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ $\frac{B(4)-B(0)}{4-0}$
- ☐ answer with justification

Solution:

B is differentiable $\Rightarrow B$ is continuous on $\left[0,\ 4\right]$.

$\frac{B(4)-B(0)}{4-0}=\frac{11-1}{4-0}=2.5$

5-1-1

By the Mean Value Theorem, there is a time t, for $0 < t < 4$, such that $B'(t) = 2.5$ miles per hour per hour.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by comparing B and C, either by a subtraction statement or by comparing $B(t)$ and $C(t)$ for some t in $\{0,\ 1,\ 2\}$.

The second point is earned with a justification that includes asserting the continuity of B and C or of $B - C$, and showing that $B(0) > C(0)$ and $B(2) < C(2)$. It is not required for the response to specifically refer to the Intermediate Value Theorem.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] considers $B(t) - C(t)$
- [ ] answer with justification

Solution:

B and C are continuous on $[0,\ 2]$, therefore $B - C$ is continuous on $[0,\ 2]$.

$B(0) - C(0) = 1 - 0 > 0$

$B(2) - C(2) = 1.5 - 2 < 0$

By the Intermediate Value Theorem, there is a time t, for $0 < t < 2$, such that$B(t) - C(t) = 0$, or $B(t) = C(t)$.

5-1-1

90. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	0.3	1.7	2.8	4
$v_P(t)$ (meters per hour)	0	55	-29	55	48

The velocity of a particle, P, moving along the x-axis is given by the differentiable function v_P, where $v_P(t)$ is measured in meters per hour and t is measured in hours. Selected values of $v_P(t)$ are shown in the table above. Particle P is at the origin at time $t = 0$.

a) Justify why there must be at least one time t, for $0.3 \le t \le 2.8$, at which $v_P{}'(t)$, the acceleration of particle P, equals 0 meters per hour per hour.

b) Use a trapezoidal sum with the three subintervals $[0, 0.3]$, $[0.3, 1.7]$, and $[1.7, 2.8]$ to approximate the value of $\int_0^{2.8} v_P(t)\,dt$.

c) A second particle, Q, also moves along the x-axis so that its velocity for $0 \le t \le 4$ is given by $v_Q(t) = 45\sqrt{t}\cos\left(0.063t^2\right)$ meters per hour. Find the time interval during which the velocity of particle Q is at least 60 meters per hour. Find the distance traveled by particle Q during the interval when the velocity of particle Q is at least 60 meters per hour.

d) At time $t = 0$, particle Q is at position $x = -90$. Using the result from part (b) and the function v_Q from part (c), approximate the distance between particles P and Q at time $t = 2.8$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for expressing $v_P(2.8) - v_P(0.3) = 0$. Because these values come from the table, expressing $55 - 55$ earns the first point.

The second point is earned for declaring v_P to be continuous and connecting $v_P{}'(t)$ to the work for the first point.

Another approach involves an Extreme Value Theorem argument. The first point is earned for declaring $v_P(0.3) > v_P(1.7)$ and $v_P(1.7) < v_P(2.8)$. The second point is earned for declaring v_P to be continuous, and using continuity to conclude that $v_P(t)$ must have a minimum at some point c, $0.3 < c < 2.8$, so $v_P{}'(c) = 0$.

5-1-1

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ $v_P\ (2.8) - v_P\ (0.3) = 0$
- ☐ justification, using Mean Value Theorem

v_P is differentiable $\Rightarrow v_P$ is continuous on $0.3 \leq t \leq 2.8$.

$$\frac{v_P(2.8)-v_P(0.3)}{2.8-0.3} = \frac{55-55}{2.5} = 0$$

By the Mean Value Theorem, there is a value c, $0.3 < c < 2.8$, such that $v_P{}'\ (c) = 0$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Simplification is not required to earn this point. The point requires substitution of function values from the table.

0	1

The student response accurately includes the criteria below.

- ☐ · answer, using trapezoidal sum

Solution:

$$\begin{aligned} \int_0^{2.8} v_P\ (t)\ dt &\approx 0.3\left(\frac{v_P(0)+v_P(0.3)}{2}\right) + 1.4\left(\frac{v_P(0.3)+v_P(1.7)}{2}\right) \\ &\quad + 1.1\left(\frac{v_P(1.7)+v_P(2.8)}{2}\right) \\ &= 0.3\left(\frac{0+55}{2}\right) + 1.4\left(\frac{55+(-29)}{2}\right) + 1.1\left(\frac{-29+55}{2}\right) \\ &= 40.75 \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for an open or closed interval. The interval must be stated explicitly analytically or verbally; using the endpoints of the interval as the limits of integration in a definite integral is not sufficient to earn the first point.

The second point is earned for a definite integral of the form $\int_{t_0}^{t_1} v_Q\ (t)\ dt$ or $\int_{t_0}^{t_1} |v_Q\ (t)|\ dt$.

The third point does not require units.

5-1-1

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ interval
- ☐ definite integral
- ☐ distance

Solution:

$v_Q(t) = 60 \;\Rightarrow\; t = A = 1.866181 \text{ or } t = B = 3.519174$

$v_Q(t) \geq 60$ for $A \leq t \leq B$

$$\int_A^B |v_Q(t)|\, dt = 106.108754$$

The distance traveled by particle Q during the interval $A \leq t \leq B$ is 106.109 (or 106.108) meters.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for the integral $\int_0^{2.8} v_Q(t)\, dt$ or $\int_0^{2.8} |v_Q(t)|\, dt$.

The third point is earned for the difference between $x_P(2.8)$ and $x_Q(2.8)$. Any incorrect value from Part B of this question, handled correctly, makes the response eligible for the third point. The third point does not require units.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ $\int_0^{2.8} v_Q(t)\, dt$
- ☐ position of particle Q
- ☐ answer

Solution:

From Part B, the position of particle P at time $t = 2.8$ is $x_P(2.8) = \int_0^{2.8} v_P(t)\, dt \approx 40.75$.

$$x_Q(2.8) = x_Q(0) + \int_0^{2.8} v_Q(t)\, dt = -90 + 135.937653 = 45.937653$$

Therefore at time $t = 2.8$, particles P and Q are approximately $45.937653 - 40.75 = 5.188$ (or 5.187) meters apart.

5-1-1

91. A particle moves along the x-axis so that its velocity at any time $t \geq 0$ is given by $v(t) = 3t^2 - 2t - 1$. The position $x(t)$ is 5 for $t = 2$.

(a) Write a polynomial expression for the position of the particle at any time $t \geq 0$.

(b) For what values of t, $0 \leq t \leq 3$, is the particle's instantaneous velocity the same as its average velocity on the closed interval [0,3]?

(c) Find the total distance traveled by the particle from time $t = 0$ until time $t = 3$.

Part A

2 points are earned for the correct answer $x(t) = t^3 - t^2 - t + C$

<-1> for error(s) in $t^3 - t^2 - t$

<-1> for no constant of integration

$$\begin{aligned} x(t) &= \int v(t)dt = \int (3t^2 - 2t - 1)dt \\ &= t^3 - t^2 - t + C \end{aligned}$$

1 point is earned for the correct evaluation of the constant of integration

$$\begin{aligned} &x(2) = 8 - 4 - 2 + C = 5;\ C = 3 \\ &x(t) = t^3 - t^2 - t + 3 \end{aligned}$$

0	1	2	3

The student response earns all of the following points:

2 points are earned for the correct answer $x(t) = t^3 - t^2 - t + C$

<-1> for error(s) in $t^3 - t^2 - t$

<-1> for no constant of integration

$$\begin{aligned} x(t) &= \int v(t)dt = \int (3t^2 - 2t - 1)dt \\ &= t^3 - t^2 - t + C \end{aligned}$$

1 point is earned for the correct evaluation of the constant of integration

$$\begin{aligned} &x(2) = 8 - 4 - 2 + C = 5;\ C = 3 \\ &x(t) = t^3 - t^2 - t + 3 \end{aligned}$$

5-1-1

Part B

1 point is earned for the correct average velocity = 5

$$\text{avg. vel.} = \frac{x(3)-x(0)}{3-0} = \frac{18-3}{3} = 5$$

1 point is earned for correctly setting $v(t)$ equal to student's average velocity OR from student's $x(t)$

$$3t^2 - 2t - 1 = 5$$

1 point is earned for the correct answer

0/1 if not solving $3t^2 - 2t - 1 =$ an avg. velocity in [0, 3]

$$t = \frac{1+\sqrt{19}}{3} \text{ or } 1.786$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for the correct average velocity = 5

$$\text{avg. vel.} = \frac{x(3)-x(0)}{3-0} = \frac{18-3}{3} = 5$$

1 point is earned for correctly setting $v(t)$ equal to student's average velocity OR from student's $x(t)$

$$3t^2 - 2t - 1 = 5$$

1 point is earned for the correct answer

0/1 if not solving $3t^2 - 2t - 1 =$ an avg. velocity in [0, 3]

$$t = \frac{1+\sqrt{19}}{3} \text{ or } 1.786$$

Part C

1 point is earned for the correct limits of 0 and 3 on an integral of $v(t)$ or $|v(t)|$ or substitutes $t = 0$ and $t = 3$ in $x(t)$

5-1-1

$$\text{distance} = \int_0^3 |v(t)|dt$$
$$= \int_0^3 \left|3t^2 - 2t - 1\right|dt = 17$$

or

$v(t) = 3t^2 - 2t - 1 = 0$
$t = -\frac{1}{3}, t = 1$
$x(0) = 3$
$x(1) = 1 - 1 - 1 + 3 = 2$
$x(3) = 27 - 9 - 3 + 3 = 18$

1 point is earned for the correct handling of the change in direction at student's turning point

1 point is earned for the correct answer

distance = $(3 - 2) + (18 - 2) = 17$

0	1	2	3 ✓

The student response earns all of the following points:

1 point is earned for the correct limits of 0 and 3 on an integral of $v(t)$ or $|v(t)|$ or substitutes $t = 0$ and $t = 3$ in $x(t)$

$$\text{distance} = \int_0^3 |v(t)|dt$$
$$= \int_0^3 \left|3t^2 - 2t - 1\right|dt = 17$$

or

$v(t) = 3t^2 - 2t - 1 = 0$
$t = -\frac{1}{3}, t = 1$
$x(0) = 3$
$x(1) = 1 - 1 - 1 + 3 = 2$
$x(3) = 27 - 9 - 3 + 3 = 18$

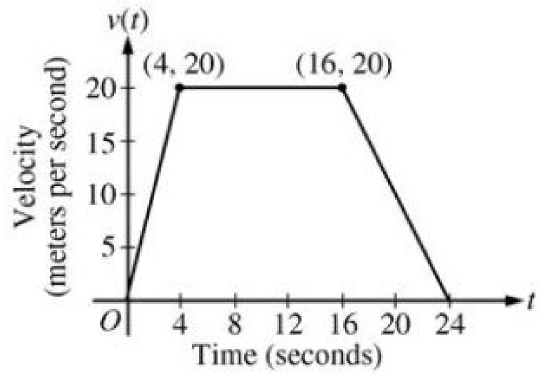
1 point is earned for the correct handling of the change in direction at student's turning point

1 point is earned for the correct answer

distance = $(3 - 2) + (18 - 2) = 17$

5-1-1

92.



A car is traveling on a straight road. For $0 \le t \le 24$ *seconds, the car's velocity* $v(t)$, in meters per second, is modeled by the piecewise-linear function defined by the graph above.

(a) Find $\int_0^{24} v\,(t)\,dt$. Using correct units, explain the meaning of $\int_0^{24} v\,(t)\,dt$.

(b) For each of $v'(4)$ and $v'(20)$, find the value or explain why it does not exist. Indicate units of measure.

(c) Let $a(t)$ be the car's acceleration at time t, in meters per second per second. For $0 < t < 24$, write a piecewise-defined function for $a(t)$.

(d) Find the average rate of change of v over the interval $8 \le t \le 20$. Does the Mean Value Theorem guarantee a value of c, for $8 < c < 20$, such that $v'(c)$ is equal to this average rate of change? Why or why not?

Part A

1 point is earned for the value

1 point is earned for the meaning with units

$$\int_0^{24} v\,(t)dt = \tfrac{1}{2}(4)\,(20) + (12)\,(20) + \tfrac{1}{2}(8)\,(20) = 360$$

The car travels 360 meters in these 24 seconds.

0	1	2

The student response earns all of the following points:

1 point is earned for the value

1 point is earned for the meaning with units

$$\int_0^{24} v\,(t)dt = \tfrac{1}{2}(4)\,(20) + (12)\,(20) + \tfrac{1}{2}(8)\,(20) = 360$$

The car travels 360 meters in these 24 seconds.

Part B

1 point is earned for: $v'(4)$ does not exist, with explanation

5-1-1

1 point is earned for: $v'(20)$

1 point is earned for the units

$v'(4)$ does not exist because

$$\lim_{t \to 4^-} \left(\frac{v(t)-v(4)}{t-4}\right) = 5 \neq 0 = \lim_{t \to 4^+} \left(\frac{v(t)-v(4)}{t-4}\right)$$
$$v'(20) = \frac{20-0}{16-24} = -\frac{5}{2}\ \mathrm{m/sec}^2$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for: $v'(4)$ does not exist, with explanation

1 point is earned for: $v'(20)$

1 point is earned for the units

$v'(4)$ does not exist because

$$\lim_{t \to 4^-} \left(\frac{v(t)-v(4)}{t-4}\right) = 5 \neq 0 = \lim_{t \to 4^+} \left(\frac{v(t)-v(4)}{t-4}\right)$$
$$v'(20) = \frac{20-0}{16-24} = -\frac{5}{2}\ \mathrm{m/sec}^2$$

Part C

1 point is earned for finds the values 5, 0, $-\frac{5}{2}$

1 point is earned for identifies constants with correct intervals

$$a(t) = \begin{cases} 5 & \text{if } 0 < t < 4 \\ 0 & \text{if } 4 < t < 16 \\ -\frac{5}{2} & \text{if } 16 < t < 24 \end{cases}$$

$a(t)$ does not exist at $t = 4$ and $t = 16$.

0	1	2

The student response earns all of the following points:

1 point is earned for finds the values 5, 0, $-\frac{5}{2}$

1 point is earned for identifies constants with correct intervals

5-1-1

$$a(t) = \begin{cases} 5 & \text{if } 0 < t < 4 \\ 0 & \text{if } 4 < t < 16 \\ -\frac{5}{2} & \text{if } 16 < t < 24 \end{cases}$$

$a(t)$ does not exist at $t = 4$ and $t = 16$.

Part D

1 point is earned for the average rate of change of v on [8, 20]

1 point is earned for the answer with explanation

The average rate of change of v on [8, 20] is

$$\frac{v(20) - v(8)}{20 - 8} = -\frac{5}{6} \text{ m/sec}^2.$$

No, the Mean Value Theorem does not apply to v on [8, 20] because v is not differentiable at

$t = 16$.

0	1	2

The student response earns all of the following points:

1 point is earned for the average rate of change of v on [8, 20]

1 point is earned for the answer with explanation

The average rate of change of v on [8, 20] is

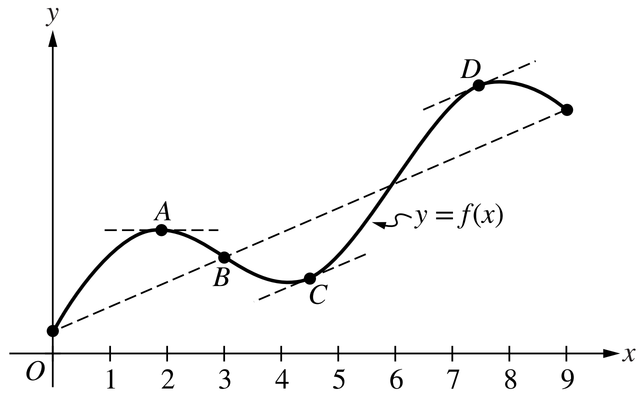
$$\frac{v(20) - v(8)}{20 - 8} = -\frac{5}{6} \text{ m/sec}^2.$$

No, the Mean Value Theorem does not apply to v on [8, 20] because v is not differentiable at

$t = 16$.

5-1-1

93.



The function f shown in the figure is continuous on the closed interval $[0, 9]$ and differentiable on the open interval $(0, 9)$. Which of the following points satisfies conclusions of both the Intermediate Value Theorem and the Mean Value Theorem for f on the closed interval $[0, 9]$?

(A) A

(B) B

(C) C

(D) D

Answer C

Correct. By the Mean Value Theorem, there is a c_1 with $0 < c_1 < 9$ such that $f'(c_1) = \frac{f(9)-f(0)}{9-0}$. This means that the line tangent to the graph of f is parallel to the secant line through the points on the graph at $x = 0$ and $x = 9$. Only the points C and D have a tangent line that is parallel to the dashed line segment connecting $(0, f(0))$ and $(9, f(9))$. By the Intermediate Value Theorem, if $f(0) \leq d \leq f(9)$, then there is a c_2 with $0 \leq c_2 \leq 9$ such that $f(c_2) = d$. Of the two points C and D, only point C has a y-coordinate that is between $f(0)$ and $f(9)$. Therefore, only point C satisfies conclusions of both the Intermediate Value Theorem and the Mean Value Theorem.

5-1-1

94. Let *f* be a continuous function defined on the interval $0 \leq x \leq 7$. The function *f* and its derivatives have the properties indicated in the table shown, where DNE indicates the *x*-values at which the derivatives of *f* do not exist.

x	$x = 0$	$0 < x < 2$	$x = 2$	$2 < x < 4$	$x = 4$	$4 < x < 7$	$x = 7$
$f(x)$	-1	Negative	-4	Negative	-2	Negative	0
$f'(x)$	DNE	Negative	DNE	Positive	0	Positive	DNE
$f''(x)$	DNE	Positive	DNE	Negative	0	Positive	DNE

Part A

On what open intervals contained in $0 < x < 7$, if any, is the graph of *f* both increasing and concave down? Give a reason for your answer.

Part B

Find the average rate of change of *f* over the interval $0 \leq x \leq 4$. Can the Mean Value Theorem be applied to *f* on the interval $0 \leq x \leq 4$ to guarantee a value of c, for $0 < c < 4$, such that $f'(c)$ is equal to this average rate of change? Give a reason for your answer.

Part C

Use a left Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\int_0^7 f(x)dx$.

Part D

Let g be the function defined by $g(x) = x \cdot f(x)$. Is g increasing or decreasing at $x = 1$? Give a reason for your answer.

Part A Point 1

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P1** is earned only by an answer of $2 < x < 4$ or this interval including one or both endpoints.
- Special case: A response of "*f* is increasing on $2 < x < 4$ and $4 < x < 7$ because $f'(x) > 0$ there" earns **P1**.
- Special case: A response of "*f* is concave down on $2 < x < 4$ because $f''(x) < 0$ there" earns **P1**.

Model Solution:

f is increasing and concave down on the interval $2 < x < 4$.

5-1-1

0	1

The response accurately includes the criteria below.

- Answer

Part A Point 2

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response must have earned **P1** to be eligible for **P2**.
- To earn **P2**, a response must correctly cite both $f'(x) > 0$ and $f''(x) < 0$.
- Special case: A response of "*f* is increasing on $2 < x < 4$ and $4 < x < 7$ because $f'(x) > 0$ there" does not earn **P2**.
- Special case: A response of "*f* is concave down on $2 < x < 4$ because $f''(x) < 0$ there" does not earn **P2**.

Model Solution:

The graph of function *f* is increasing on an interval when $f'(x) > 0$ and concave down on an interval when $f''(x) < 0$.

0	1

The response accurately includes the criteria below.

- Reason

Part B Point 3

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P3**, a response must include a difference and a quotient. Any of the following is sufficient to earn **P3**: $\frac{f(4)-f(0)}{4-0} = -\frac{1}{4}$, $\frac{-2-(-1)}{4}$, or $\frac{-1}{4-0}$.

Model Solution:

$$\frac{f(4)-f(0)}{4-0} = \frac{-2-(-1)}{4} = -\frac{1}{4}$$

0	1

The response accurately includes the criteria below.

- Average rate of change

5-1-1

Part B Point 4

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response does not need to have earned **P3** to be eligible for **P4**.
- To earn **P4**, a response must note that f is not differentiable and conclude "no."
- To earn **P4**, a response need not explicitly name the Mean Value Theorem, but if a theorem is named, it must be correct.

Model Solution:

No, the Mean Value Theorem cannot be applied to *f* on the interval $0 \leq x \leq 4$ because *f* is not differentiable at $x = 2$.

0	1

The response accurately includes the criteria below.

- Answer with reason

Part C Point 5

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- Read "=" as "≈" for **P5**.
- The form of a left Riemann sum includes three terms, each of which includes a product of two factors. To earn **P5**, at least five of the six factors must be correct.
- A response of $(f(0))(2) + (f(2))(2) + (f(4))(3)$ earns **P5**.
- A response of $(-1)(2) + (-4)(2) + (-2)(3)$ earns **P5**.
- An unsupported answer of -16 does not earn **P5**.
- A completely correct right Riemann sum with accompanying work (e.g., $(-4)(2) + (-2)(2) + (0)(3) = -12$) earns **P5**. A response with any errors or missing factors in the right Riemann sum does not earn **P5**.

Model Solution:

$$\int_0^7 f(x)dx \approx f(0)(2-0) + f(2)(4-2) + f(4)(7-4)$$

0	1

The response accurately includes the criteria below.

- Form of left Riemann sum

Part C Point 6

5-1-1

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response of $(f(0))(2)+(f(2))(2)+(f(4))(3)$ is eligible to earn **P6** with correct values for $f(0)$, $f(2)$, and $f(4)$ imported from the table. If any of the six factors is incorrect, the response does not earn **P6**.
- A response of $(-1)(2)+(-4)(2)+(-2)(3)$ earns **P6**.
- A response that includes $(-1)(2)+(-4)(2)+(-2)(3)$ or equivalent banks **P6** (i.e., subsequent errors in simplification will not be considered in scoring for **P6**).
- An unsupported answer of -16 does not earn **P6**.
- A right Riemann sum (e.g., $(-4)(2)+(-2)(2)+(0)(3)=-12$) does not earn **P6**.

Model Solution:

$(-1)(2)+(-4)(2)+(-2)(3)=-16$

0	1

The response accurately includes the criteria below.

- Answer with supporting work

Part D Point 7

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P7** is earned for considering either $g'(x)$ or $g'(1)$.

Model Solution:

$g'(x)=1\cdot f(x)+x\cdot f'(x)$

$g'(1)=f(1)+1\cdot f'(1)$

0	1

The response accurately includes the criteria below.

- Considers $g'(x)$

Part D Point 8

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P8** is earned with either $g'(x)=f(x)+x\cdot f'(x)$ or $g'(1)=f(1)+1\cdot f'(1)$.

5-1-1

Model Solution:

$g'(x) = 1 \cdot f(x) + x \cdot f'(x)$

$g'(1) = f(1) + 1 \cdot f'(1)$

0	1

The response accurately includes the criteria below.

- $g'(x)$ or $g'(1)$.

Part D Point 9

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response must have earned **P7** and **P8** to be eligible for **P9**.
- An eligible response that states either "decreasing because $g'(1) < 0$" or "decreasing because $g'(x) < 0$ for $0 < x < 2$" earns **P9**.
- A response that states "$g(1)$ is decreasing" does not earn **P9**.
- A response that estimates negative values for $f(1)$ and/or $f'(1)$ using the table and reaches a correct conclusion earns **P9**.

Model Solution:

Because $g'(1) < 0$ and g' is continuous on $0 < x < 2$, g is decreasing on an interval containing $x = 1$.

0	1

The response accurately includes the criteria below.

- Answer with reason

5-1-1

95. The table shown gives selected values of f', the first derivative of the twice-differentiable function *f*.

x	0	1	3	7	11
$f'(x)$	16	-10	-4	-1	2

In which of the following intervals does the Mean Value Theorem guarantee a number *c* such that $f''(c) = 3$?

(A) $(0, 1)$

(B) $(1, 3)$

(C) $(3, 7)$

(D) $(7, 11)$

Answer B

Correct. Because *f* is twice differentiable, the derivative f' is continuous on the interval $[1, 3]$ and differentiable on the interval $(1, 3)$. By the Mean Value Theorem, there is a number *c* with $1 < c < 3$ such that $f''(c) = \frac{f'(3)-f'(1)}{3-1} = \frac{-4-(-10)}{2} = \frac{6}{2} = 3$.

5-1-1

96.

x	-2	$-2 < x < -1$	-1	$-1 < x < 1$	1	$1 < x < 3$	3
$f(x)$	12	Positive	8	Positive	2	Positive	7
$f'(x)$	-5	Negative	0	Negative	0	Positive	$\frac{1}{2}$
$g(x)$	-1	Negative	0	Positive	3	Positive	1
$g'(x)$	2	Positive	$\frac{3}{2}$	Positive	0	Negative	-2

The twice-differentiable functions f and g are defined for all real numbers x. Values of f, f', g, and g' for various values of x are given in the table above.

(a) Find the x-coordinate of each relative minimum of f on the interval [-2, 3]. Justify your answers.

(b) Explain why there must be a value c, for $-1 < c < 1$, such that $f''(c) = 0$.

(c) The function h is defined by $h(x) = \ln(f(x))$. Find $h'(3)$. Show the computations that lead to your answer.

(d) Evaluate $\int_{-2}^{3} f'(g(x))g'(x)\,dx$.

Part A

1 point is earned for the answer with justification

$x = 1$ is the only critical point at which f' changes sign from negative to positive. Therefore, f has a relative minimum at $x = 1$.

0	1

The student response earns all of the following points:

1 point is earned for the answer with justification

$x = 1$ is the only critical point at which f' changes sign from negative to positive. Therefore, f has a relative minimum at $x = 1$.

Part B

1 point is earned for $f'(1) - f'(-1) = 0$

1 point is earned for the explanation, using Mean Value Theorem

f' is differentiable $\Rightarrow$ f' is continuous on the interval $-1 \leq x \leq 1$

$$\frac{f'(1)-f'(-1)}{1-(-1)} = \frac{0-0}{2} = 0$$

Therefore, by the Mean Value Theorem, there is at least one value c, $-1 < c < 1$, such that $f''(c) = 0$.

0	1	2

The student response earns all of the following points:

5-1-1

1 point is earned for $f'(1) - f'(-1) = 0$

1 point is earned for the explanation, using Mean Value Theorem

f' is differentiable ⇒ f' is continuous on the interval $-1 \leq x \leq 1$

$$\frac{f'(1)-f'(-1)}{1-(-1)} = \frac{0-0}{2} = 0$$

Therefore, by the Mean Value Theorem, there is at least one value c, $-1 < c < 1$, such that $f''(c) = 0$.

Part C

2 points are earned for $h'(x)$

1 point is earned for the answer

$$h'(x) = \frac{1}{f(x)} \cdot f'(x)$$
$$h'(3) = \frac{1}{f(3)} \cdot f'(3) = \frac{1}{7} \cdot \frac{1}{2} = \frac{1}{14}$$

0	1	2	3 ✓

The student response earns all of the following points:

2 points are earned for $h'(x)$

1 point is earned for the answer

$$h'(x) = \frac{1}{f(x)} \cdot f'(x)$$
$$h'(3) = \frac{1}{f(3)} \cdot f'(3) = \frac{1}{7} \cdot \frac{1}{2} = \frac{1}{14}$$

Part D

2 points are earned for stating Fundamental Theorem of Calculus

1 point is earned for the answer

$$\begin{aligned}\int_{-2}^{3} f'(g(x))g'(x)\,dx &= \Big[f(g(x))\Big]_{x=-2}^{x=3} \\ &= f(g(3)) - f(g(-2)) \\ &= f(1) - f(-1) \\ &= 2 - 8 = -6\end{aligned}$$

5-1-1

0	1	2	3

The student response earns all of the following points:

2 points are earned for stating Fundamental Theorem of Calculus

1 point is earned for the answer

$$\begin{aligned}\int_{-2}^{3} f'(g(x))g'(x)\,dx &= [f(g(x))]_{x=-2}^{x=3} \\ &= f(g(3)) - f(g(-2)) \\ &= f(1) - f(-1) \\ &= 2 - 8 = -6\end{aligned}$$

5-1-1

97.

t (min)	0	5	10	15	20	25	30	35	40
$v(t)$ (mpm)	7.0	9.2	9.5	7.0	4.5	2.4	2.4	4.3	7.3

A test plane flies in a straight line with positive velocity $v(t)$, in miles per minute at time t minutes, where v is a differentiable function of t. Selected values of $v(t)$ for $0 \le t \le 40$ are shown in the table above.

(a) Use a midpoint Riemann sum with four subintervals of equal length and values from the table to approximate $\int_0^{40} v(t)dt$. Show the computations that lead to your answer. Using correct units, explain the meaning of $\int_0^{40} v(t)dt$ in terms of the plane's flight.

(b) Based on the values in the table, what is the smallest number of instances at which the acceleration of the plane could equal zero on the open interval $0 < t < 40$? Justify your answer.

(c) The function f, defined by $f(t) = 6 + \cos\left(\frac{t}{10}\right) + 3\sin\left(\frac{7t}{40}\right)$, is used to model the velocity of the plane, in miles per minute, for $0 \le t \le 40$. According to this model, what is the acceleration of the plane at $t = 23$? Indicate units of measure.

(d) According to the model f, given in part (c), what is the average velocity of the plane, in miles per minute, over the time interval $0 \le t \le 40$?

Part A

1 point is earned for $v(5) + v(15) + v(25) + v(35)$

1 point is earned for the answer

1 point is earned for the meaning with units

Midpoint Riemann sum is $10 \cdot [v(5) + v(15) + v(25) + v(35)] = 10 \cdot [9.2 + 7.0 + 2.4 + 4.3] = 229$

The integral gives the total distance in miles that the plane flies during the 40 minutes.

0	1	2	3

The student response earns all of the following points:

1 point is earned for $v(5) + v(15) + v(25) + v(35)$

1 point is earned for the answer

1 point is earned for the meaning with units

Midpoint Riemann sum is

$$10 \cdot [v(5) + v(15) + v(25) + v(35)] = 10 \cdot [9.2 + 7.0 + 2.4 + 4.3] = 229$$

The integral gives the total distance in miles that the plane flies during the 40 minutes.

Part B

1 point is earned for the two instances

5-1-1

1 point is earned for the justification

By the Mean Value Theorem, $v'(t) = 0$ somewhere in the interval (0,15) and somewhere in the interval (25, 30). Therefore the acceleration will equal 0 for at least two values of t.

0	1	2

The student response earns all of the following points:

1 point is earned for the two instances

1 point is earned for the justification

By the Mean Value Theorem, $v'(t) = 0$ somewhere in the interval (0,15) and somewhere in the interval (25, 30). Therefore the acceleration will equal 0 for at least two values of t.

Part C

1 point is earned for the answer with units

$f'(23) = -0.407$ or -0.408 miles per minute^2

0	1

The student response earns all of the following points:

1 point is earned for the answer with units

$f'(23) = -0.407$ or -0.408 miles per minute^2

Part D

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Average velocity} = \frac{1}{40}\int_0^{40} f(t)dt$$

= 5.916 miles per minute

0	1	2	3

5-1-1

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Average velocity} = \frac{1}{40}\int_0^{40} f(t)dt$$

= 5.916 miles per minute

5-1-1

98.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	6	4	2	5
2	9	2	3	1
3	10	-4	4	2
4	-1	3	6	7

The functions f and g are differentiable for all real numbers, and g is strictly increasing. The table above gives values of the functions and their first derivatives at selected values of x. The function h is given by $h(x) = f(g(x)) - 6$.

(a) Explain why there must be a value r for $1 < r < 3$ such that $h(r) = -5$.

(b) Explain why there must be a value c for $1 < c < 3$ such that $h'(c) = -5$.

(c) Let w be the function given by $w(x) = \int_1^{g(x)} f(t)dt$. Find the value of $w'(3)$.

(d) If g^{-1} is the inverse function of g, write an equation for the line tangent to the graph of $y = g^{-1}(x)$ at $x = 2$.

Part A

1 point is earned for $h(1)$ and $h(3)$

1 point is earned for the conclusion, using IVT

$h(1) = f(g(1)) - 6 = f(2) - 6 = 9 - 6 = 3$

$h(3) = f(g(3)) - 6 = f(4) - 6 = -1 - 6 = -7$

Since $h(3) < -5 < h(1)$ and h is continuous, by the Intermediate Value Theorem, there exists a value r, $1 < r < 3$, such that $h(r) = -5$.

0	1	2

The student response earns all of the following points:

1 point is earned for $h(1)$ and $h(3)$

1 point is earned for the conclusion, using IVT

$h(1) = f(g(1)) - 6 = f(2) - 6 = 9 - 6 = 3$

$h(3) = f(g(3)) - 6 = f(4) - 6 = -1 - 6 = -7$

Since $h(3) < -5 < h(1)$ and h is continuous, by the Intermediate Value Theorem, there exists a value r, $1 < r < 3$, such that $h(r) = -5$.

Part B

1 point is earned for: $\dfrac{h(3) - h(1)}{3 - 1}$

1 point is earned for the conclusion, using MVT

5-1-1

$$\frac{h(3)-h(1)}{3-1}=\frac{-7-3}{3-1}=-5$$

Since h is continuous and differentiable, by the Mean Value Theorem, there exists a value c, $1 < c < 3$, such that $h'(c) = -5$.

0	1	2

The student response earns all of the following points:

1 point is earned for: $\frac{h(3)-h(1)}{3-1}$

1 point is earned for the conclusion, using MVT

$$\frac{h(3)-h(1)}{3-1}=\frac{-7-3}{3-1}=-5$$

Since h is continuous and differentiable, by the Mean Value Theorem, there exists a value c, $1 < c < 3$, such that $h'(c) = -5$.

Part C

1 point is earned for applying chain rule

1 point is earned for correct answer

$$w'(3)=f(g(3))\cdot g'(3)=f(4)\cdot 2=-2$$

0	1	2

The student response earns all of the following points:

1 point is earned for applying chain rule

1 point is earned for correct answer

$$w'(3)=f(g(3))\cdot g'(3)=f(4)\cdot 2=-2$$

Part D

1 point is earned for $g^{-1}(2)$

5-1-1

1 point is earned for $(g^{-1})'(2)$

1 point is earned for the tangent line equation

$g(1) = 2$, so $g^{-1}(2) = 1$.

$(g^{-1})'(2) = \frac{1}{g'(g^{-1}(2))} = \frac{1}{g'(1)} = \frac{1}{5}$

An equation of the tangent line is $y - 1 = \frac{1}{5}(x - 2)$.

0	1	2	3 ✓

The student response earns all of the following points:

1 point is earned for $g^{-1}(2)$

1 point is earned for $(g^{-1})'(2)$

1 point is earned for the tangent line equation

$g(1) = 2$, so $g^{-1}(2) = 1$.

$(g^{-1})'(2) = \frac{1}{g'(g^{-1}(2))} = \frac{1}{g'(1)} = \frac{1}{5}$

An equation of the tangent line is $y - 1 = \frac{1}{5}(x - 2)$.

5-1-1

99.

t (hours)	$R(t)$ (gallons per hour)
0	9.6
3	10.4
6	10.8
9	11.2
12	11.4
15	11.3
18	10.7
21	10.2
24	9.6

The rate at which water flows out of a pipe, in gallons per hour, is given by a differentiable function R of time t. The table above shows the rate as measured every 3 hours for a 24-hour period.

(a) Use a midpoint Riemann sum with 4 subdivisions of equal length to approximate $\int_0^{24} R(t)dt$. Using correct units, explain the meaning of your answer in terms of water flow.

(b) Is there some time t, $0 < t < 24$, such that $R'(t) = 0$? Justify your answer.

(c) The rate of water flow $R(t)$ can be approximated by $Q(t) = \frac{1}{79}\left(768 + 23t - t^2\right)$. Use $Q(t)$ to approximate the average rate of water flow during the 24-hour time period. Indicate units of measure.

Part A

1 point is earned for: $R(3) + R(9) + R(15) + R(21)$

1 point is earned for the answer

1 point is earned for the explanation

$$\begin{aligned}\int_0^{24} R(t)dt &\approx 6\left[R(3) + R(9) + R(15) + R(21)\right] \\ &= 6\left[10.4 + 11.2 + 11.3 + 10.2\right] \\ &= 258.6 \text{ gallons}\end{aligned}$$

This is an approximation to the total flow in gallons of water from the pipe in the 24-hour period.

0	1	2	3

The student response earns all of the following points:

1 point is earned for: $R(3) + R(9) + R(15) + R(21)$

1 point is earned for the answer

1 point is earned for the explanation

5-1-1

$$\int_0^{24} R(t)dt \approx 6\left[R(3)+R(9)+R(15)+R(21)\right]$$
$$= 6\left[10.4+11.2+11.3+10.2\right]$$
$$= 258.6 \text{ gallons}$$

This is an approximation to the total flow in gallons of water from the pipe in the 24-hour period.

Part B

1 point is earned for the answer

1 point is earned for MVT or equivalent

Yes;

Since $R(0) = R(24) = 9.6$, the Mean Value Theorem guarantees that there is a t, $0 < t < 24$, such that $R'(t) = 0$.

0	1	2

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for MVT or equivalent

Yes;

Since $R(0) = R(24) = 9.6$, the Mean Value Theorem guarantees that there is a t, $0 < t < 24$, such that $R'(t) = 0$.

Part C

1 point is earned for the limits and average value constant

1 point is earned for: $Q(t)$ as integrand

1 point is earned for the answer

Average rate of flow

$\approx$ average value of $Q(t)$

$$= \frac{1}{24}\int_0^{24} \frac{1}{79}\left(768+23t-t^2\right)dt$$
$$= 10.785 \text{ gal/hr} \quad \text{or} \quad 10.784 \text{ gal/hr}$$

1 point is earned for units

(units) Gallons in part (a) and gallons/hr in part (c), or equivalent.

5-1-1

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for the limits and average value constant

1 point is earned for: $Q(t)$ as integrand

1 point is earned for the answer

Average rate of flow

$$\approx \text{average value of } Q(t)$$
$$= \frac{1}{24}\int_0^{24} \frac{1}{79}\left(768 + 23t - t^2\right) dt$$
$$= 10.785 \text{ gal/hr} \;\text{ or }\; 10.784 \text{ gal/hr}$$

1 point is earned for units

(units) Gallons in part (a) and gallons/hr in part (c), or equivalent.

5-1-1

100. The wind chill is the temperature, in degrees Fahrenheit (°F), a human feels based on the air temperature, in degrees Fahrenheit, and the wind velocity v, in miles per hour (mph). If the air temperature is 32°F, then the wind chill is given by $W(v) = 55.6 - 22.1v^{0.16}$ and is valid for $5 \le v \le 60$.

(a) Find $W'(20)$. Using correct units, explain the meaning of $W'(20)$ in terms of the wind chill.

(b) Find the average rate of change of W over the interval $5 \le v \le 60$. Find the value of v at which the instantaneous rate of change of W is equal to the average rate of change of W over the interval $5 \le v \le 60$.

(c) Over the time interval $0 \le t \le 4$ hours, the air temperature is a constant 32°F. At time $t = 0$, the wind velocity is $v = 20$ mph. If the wind velocity increases at a constant rate of 5 mph per hour, what is the rate of change of the wind chill with respect to time at $t = 3$ hours? Indicate units of measure.

Part A

1 point is earned for the value

1 point is earned for the explanation

$$W'(20) = -22.1 \cdot 0.16 \cdot 20^{-0.84} = -0.285 \text{ or } -0.286$$

When $v = 20$ mph, the wind chill is decreasing at 0.286 $^\circ\mathrm{F}$ /mph.

1 point is earned for units in (a) and (c)

Units of °F/mph in (a) and °F/hr in (c)

0	1	2	3

The student response earns all of the following points:

1 point is earned for the value

1 point is earned for the explanation

$$W'(20) = -22.1 \cdot 0.16 \cdot 20^{-0.84} = -0.285 \text{ or } -0.286$$

When $v = 20$ mph, the wind chill is decreasing at 0.286 $^\circ\mathrm{F}$ /mph.

1 point is earned for units in (a) and (c)

Units of °F/mph in (a) and °F/hr in (c)

Part B

1 point is earned for the average rate of change.

1 point is earned for $W'(v)$ = average rate of change

1 point is earned for the value of v.

5-1-1

The average rate of change of W over the interval $5 \le v \le 60$ is $\dfrac{W(60) - W(5)}{60 - 5}$ = -0.253 or -0.254.

$W'(v) = \dfrac{W(60) - W(5)}{60 - 5}$ when $v = 23.011$.

0	1	2	3

The student response earns all of the following points:

1 point is earned for the average rate of change.

1 point is earned for $W'(v)$ = average rate of change

1 point is earned for the value of v.

The average rate of change of W over the interval $5 \le v \le 60$ is $\dfrac{W(60) - W(5)}{60 - 5}$ = -0.253 or -0.254.

$W'(v) = \dfrac{W(60) - W(5)}{60 - 5}$ when $v = 23.011$.

Part C

1 point is earned for $\dfrac{dv}{dt} = 5$

1 point is earned if uses $v(3) = 35$ or uses $v(5) = 20 + 5t$

1 point is earned for answer

$$\left.\frac{dW}{dt}\right|_{t=3} = \left.\left(\frac{dW}{dv} \cdot \frac{dv}{dt}\right)\right|_{t=3} = W'(35) \cdot 5 = -0.892^{\circ}\mathrm{F/hr}$$

OR

$$W = 55.6 - 22.1(20 + 5t)^{0.16}$$

$$\left.\frac{dW}{dt}\right|_{t=3} = -0.892\ ^{\circ}\mathrm{F/hr}$$

1 point is earned for units in (a) and (c)

5-1-1

Units of °F/mph in (a) and °F/hr in (c)

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for $\frac{dv}{dt} = 5$

1 point is earned if uses $v(3) = 35$ or uses $v(5) = 20 + 5t$

1 point is earned for answer

$$\left.\frac{dW}{dt}\right|_{t=3} = \left.\left(\frac{dW}{dv} \cdot \frac{dv}{dt}\right)\right|_{t=3} = W'(35) \cdot 5 = -0.892^{\circ}\mathrm{F/hr}$$

OR

$$W = 55.6 - 22.1(20 + 5t)^{0.16}$$

$\left.\frac{dW}{dt}\right|_{t=3}$ = -0.892 °F/hr

1 point is earned for units in (a) and (c)

Units of °F/mph in (a) and °F/hr in (c)